

Admissibility and the Graybill–Deal Estimator

A guided tour through the Graybill–Deal admissibility problem

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Preface

This textbook is the result of a summer of independent research that I undertook between graduating from undergraduate study and beginning graduate school. My goal was to provide a comprehensive, explanatory, and above all accessible survey of the existing work that has been done in pursuit of a better understanding of the Graybill–Deal estimator and the open question of its admissibility.

I began writing this as a way to organize what I was learning during the project. The literature contains an unusually rich collection of partial answers, but those answers live in different mathematical settings and do not always fit together nicely. Some concern finite-sample risk, some restrict the class of competing estimators, some use Bayes or nonstandard-analysis arguments, and others study only the first meaningful terms of a large-sample expansion. My aim here is to put those results into one continuous story and to make clear both what they establish and what they leave open.

The chapters begin with the common-mean normal model and the practical motivation for precision weighting, then build the decision-theoretic language of loss, risk, domination, and admissibility. From there, the discussion turns to the Graybill–Deal estimator itself and to the central technical obstacle: the natural sufficient statistic is not complete, so the usual Rao–Blackwell and Lehmann–Scheffé route cannot resolve the problem. The middle chapters explore the main ways researchers have worked around that obstacle: restricted equivariant classes, generalized Bayes and Blyth arguments, a nonstandard version of Blyth’s method using infinitesimal priors, and second-order asymptotic comparisons. The final chapters place these results beside alternative estimators and other notions of optimality before reviewing the current state of the art.

I hope that this can be an explanatory resource for those who are trying to quickly familiarize themselves with the current state of the admissibility question for the Graybill–Deal estimator.

All yours,
Aidan J. Wagner

A stylized, handwritten signature in black ink, appearing to read 'AJW' with a large, sweeping flourish underneath.

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Chapter 1:

Introduction to the Common Mean Problem

Suppose that you check the weather for today and one website says that there is a 90% chance of thunderstorms, but a different website reports clear skies all day. Each website estimates the probability that it will rain today, but these two estimates do not agree with one another. A natural next question is then: how do you combine these estimates into a single estimated probability that it rains today, or at least how do you combine them in an empirically responsible way? As statisticians, our first inclination may be to average the two probabilities by dividing their sum by two. When we average the two estimates like this, what we are essentially doing is giving each of the two estimates equal weight; 50% of the average comes from the first source and the remaining 50% comes from the second. This only makes sense to do when each of the two estimates is equally reliable as the other. Intuitively, if we know that the first website tends to predict the weather more accurately than the second, then it does not make sense to give each site equal say in the average:

More precise estimate $\Rightarrow$ More weight in the average

Less precise estimate $\Rightarrow$ Less weight in the average

To achieve this effect, we weight each estimate by its *precision*, which is defined as the reciprocal of its variance. If $\hat{\mu}_1$ and $\hat{\mu}_2$ are our two estimates with variances σ_1^2 and σ_2^2 , then the weighted combination

$$\hat{\mu} = \frac{(1/\sigma_1^2)\hat{\mu}_1 + (1/\sigma_2^2)\hat{\mu}_2}{1/\sigma_1^2 + 1/\sigma_2^2}$$

gives more say to whichever website has been historically more accurate, and it does so automatically, just by plugging in the right numbers. Using this weighted average technique, we obtain a single estimate that effectively combines both provided estimates while accounting for their reliability. But this assumes that the variances of the estimates are known, which is, unfortunately, rarely the case.

In situations where the true variance of our estimates is unknown, we have to do the best we can with what we have by estimating each variance from the same data that we are using to estimate the common mean. This creates a problem: the sample variances themselves are random. That is, a sample may look more precise than it actually is just by chance, or conversely a sample may appear excessively noisy even though it still contains useful information. Computing a sample variance from the same data that we used to compute our sample mean causes these two sample estimates to become entangled with one another. This makes determining the proper weights for each sample mean much more difficult, because the weights we want to use are themselves something we have to guess at.

The weather forecasting example is a useful warm-up, but it is also a little too informal to build a theory on; we don't typically get to write down a clean probability distribution for what a weather website reports. The same basic problem, though, shows up constantly in settings where we can. A lab might run the same assay on two different machines and get two different readings, each with its own unknown amount of noise. An engineer might pull samples from two batches of material that, according to the underlying theory, should have identical mean properties, even though one batch

was produced under tighter process control than the other. Whenever we have several independent samples that are all estimating the same unknown mean, but where the spread of each sample is unknown and possibly different from the others, we are looking at an instance of what statisticians call the *common mean problem*.

1.1 The Statistical Setup: Defining the Mathematical Environment The common mean problem begins with k independent populations: data drawn from one population tells us nothing about the probability law governing any other.

Definition – Common mean normal model

For $i = 1, \dots, k$, the i th population produces an independent sample

$$\mathbf{X}_{i1}, \dots, \mathbf{X}_{in_i} \stackrel{\text{iid}}{\sim} N(\boldsymbol{\mu}, \boldsymbol{\sigma}_i^2), \quad n_i \geq 2.$$

The parameter of interest is the common mean $\boldsymbol{\mu} \in \mathbb{R}$. The nuisance parameters are the variances

$$\boldsymbol{\sigma}_1^2, \dots, \boldsymbol{\sigma}_k^2 > 0,$$

which are completely unknown and may be unequal. Thus the full parameter is

$$\boldsymbol{\theta} = (\boldsymbol{\mu}, \boldsymbol{\sigma}_1^2, \dots, \boldsymbol{\sigma}_k^2) \in \mathbb{R} \times (0, \infty)^k,$$

but the quantity we want to estimate is only its first coordinate.

This general notation is useful because the common-mean literature includes many populations, unequal sample sizes, and several variance structures. The main development in this text, however, deliberately returns again and again to the equal-size two-sample case

$$k = 2, \quad n_1 = n_2 = n.$$

That case is the smallest setting in which the nuisance-variance difficulty is real: there is already a nontrivial variance ratio, the Graybill–Deal weight is random, and the admissibility question is not settled by the usual complete-class shortcuts. The later chapters use this two-sample model as the canonical proving ground. Broader k -sample versions are mentioned when they clarify the literature, but the detailed finite-sample, restricted-class, Bayes, and second-order arguments are developed for this canonical case.

For each population, the raw observations reduce to two familiar summaries:

$$\bar{\mathbf{X}}_i = \frac{1}{n_i} \sum_{j=1}^{n_i} \mathbf{X}_{ij}, \quad \mathbf{S}_i^2 = \frac{1}{n_i - 1} \sum_{j=1}^{n_i} (\mathbf{X}_{ij} - \bar{\mathbf{X}}_i)^2.$$

Here $\mathbf{S}_i^2$ denotes the unbiased sample variance. The quantity $\mathbf{S}_i$ is the corresponding sample error scale, so the estimated standard error of $\bar{\mathbf{X}}_i$ is $\mathbf{S}_i/\sqrt{n_i}$. The normal model gives the exact distributional facts

$$\bar{\mathbf{X}}_i \sim N\left(\boldsymbol{\mu}, \frac{\boldsymbol{\sigma}_i^2}{n_i}\right), \quad \frac{(n_i - 1)\mathbf{S}_i^2}{\boldsymbol{\sigma}_i^2} \sim \chi_{n_i-1}^2,$$

and, within each population, $\bar{\mathbf{X}}_i$ is independent of $\mathbf{S}_i^2$. Across different populations all these pairs are independent as well. Consequently, the sample means carry direct information about $\boldsymbol{\mu}$, while the sample variances carry information about how noisy those sample means are.

Definition – Sufficient statistic

Let $\{P_{\boldsymbol{\theta}} : \boldsymbol{\theta} \in \Theta\}$ be a statistical model. A statistic $\mathbf{T}(\mathbf{X})$ is **sufficient** for $\boldsymbol{\theta}$ if the conditional distribution of the full sample $\mathbf{X}$, given $\mathbf{T}(\mathbf{X})$, does not depend on $\boldsymbol{\theta}$. Equivalently, $\mathbf{T}$ retains all sample information about the parameter that is relevant to the likelihood.

Definition – Minimal sufficient statistic

Let $\{P_{\boldsymbol{\theta}} : \boldsymbol{\theta} \in \Theta\}$ be a statistical model. A statistic $\mathbf{T}(\mathbf{X})$ is **minimal sufficient** if the following two conditions hold:

- $\mathbf{T}(\mathbf{X})$ is sufficient for $\boldsymbol{\theta}$.
- For every sufficient statistic $\mathbf{U}(\mathbf{X})$, there exists a measurable function h such that $\mathbf{T}(\mathbf{X}) = h(\mathbf{U}(\mathbf{X}))$ almost surely.

Thus a minimal sufficient statistic is the coarsest statistic that still retains all information about $\boldsymbol{\theta}$ carried by the sample.

Now return to the common mean normal model and derive the statistic used in this problem. The joint density of the full sample is

$$f_{\boldsymbol{\theta}}(\mathbf{x}) = \prod_{i=1}^k (2\pi\boldsymbol{\sigma}_i^2)^{-n_i/2} \exp \left\{ -\frac{1}{2\boldsymbol{\sigma}_i^2} \sum_{j=1}^{n_i} (x_{ij} - \boldsymbol{\mu})^2 \right\}.$$

For each population, decompose the sum of squares into the part measuring the location of the sample mean and the part measuring spread around that sample mean:

$$\sum_{j=1}^{n_i} (x_{ij} - \boldsymbol{\mu})^2 = \sum_{j=1}^{n_i} (x_{ij} - \bar{\mathbf{x}}_i)^2 + n_i(\bar{\mathbf{x}}_i - \boldsymbol{\mu})^2 = (n_i - 1)\mathbf{s}_i^2 + n_i(\bar{\mathbf{x}}_i - \boldsymbol{\mu})^2.$$

Substituting this identity into the joint density gives

$$f_{\boldsymbol{\theta}}(\mathbf{x}) = \prod_{i=1}^k (2\pi\boldsymbol{\sigma}_i^2)^{-n_i/2} \exp \left\{ -\frac{(n_i - 1)\mathbf{s}_i^2 + n_i(\bar{\mathbf{x}}_i - \boldsymbol{\mu})^2}{2\boldsymbol{\sigma}_i^2} \right\}.$$

After this rewrite, the data enter the likelihood only through

$$\mathbf{T} = (\bar{\mathbf{X}}_1, \mathbf{S}_1^2, \dots, \bar{\mathbf{X}}_k, \mathbf{S}_k^2).$$

By the factorization criterion, $\mathbf{T}$ is sufficient for $(\boldsymbol{\mu}, \boldsymbol{\sigma}_1^2, \dots, \boldsymbol{\sigma}_k^2)$. Standard likelihood-ratio arguments further show that this statistic is minimal sufficient: two samples give the same value of $\mathbf{T}$ exactly when their likelihood ratio is free of $(\boldsymbol{\mu}, \boldsymbol{\sigma}_1^2, \dots, \boldsymbol{\sigma}_k^2)$ [12]. Once these values are known,

the remaining arrangement of observations inside each sample contributes no further parameter information.

A useful way to picture the situation: each sample mean is a vote for where μ lies, and each sample variance sets how loudly that vote should count. If the variances were known, the volume of each vote would be fixed once and for all. Because the variances must instead be estimated, the volume knobs are themselves random, and that randomness is the source of nearly every difficulty that follows.

To see the natural target, temporarily pretend that $\sigma_1^2, \dots, \sigma_k^2$ are known. The likelihood contribution involving μ is, up to factors not depending on μ ,

$$\prod_{i=1}^k \exp \left\{ -\frac{1}{2\sigma_i^2} \sum_{j=1}^{n_i} (\mathbf{X}_{ij} - \mu)^2 \right\}.$$

Maximizing this likelihood is the same as minimizing

$$\sum_{i=1}^k \frac{1}{\sigma_i^2} \sum_{j=1}^{n_i} (\mathbf{X}_{ij} - \mu)^2.$$

Differentiate with respect to μ :

$$\frac{d}{d\mu} \sum_{i=1}^k \frac{1}{\sigma_i^2} \sum_{j=1}^{n_i} (\mathbf{X}_{ij} - \mu)^2 = -2 \sum_{i=1}^k \frac{n_i}{\sigma_i^2} (\bar{\mathbf{X}}_i - \mu).$$

Setting this derivative equal to zero gives

$$\sum_{i=1}^k \frac{n_i}{\sigma_i^2} \bar{\mathbf{X}}_i = \mu \sum_{i=1}^k \frac{n_i}{\sigma_i^2},$$

so the known-variance maximum likelihood estimator has the following form.

Theorem – Known-variance MLE

If $\sigma_1^2, \dots, \sigma_k^2$ are known, then the maximum likelihood estimator of the common mean is the precision-weighted average

$$\hat{\mu}(\sigma_1^2, \dots, \sigma_k^2) = \frac{\sum_{i=1}^k (n_i / \sigma_i^2) \bar{\mathbf{X}}_i}{\sum_{i=1}^k n_i / \sigma_i^2}.$$

The weight on $\bar{\mathbf{X}}_i$ is proportional to n_i / σ_i^2 , the precision of the i th sample mean, and this formula anchors the Graybill–Deal idea. Since the true variances are unavailable, replacing them by their sample estimates gives

$$\hat{\mu}_{\text{GD}} = \frac{\sum_{i=1}^k (n_i / \mathbf{S}_i^2) \bar{\mathbf{X}}_i}{\sum_{i=1}^k n_i / \mathbf{S}_i^2}.$$

Chapter 3 studies this estimator in detail; for now the important point is just where it comes from. Graybill–Deal is the plug-in version of the estimator we would use if the variances happened to be known.

1.2 Practical Motivation: Why the Common Mean Matters Graybill and Deal were concerned specifically with the following dilemma: when two independent, unbiased pieces of information estimate the same scientific quantity, how should they be combined if their precisions are unknown [7]? The point is easiest to see through the kind of experimental-design problem that made the question important.

Suppose an agricultural experiment is run as a *balanced incomplete block design* (BIBD) with *random block effects*. In such a design, experimental units are grouped into blocks such as field plots with similar growing conditions; each block contains only a subset of the treatments, but the design is balanced so that treatment pairs appear together symmetrically across blocks. The block-to-block differences are treated as random variation with their own variance component rather than as fixed constants.

The investigator wants a treatment effect τ_i , but the design supplies two routes to it:

the **intra-block estimate**, built from comparisons inside blocks

the **inter-block estimate**, which recovers information from how block-level averages move

These estimates arise from different parts of the experimental structure, so it is reasonable in the reduced analysis to treat them as independent. Both are aimed at the same target τ_i , but they need not be equally reliable [2, 12]; in simplified notation,

$$\hat{\tau}_{\text{intra}} \sim N(\tau_i, \sigma_{\text{intra}}^2), \quad \hat{\tau}_{\text{inter}} \sim N(\tau_i, \sigma_{\text{inter}}^2),$$

with the two estimates independent.

At first glance, one might try to settle the matter by choosing a constant weight in advance. For example, perhaps the intra-block estimate should always get 70% of the weight and the inter-block estimate 30%. But that kind of rule treats the relative usefulness of the two estimates as fixed, when it is not. The block effects are themselves random, so the amount of information carried by the inter-block estimate depends on the variance of those random block effects. When block-to-block variation is large, block averages contain different information than when block-to-block variation is small. Thus no single constant weight can be right across designs or data sets. The appropriate weight depends on the unknown variance components: if σ_{intra}^2 is small, the intra-block estimate should receive more weight; if σ_{inter}^2 is small, the inter-block estimate should receive more weight. The analyst does not know in advance which case holds.

Write the two independent unbiased estimators more abstractly as $\hat{\alpha}_1$ and $\hat{\alpha}_2$, both estimating the same quantity α , with variances v_1 and v_2 . A fixed-weight combination is

$$\hat{\alpha}_c = c\hat{\alpha}_1 + (1 - c)\hat{\alpha}_2.$$

It is unbiased whenever $\mathbf{c} + (1 - \mathbf{c}) = 1$, and independence gives

$$\mathbb{V}(\hat{\alpha}_{\mathbf{c}}) = \mathbf{c}^2 \mathbf{v}_1 + (1 - \mathbf{c})^2 \mathbf{v}_2.$$

One might hope to choose a single constant $\mathbf{c}$ that always improves on both original estimators. That would require

$$\mathbf{c}^2 \mathbf{v}_1 + (1 - \mathbf{c})^2 \mathbf{v}_2 \leq \mathbf{v}_1 \quad \text{and} \quad \mathbf{c}^2 \mathbf{v}_1 + (1 - \mathbf{c})^2 \mathbf{v}_2 \leq \mathbf{v}_2$$

for every possible pair $\mathbf{v}_1, \mathbf{v}_2 > 0$, with a strict improvement somewhere. But no such constant exists.

To see why, compare the combined estimator with each original estimator in the two extreme variance cases:

$$\mathbb{V}(\hat{\alpha}_{\mathbf{c}}) \leq \mathbf{v}_1 \implies \mathbf{c}^2 \mathbf{v}_1 + (1 - \mathbf{c})^2 \mathbf{v}_2 \leq \mathbf{v}_1 \quad \text{for all } \mathbf{v}_1, \mathbf{v}_2 > 0.$$

Now let $\mathbf{v}_2 \rightarrow \infty$. If $(1 - \mathbf{c})^2 > 0$, then the left-hand side becomes arbitrarily large, so the inequality cannot hold. Hence

$$(1 - \mathbf{c})^2 = 0 \implies \mathbf{c} = 1.$$

Thus, any constant weight that uniformly improves on $\hat{\alpha}_1$ must put all its weight on $\hat{\alpha}_1$.

Repeating the same comparison with $\hat{\alpha}_2$,

$$\mathbb{V}(\hat{\alpha}_{\mathbf{c}}) \leq \mathbf{v}_2 \implies \mathbf{c}^2 \mathbf{v}_1 + (1 - \mathbf{c})^2 \mathbf{v}_2 \leq \mathbf{v}_2 \quad \text{for all } \mathbf{v}_1, \mathbf{v}_2 > 0.$$

Now let $\mathbf{v}_1 \rightarrow \infty$. If $\mathbf{c}^2 > 0$, then the left-hand side becomes arbitrarily large, so the inequality cannot hold. Hence

$$\mathbf{c}^2 = 0 \implies \mathbf{c} = 0.$$

Uniform improvement over $\hat{\alpha}_1$ forces $\mathbf{c} = 1$, while uniform improvement over $\hat{\alpha}_2$ forces $\mathbf{c} = 0$. A single constant cannot satisfy both conditions, so fixed weights cannot solve the problem across all unknown variance configurations.

This calculation explains why the Graybill–Deal idea is so natural. The combining rule must adapt its weights to the data, because which source is more precise is itself unknown ahead of time. Graybill–Deal does this by making the weights functions of the observed sample variances. In the BIBD story, that means the intra-block and inter-block estimates are not combined by habit or by a fixed design convention; they are combined according to the precision suggested by the realized data.

$$\underbrace{\hat{\alpha}_{\text{known}} = \frac{\frac{1}{\mathbf{v}_1} \hat{\alpha}_1 + \frac{1}{\mathbf{v}_2} \hat{\alpha}_2}{\frac{1}{\mathbf{v}_1} + \frac{1}{\mathbf{v}_2}}}_{\text{if } \mathbf{v}_1, \mathbf{v}_2 \text{ were known}} \implies \left\{ \begin{array}{ll} \text{Fixed constant weight:} & \hat{\alpha}_{\mathbf{c}} = \mathbf{c} \hat{\alpha}_1 + (1 - \mathbf{c}) \hat{\alpha}_2 \quad \text{X} \\ \text{Graybill–Deal} & \\ \text{estimated precision weights:} & \hat{\alpha}_{\text{GD}} = \frac{\frac{1}{\hat{\mathbf{v}}_1} \hat{\alpha}_1 + \frac{1}{\hat{\mathbf{v}}_2} \hat{\alpha}_2}{\frac{1}{\hat{\mathbf{v}}_1} + \frac{1}{\hat{\mathbf{v}}_2}} \quad \checkmark \end{array} \right.$$

The same structure appears outside agricultural designs. In a classic consulting-style example often attributed to S. Zacks, a soil engineer samples two different soils whose true mean property is known from scientific theory to be the same. The observed samples may have very different variabilities because the soils differ in texture, sampling conditions, or measurement stability. Treating both sample means equally ignores precision, but trusting the more stable-looking sample too aggressively risks being misled by a finite-sample variance estimate. This is the same common mean problem in a different costume.

Example 1 (Two instruments). Imagine two measuring devices calibrated to estimate the same physical constant. Device 1 gives tightly clustered measurements in some runs and noisy measurements in others; device 2 behaves similarly. If the long-run variances were known, the better device would receive the larger weight. In practice, each run only gives a finite estimate of each variance. A method that overreacts to an unusually small observed variance can put too much confidence in a lucky source. The common mean problem is the formal version of this tension.

A worked numerical illustration. Suppose the two devices are pointed at the same physical constant μ , and each is run $n_1 = n_2 = 5$ times, producing

Device 1	12.1	11.9	12.0	12.2	11.8
Device 2	11.0	14.0	13.0	12.0	15.0

By eye, Device 1's readings huddle close together while Device 2's readings are scattered over a much wider range; we should expect Device 1 to deserve more say in the combined estimate. The sample means and unbiased sample variances make this precise. For Device 1,

$$\begin{aligned}\overline{X}_1 &= \frac{12.1 + 11.9 + 12.0 + 12.2 + 11.8}{5} = 12.0, \\ \mathbf{S}_1^2 &= \frac{(0.1)^2 + (-0.1)^2 + (0)^2 + (0.2)^2 + (-0.2)^2}{5 - 1} = \frac{0.10}{4} = 0.025.\end{aligned}$$

For Device 2,

$$\begin{aligned}\overline{X}_2 &= \frac{11.0 + 14.0 + 13.0 + 12.0 + 15.0}{5} = 13.0, \\ \mathbf{S}_2^2 &= \frac{(-2)^2 + (1)^2 + (0)^2 + (-1)^2 + (2)^2}{5 - 1} = \frac{10}{4} = 2.5.\end{aligned}$$

So Device 1 is roughly 100 times more precise than Device 2 in this run: $\mathbf{S}_2^2 / \mathbf{S}_1^2 = 2.5 / 0.025 = 100$.

Naive average. Ignoring precision altogether and simply averaging the two readings gives

$$\frac{\overline{X}_1 + \overline{X}_2}{2} = \frac{12.0 + 13.0}{2} = 12.5,$$

which lets the noisy Device 2 pull the combined estimate a full half-unit away from Device 1, even though Device 1's measurements look far more trustworthy.

Graybill–Deal estimate. Weighting each sample mean by its estimated precision $n_i/\mathbf{S}_i^2$ instead gives

$$\frac{n_1}{\mathbf{S}_1^2} = \frac{5}{0.025} = 200, \quad \frac{n_2}{\mathbf{S}_2^2} = \frac{5}{2.5} = 2,$$

so that

$$\hat{\mu}_{\text{GD}} = \frac{(n_1/\mathbf{S}_1^2)\overline{\mathbf{X}}_1 + (n_2/\mathbf{S}_2^2)\overline{\mathbf{X}}_2}{n_1/\mathbf{S}_1^2 + n_2/\mathbf{S}_2^2} = \frac{200(12.0) + 2(13.0)}{200 + 2} = \frac{2400 + 26}{202} = \frac{2426}{202} \approx 12.01.$$

Because Device 1's estimated precision so dominates Device 2's, $\hat{\mu}_{\text{GD}}$ lands almost on top of $\overline{\mathbf{X}}_1$ and barely moves toward $\overline{\mathbf{X}}_2$, exactly the behavior we wanted when we first set up the precision-weighted average.

Why the weighting matters. The estimated variance of the combined estimate also tells a stark story. Treating $\mathbf{S}_i^2/n_i$ as a stand-in for $\mathbb{V}(\overline{\mathbf{X}}_i)$, the naive average has estimated variance

$$\frac{1}{4} \left(\frac{\mathbf{S}_1^2}{n_1} + \frac{\mathbf{S}_2^2}{n_2} \right) = \frac{1}{4} \left(\frac{0.025}{5} + \frac{2.5}{5} \right) = \frac{1}{4} (0.005 + 0.5) \approx 0.126,$$

for a standard error of about 0.355, while the precision-weighted estimate has estimated variance

$$\frac{1}{n_1/\mathbf{S}_1^2 + n_2/\mathbf{S}_2^2} = \frac{1}{202} \approx 0.00495,$$

for a standard error of about 0.070 – roughly a fivefold reduction. By letting the data decide which device to trust, $\hat{\mu}_{\text{GD}}$ not only lands closer to the more reliable device's reading but also comes with a much tighter estimated standard error than treating both devices as equally informative.

This is, of course, only a single run of five measurements per device, so $\mathbf{S}_1^2$ and $\mathbf{S}_2^2$ are themselves noisy estimates of the true long-run variances. Had Device 1's five readings happened to scatter more than usual just by chance, $\mathbf{S}_1^2$ would look larger than its long-run value, and $\hat{\mu}_{\text{GD}}$ would (incorrectly) shift weight away from the better instrument. This is exactly the entanglement problem described above: the weights in $\hat{\mu}_{\text{GD}}$ are built from the same data used to estimate μ , so the weights inherit sampling variability of their own. Quantifying how much that variability affects $\hat{\mu}_{\text{GD}}$ is the central question that Chapters 2 and 3, work toward answering.

Chapter 2:

First Principles of Statistical Decision Theory

2.1 Evaluating Performance: The Foundation of Risk Statistical decision theory separates three ingredients: the unknown state of nature, the action taken by the statistician, and the loss incurred when that action is wrong [1]. In the common mean problem the state of nature is $\theta = (\mu, \sigma_1^2, \dots, \sigma_k^2)$, and the action is a numerical estimate of μ .

Definition – Decision rule

A decision rule is a measurable function of the observed data. If the full sample is denoted by $\mathbf{X}$, then a rule for estimating μ has the form

$$\delta(\mathbf{X}) \in \mathbb{R}.$$

Measurability is a technical regularity condition, not a substantive restriction: it ensures that probabilities and expectations involving $\delta(\mathbf{X})$ are well defined. Every ordinary estimator built from sample means, sample variances, and algebraic operations satisfies it. Because Chapter 1 identified

$$\mathbf{T} = (\bar{\mathbf{X}}_1, \mathbf{S}_1^2, \dots, \bar{\mathbf{X}}_k, \mathbf{S}_k^2)$$

as sufficient, many rules of interest can be written as $\delta(\mathbf{T})$, but the definition of a decision rule allows any function of the observed sample.

Definition – Squared-error loss

A loss function says how costly an action is when the true parameter is θ . For estimation of μ , squared-error loss is

$$L(\theta, a) = (a - \mu)^2.$$

This loss is zero when the estimate is exactly correct, positive otherwise, and grows quadratically as the error grows. The quadratic growth gives severe penalties to large errors. It is also mathematically convenient: it connects directly to variance calculations, least squares, orthogonal projection, and conditional expectation.

Once a decision rule is chosen, the loss is no longer a fixed number. It is the random variable

$$L(\theta, \delta(\mathbf{X})) = (\delta(\mathbf{X}) - \mu)^2,$$

because $\delta(\mathbf{X})$ depends on random data. The frequentist way to summarize long-run performance at a fixed parameter value is to average this random loss under the sampling distribution determined by that parameter.

Definition – Risk function

The risk function of a decision rule δ is its expected loss:

$$\mathbf{R}(\theta; \delta) = \mathbb{E}_\theta\{L(\theta, \delta(\mathbf{X}))\} = \mathbb{E}_{\mu, \sigma_1^2, \dots, \sigma_k^2}\{(\delta(\mathbf{X}) - \mu)^2\}.$$

Risk should therefore be thought of as a function of the whole parameter vector rather than a single universal score.

Under squared-error loss, risk decomposes into variance plus squared bias:

$$\mathbf{R}(\boldsymbol{\theta}; \boldsymbol{\delta}) = \mathbb{V}_{\boldsymbol{\theta}}(\boldsymbol{\delta}) + \{\mathbb{E}_{\boldsymbol{\theta}}(\boldsymbol{\delta}) - \boldsymbol{\mu}\}^2.$$

This identity follows by writing

$$\boldsymbol{\delta} - \boldsymbol{\mu} = \{\boldsymbol{\delta} - \mathbb{E}_{\boldsymbol{\theta}}(\boldsymbol{\delta})\} + \{\mathbb{E}_{\boldsymbol{\theta}}(\boldsymbol{\delta}) - \boldsymbol{\mu}\},$$

squaring, and taking expectation. The cross term has expectation zero because $\mathbb{E}_{\boldsymbol{\theta}}\{\boldsymbol{\delta} - \mathbb{E}_{\boldsymbol{\theta}}(\boldsymbol{\delta})\} = 0$. If $\boldsymbol{\delta}$ is unbiased for $\boldsymbol{\mu}$, then $\mathbb{E}_{\boldsymbol{\theta}}(\boldsymbol{\delta}) = \boldsymbol{\mu}$ for every $\boldsymbol{\theta}$, and the risk simplifies to

$$\mathbf{R}(\boldsymbol{\theta}; \boldsymbol{\delta}) = \mathbb{V}_{\boldsymbol{\theta}}(\boldsymbol{\delta}).$$

Decision theorists compare estimators by evaluating their risk functions across all possible parameter values. A simulation at $\boldsymbol{\sigma}_1^2 = \boldsymbol{\sigma}_2^2$ may show that one estimator looks excellent when the two populations have equal precision. But admissibility asks for all variance ratios, from nearly equal variances to cases where one source is vastly more reliable than the other.

2.2 The Gold Standard of Optimality: Admissibility

Risk functions can be partially ordered.

Definition – **R**-better than

Let $\boldsymbol{\delta}_1$ and $\boldsymbol{\delta}_2$ be two decision rules for estimating $\boldsymbol{\mu}$. We say $\boldsymbol{\delta}_1$ is **R**-better than $\boldsymbol{\delta}_2$ if

$$\mathbf{R}(\boldsymbol{\theta}; \boldsymbol{\delta}_1) \leq \mathbf{R}(\boldsymbol{\theta}; \boldsymbol{\delta}_2) \quad \text{for every } \boldsymbol{\theta},$$

and if the inequality is strict for at least one parameter value.

This ordering is only partial: many pairs of estimators cannot be compared this way, since one may have smaller risk for some variance ratios and larger risk for others.

Definition – Admissibility

An estimator $\boldsymbol{\delta}_0$ is admissible in a class $\mathcal{D}$ if there is no estimator $\boldsymbol{\delta} \in \mathcal{D}$ that is **R**-better than $\boldsymbol{\delta}_0$. If such a $\boldsymbol{\delta}$ exists, then $\boldsymbol{\delta}_0$ is inadmissible and $\boldsymbol{\delta}$ dominates it.

Equivalently, $\boldsymbol{\delta}_0$ is inadmissible if some competitor guarantees equal or smaller risk everywhere in the parameter space and strictly smaller risk somewhere. Using an inadmissible rule is difficult to defend once the dominating rule is known: the dominated rule never wins and sometimes loses.

Admissibility is nevertheless not the same thing as practical optimality. It is a weak criterion in one sense: it only says no estimator is everywhere better, so many admissible estimators are still unattractive if they perform poorly in important regions of the parameter space. It is an exacting criterion in another sense: “everywhere” means the entire parameter space, including every possible

configuration of the nuisance variances. The phrase “strict inequality for at least one parameter value” also matters. A competitor that ties δ_0 almost everywhere and wins only on a small region still counts as domination, while an estimator that does better at many parameter values but worse at even one does not dominate at all. Admissibility ranks risk functions the way the Pareto criterion ranks allocations, not the way an average-performance score would.

For the Graybill–Deal estimator, this exact criterion is unusually hard to settle. The estimator is natural, unbiased, and supported by important finite-sample and asymptotic results. But proving exact admissibility among all estimators under ordinary squared-error loss is a much stronger claim than showing that it beats a few obvious competitors. The difficulty of that exact claim is one of the central themes of this textbook.

2.3 The Classical Toolkit The usual route to optimal unbiased estimation uses three classical ideas: sufficiency, Rao–Blackwellization, and completeness.

Recall from Chapter 1 that a sufficient statistic retains all parameter-relevant information in the sample: once it is known, the conditional distribution of the remaining raw data does not depend on the unknown parameter. In the common mean model,

$$\mathbf{T} = (\bar{\mathbf{X}}_1, \mathbf{S}_1^2, \dots, \bar{\mathbf{X}}_k, \mathbf{S}_k^2)$$

is sufficient for $(\boldsymbol{\mu}, \boldsymbol{\sigma}_1^2, \dots, \boldsymbol{\sigma}_k^2)$. If an estimator uses details of the raw sample beyond $\mathbf{T}$, those details may add randomness but cannot add parameter information.

The Rao–Blackwell theorem turns that observation into an improvement procedure. Start with an estimator $\delta_0(\mathbf{X})$, and condition it on a sufficient statistic:

$$\delta_1(\mathbf{T}) = \mathbb{E}_{\boldsymbol{\theta}}\{\delta_0(\mathbf{X}) \mid \mathbf{T}\}.$$

Although the notation contains $\boldsymbol{\theta}$, sufficiency ensures that the conditional distribution used in this expectation does not depend on the unknown parameter, so δ_1 is a genuine statistic.

Theorem – Rao–Blackwell theorem

If $\mathbf{T}$ is sufficient and $\delta_1(\mathbf{T}) = \mathbb{E}_{\boldsymbol{\theta}}\{\delta_0(\mathbf{X}) \mid \mathbf{T}\}$, then δ_1 has risk no larger than δ_0 under squared-error loss. If δ_0 is unbiased for $\boldsymbol{\mu}$, then δ_1 is also unbiased.

Proof

Because $\mathbf{T}$ is sufficient, the conditional distribution of $\mathbf{X}$ given $\mathbf{T}$ does not depend on $\boldsymbol{\theta}$, so $\delta_1(\mathbf{T}) = \mathbb{E}_{\boldsymbol{\theta}}\{\delta_0(\mathbf{X}) \mid \mathbf{T}\}$ is a genuine statistic and not merely a formal expression involving the unknown parameter.

Unbiasedness is immediate from the tower property of conditional expectation:

$$\mathbb{E}_{\boldsymbol{\theta}}\{\delta_1(\mathbf{T})\} = \mathbb{E}_{\boldsymbol{\theta}}[\mathbb{E}_{\boldsymbol{\theta}}\{\delta_0(\mathbf{X}) \mid \mathbf{T}\}] = \mathbb{E}_{\boldsymbol{\theta}}\{\delta_0(\mathbf{X})\} = \boldsymbol{\mu},$$

so if δ_0 is unbiased for $\boldsymbol{\mu}$ then so is δ_1 .

For the risk inequality, write $\delta_0(\mathbf{X}) - \mu$ as

$$\delta_0(\mathbf{X}) - \mu = \{\delta_0(\mathbf{X}) - \delta_1(\mathbf{T})\} + \{\delta_1(\mathbf{T}) - \mu\}.$$

Squaring and taking conditional expectation given $\mathbf{T}$,

$$\mathbb{E}_\theta\{(\delta_0(\mathbf{X}) - \mu)^2 \mid \mathbf{T}\} = \mathbb{E}_\theta\{(\delta_0(\mathbf{X}) - \delta_1(\mathbf{T}))^2 \mid \mathbf{T}\} + (\delta_1(\mathbf{T}) - \mu)^2,$$

since the cross term vanishes: $\delta_1(\mathbf{T})$ is a function of $\mathbf{T}$, and $\mathbb{E}_\theta\{\delta_0(\mathbf{X}) - \delta_1(\mathbf{T}) \mid \mathbf{T}\} = 0$ by the definition of δ_1 . The first term on the right is nonnegative, so

$$\mathbb{E}_\theta\{(\delta_0(\mathbf{X}) - \mu)^2 \mid \mathbf{T}\} \geq (\delta_1(\mathbf{T}) - \mu)^2.$$

Taking expectation over $\mathbf{T}$ and applying the tower property once more,

$$\mathbb{E}_\theta\{(\delta_0(\mathbf{X}) - \mu)^2\} \geq \mathbb{E}_\theta\{(\delta_1(\mathbf{T}) - \mu)^2\},$$

that is, $\mathbf{R}(\theta; \delta_0) \geq \mathbf{R}(\theta; \delta_1)$ for every θ . Hence δ_1 has risk no larger than δ_0 everywhere, which is the claimed conclusion. ■

The unbiasedness part also follows from iterated expectation:

$$\mathbb{E}_\theta\{\delta_1(\mathbf{T})\} = \mathbb{E}_\theta[\mathbb{E}_\theta\{\delta_0(\mathbf{X}) \mid \mathbf{T}\}] = \mathbb{E}_\theta\{\delta_0(\mathbf{X})\} = \mu,$$

so Rao–Blackwellization preserves unbiasedness.

Under squared-error loss it also cannot increase risk. The key identity is conditional Jensen's inequality, or equivalently the projection property of conditional expectation:

$$\mathbb{E}_\theta\{(\delta_1(\mathbf{T}) - \mu)^2\} \leq \mathbb{E}_\theta\{(\delta_0(\mathbf{X}) - \mu)^2\}.$$

Thus δ_1 is $\mathbf{R}$ -equivalent to or $\mathbf{R}$ -better than δ_0 . The practical lesson is simple: once a sufficient statistic is available, look for good estimators among functions of it.

Rao–Blackwellization improves estimators, but it does not automatically identify a unique best one. For that, the classical theory adds completeness.

Definition – Completeness

A statistic $\mathbf{T}$ is complete for a family of distributions $\{P_\theta : \theta \in \Theta\}$ if, whenever a measurable function g satisfies

$$\mathbb{E}_\theta\{g(\mathbf{T})\} = 0 \quad \text{for every } \theta \in \Theta,$$

then

$$g(\mathbf{T}) = 0 \quad \text{almost surely for every } \theta \in \Theta.$$

In words, no nonzero function of $\mathbf{T}$ has expectation zero for every parameter value: there is no hidden unbiased wiggle room. If two unbiased estimators are both functions of a complete sufficient

statistic, their difference has expectation zero for every θ , so completeness forces that difference to vanish almost surely.

This is the content behind the Lehmann–Scheffe theorem [10].

Theorem – Lehmann–Scheffe theorem

If T is complete and sufficient, then any unbiased estimator that is a function of T is the unique uniformly minimum variance unbiased estimator, or UMVUE.

Proof

Let $\delta^* = \delta^*(T)$ be an unbiased estimator of μ that is a function of the complete sufficient statistic T . We must show two things: that δ^* has risk no larger than any other unbiased estimator at every θ , and that any other unbiased function of T must coincide with δ^* almost surely.

Minimum variance: Let $\delta_0(X)$ be any unbiased estimator of μ , not necessarily a function of T , and set $\delta_1(T) = \mathbb{E}_\theta\{\delta_0(X) \mid T\}$. Since T is sufficient, this conditional expectation does not depend on θ and so defines a genuine statistic, and by the Rao–Blackwell theorem δ_1 is unbiased for μ with $R(\theta; \delta_1) \leq R(\theta; \delta_0)$ for every θ . Thus it suffices to compare δ^* against unbiased estimators that are themselves functions of T , since any unbiased estimator is weakly dominated by its Rao–Blackwellization.

Uniqueness: Suppose $\delta^{**}(T)$ is any other unbiased estimator of μ that is a function of T . Define $g(T) = \delta^*(T) - \delta^{**}(T)$. Then g is a measurable function of T and, by unbiasedness of both estimators,

$$\mathbb{E}_\theta\{g(T)\} = \mathbb{E}_\theta\{\delta^*(T)\} - \mathbb{E}_\theta\{\delta^{**}(T)\} = \mu - \mu = 0 \quad \text{for every } \theta.$$

Completeness of T then forces

$$g(T) = \delta^*(T) - \delta^{**}(T) = 0 \quad \text{almost surely, for every } \theta.$$

Hence $\delta^{**} = \delta^*$ almost surely: there is at most one unbiased function of T . Combined with the minimum-variance argument above, every unbiased estimator is weakly dominated by the unique unbiased function of T , so δ^* is the UMVUE. ■

The theorem combines Rao–Blackwell’s risk improvement with completeness’s uniqueness: conditioning any unbiased estimator on T cannot increase variance, and completeness prevents two different unbiased functions of T from competing.

The common mean problem reaches its first breaking point here. Chapter 1 identified the minimal sufficient statistic

$$T = (\bar{X}_1, S_1^2, \dots, \bar{X}_k, S_k^2),$$

but this statistic is not complete. Therefore the Lehmann–Scheffe theorem cannot be used to declare a unique UMVUE for μ . We can reduce attention to the sufficient statistic, but we cannot then

say, “and now the unbiased estimator is unique.” The problem lives in the gap between sufficiency and completeness.

That structural failure is what motivates the methods developed later in this book. Restricted classes, equivariance, generalized Bayes arguments, Blyth-type approximations, second-order admissibility, and nonstandard-analysis perspectives on unbiased admissibility all enter the picture for the same reason: the classical tools have not decisively answered the question of admissibility for the Graybill–Deal estimator.

Chapter 3:

The Genesis and Properties of the Graybill–Deal Estimator

3.1 The Baseline As derived in Chapter 1, if $\sigma_1^2, \dots, \sigma_k^2$ are known, the likelihood equation for μ gives

$$\hat{\mu}(\sigma_1^2, \dots, \sigma_k^2) = \frac{\sum_{i=1}^k (n_i / \sigma_i^2) \bar{X}_i}{\sum_{i=1}^k n_i / \sigma_i^2}.$$

This estimator is the MLE, the BLUE without normality, and the UMVUE under the normal model [12, 11]. Its logic is simple: weight each mean by its precision n_i / σ_i^2 .

One can derive the same formula without likelihoods. Among linear unbiased estimators $\sum_i a_i \bar{X}_i$, unbiasedness requires $\sum_i a_i = 1$, while the variance is $\sum_i a_i^2 \sigma_i^2 / n_i$. A Lagrange multiplier calculation gives $a_i \propto n_i / \sigma_i^2$. Thus the known-variance estimator is not merely a maximum-likelihood artifact; it is what the geometry of independent noisy measurements demands.

3.2 The Unknown Variance Dilemma When the variances are unknown, the likelihood equations become

$$\sum_{i=1}^k \frac{n_i (\bar{X}_i - \mu)}{\sigma_i^2} = 0, \quad \sigma_i^2 = \frac{1}{n_i} \sum_{j=1}^{n_i} (X_{ij} - \mu)^2.$$

The variance equations themselves contain μ , so substitution does not give a clean closed-form MLE for μ . The natural known-variance formula is still tempting, but the true precisions are unavailable.

3.3 Constructing the Estimator Graybill and Deal’s move is the most direct possible plug-in idea: replace each unknown σ_i^2 by the corresponding sample variance S_i^2 . The result is

$$\hat{\mu}_{\text{GD}} = \frac{\sum_{i=1}^k (n_i / S_i^2) \bar{X}_i}{\sum_{i=1}^k n_i / S_i^2}.$$

Because the random weights are functions only of $S_1^2, \dots, S_k^2$, and because the S_i^2 ’s are independent of the $\bar{X}_i$ ’s, conditioning on the sample variances shows

$$\mathbb{E}(\hat{\mu}_{\text{GD}} \mid S_1^2, \dots, S_k^2) = \mu.$$

Thus $\hat{\mu}_{\text{GD}}$ is unbiased.

For $k = 2$ and equal sample sizes, this becomes especially transparent:

$$\hat{\mu}_{\text{GD}} = \bar{X}_1 + (\bar{X}_2 - \bar{X}_1) \frac{S_1^2}{S_1^2 + S_2^2}.$$

If S_1^2 is large, the first sample looks noisy, so the estimator moves toward $\bar{X}_2$; if S_2^2 is large, it moves toward $\bar{X}_1$.

3.4 Finite-Sample Variance Bounds The pleasant unbiasedness calculation does not automatically imply good risk. Graybill and Deal proved a sharp two-population result. If the two samples have degrees of freedom $m_i = n_i - 1$, then $\hat{\mu}_{\text{GD}}$ has uniformly smaller variance than both individual sample means if and only if m_1 and m_2 both exceed nine [7]. In the usual sample-size notation this means $n_1, n_2 \geq 11$. The threshold is a degrees-of-freedom threshold, stated here in sample-size notation only because this textbook writes the sample variance with $n_i - 1$ degrees of freedom.

Theorem – Graybill–Deal variance threshold

Assume $k = 2$. Let $\bar{X}_1$ and $\bar{X}_2$ be independent normal sample means with common mean μ and variances σ_1^2/n_1 and σ_2^2/n_2 . Let $(n_i - 1)S_i^2/\sigma_i^2 \sim \chi_{n_i-1}^2$, independently of the sample means and of the other sample variance. Then

$$\mathbb{V}(\hat{\mu}_{\text{GD}}) < \frac{\sigma_i^2}{n_i}, \quad i = 1, 2,$$

for every $\sigma_1^2, \sigma_2^2 > 0$ if and only if $n_1, n_2 \geq 11$.

Proof sketch

Set $m_i = n_i - 1$. The two-sample estimator can be written as

$$\hat{\mu}_{\text{GD}} = \gamma \bar{X}_1 + (1 - \gamma) \bar{X}_2, \quad \gamma = \frac{n_1 S_2^2}{n_1 S_2^2 + n_2 S_1^2}.$$

The random weight γ is independent of $(\bar{X}_1, \bar{X}_2)$, so

$$\mathbb{V}(\hat{\mu}_{\text{GD}}) = \frac{\sigma_1^2}{n_1} \mathbb{E}(\gamma^2) + \frac{\sigma_2^2}{n_2} \mathbb{E}\{(1 - \gamma)^2\}.$$

This identity is the whole reason the proof reduces to degrees of freedom. The variance of $\hat{\mu}_{\text{GD}}$ is controlled by the second moments of the random weight, and those moments are determined by an F -ratio of the two sample variances.

To compare $\hat{\mu}_{\text{GD}}$ with $\bar{X}_1$, first consider the case $\sigma_1^2/n_1 \leq \sigma_2^2/n_2$. Define

$$a = \frac{n_2 \sigma_1^2}{n_1 \sigma_2^2}, \quad V = \frac{S_1^2/\sigma_1^2}{S_2^2/\sigma_2^2}.$$

Then $0 < a \leq 1$, $V \sim F_{m_1, m_2}$, and

$$\gamma = \frac{1}{1 + aV}.$$

Since $\sigma_2^2/n_2 = (\sigma_1^2/n_1)/a$, the variance identity becomes

$$\mathbb{V}(\hat{\mu}_{\text{GD}}) = \frac{\sigma_1^2}{n_1} \mathbb{E} \left\{ \frac{1 + aV^2}{(1 + aV)^2} \right\}.$$

Thus $\mathbb{V}(\hat{\mu}_{\text{GD}}) < \mathbb{V}(\bar{X}_1)$ is equivalent to

$$\mathbb{E}h_a(V) < 1, \quad h_a(v) = \frac{1 + av^2}{(1 + av)^2}, \quad 0 < a \leq 1.$$

The same calculation with the two samples interchanged gives the comparison with $\bar{X}_2$. Hence the theorem is a question about when both F_{m_1, m_2} and F_{m_2, m_1} make this displayed expectation less than one for every allowed value of a .

Now the degrees of freedom enter through the second moment of the F -ratio. For $V \sim F_{m_1, m_2}$,

$$\mathbb{E}V = \frac{m_2}{m_2 - 2}, \quad \mathbb{E}V^2 = \frac{m_2^2(m_1 + 2)}{m_1(m_2 - 2)(m_2 - 4)}$$

when these moments exist. Graybill and Deal compare $h_a(v)$ with the parabola

$$u_a(v) = \frac{1 + a(v - 1)^2}{1 + a},$$

which lies above $h_a(v)$ for $v > 0$ and $0 < a \leq 1$. Therefore

$$\mathbb{E}h_a(V) \leq \mathbb{E}u_a(V) = \frac{1 + a \mathbb{E}(V^2 - 2V + 1)}{1 + a}.$$

If $m_1, m_2 \geq 10$, then

$$\mathbb{E}(V^2 - 2V) \leq 0,$$

with the same inequality after interchanging m_1 and m_2 . Consequently $\mathbb{E}h_a(V) < 1$ for $0 < a \leq 1$, and the two comparisons give $\mathbb{V}(\hat{\mu}_{\text{GD}}) < \mathbb{V}(\bar{X}_1)$ and $\mathbb{V}(\hat{\mu}_{\text{GD}}) < \mathbb{V}(\bar{X}_2)$ for all positive σ_1^2, σ_2^2 .

The converse explains why the boundary is exact. If one of the degrees of freedom is below ten, Graybill and Deal return to the same one-dimensional expectation and choose a variance ratio for which it exceeds one. The most transparent part of the argument occurs near the edge of the parameter space. In the notation above, this means taking a close to zero after choosing the ordering of the two samples. Since $h_0(V) = 1$, the first derivative of $\mathbb{E}h_a(V)$ at $a = 0$ is

$$\left. \frac{d}{da} \mathbb{E}h_a(V) \right|_{a=0} = \mathbb{E}(V^2 - 2V).$$

Differentiating under the expectation is justified in the representative boundary case $m_1 = m_2 = 9$, because $F_{9,9}$ has a finite third moment. For $V \sim F_{9,9}$,

$$\mathbb{E}V = \frac{9}{7}, \quad \mathbb{E}V^2 = \frac{9^2(9 + 2)}{9(9 - 2)(9 - 4)} = \frac{99}{35}.$$

Therefore

$$\left. \frac{d}{da} \mathbb{E}h_a(V) \right|_{a=0} = \frac{99}{35} - 2 \cdot \frac{9}{7} = \frac{9}{35} > 0.$$

Because $\mathbb{E}h_0(V) = 1$ and the derivative at zero is positive, there is a small $a > 0$ for which $\mathbb{E}h_a(V) > 1$. When $m_1 = m_2 = 9$, this means that for a sufficiently small positive variance ratio $a = \sigma_1^2/\sigma_2^2$, the variance of $\hat{\mu}_{\text{GD}}$ is larger than the variance of $\bar{X}_1$. Thus the sample-size pair $n_1 = n_2 = 10$ already fails the uniform comparison.

The same derivative test handles many other subthreshold pairs, possibly after interchanging the two samples. The remaining boundary cases are checked by the same integral representation, with a no longer taken only near zero. In each case there is a positive variance ratio at which the Graybill–Deal estimator has larger variance than one of the individual sample means. This proves the necessity of the $m_i \geq 10$ condition and therefore the stated $n_i \geq 11$ threshold. ■

Later work corrected and generalized the boundary for several populations: for k populations the necessary-and-sufficient pattern includes either $n_i \geq 11$ for all i , or one sample size equal to 10 with all the others at least 19 [12]. The proof for several populations has the same source of difficulty, but with many variance ratios instead of one. The two-sample proof above is the part that carries the main lesson: few degrees of freedom make inverse sample-variance weights too volatile.

This result is often the first surprise. The estimator that imitates the known-variance optimum is not uniformly variance-improving at very small sample sizes. Estimating a variance from few observations is noisy enough that the random weight can become a liability.

For teaching purposes, this is the first place to separate three ideas that are easy to blur:

- (a) $\hat{\mu}_{\text{GD}}$ is unbiased for every admissible sample size.
- (b) $\hat{\mu}_{\text{GD}}$ can have smaller variance than each individual sample mean under sufficiently large sample sizes.
- (c) Neither statement proves exact admissibility among all estimators.

Unbiasedness is an expectation identity. The finite-sample variance result compares $\hat{\mu}_{\text{GD}}$ only with the individual sample means. Admissibility compares $\hat{\mu}_{\text{GD}}$ with the entire universe of estimators.

Chapter 4:

The Theoretical Roadblock: Incompleteness and Its Consequences

4.1 The Minimal Sufficient Statistic As established in Chapter 1, the minimal sufficient statistic for $(\mu, \sigma_1^2, \dots, \sigma_k^2)$ is

$$\mathbf{T} = (\bar{X}_1, S_1^2, \dots, \bar{X}_k, S_k^2).$$

This statistic records, for each population, the observed sample mean and the observed within-sample variation. The statistic is sufficient because, after conditioning on these quantities, the remaining within-sample directions carry no information about the common mean or the variances. It is minimal because no coarser statistic can retain the likelihood ratios for all choices of $(\mu, \sigma_1^2, \dots, \sigma_k^2)$. Pal and Sinha state this point directly in their review: the collection of sample means and sample variances is minimal sufficient, even though it is not complete [12].

This distinction is the first roadblock. Rao–Blackwellization says that, for any unbiased estimator, conditioning on $\mathbf{T}$ produces another unbiased estimator with no larger variance at any parameter value. Thus it is reasonable to restrict attention to functions of $\mathbf{T}$ when one studies unbiased procedures. The restriction does not identify a single best function. To get uniqueness from the classical Rao–Blackwell–Lehmann–Scheffe route, the family of distributions of $\mathbf{T}$ must also be complete. Lehmann and Scheffe use completeness to turn an expectation identity into an almost-sure identity, which is the step that gives a unique minimum-variance unbiased function of a sufficient statistic [10].

The common-mean problem has sufficiency without that second property. The next section exhibits two nonzero functions of $\mathbf{T}$ whose expectations vanish throughout the parameter space. The first witness proves incompleteness as quickly as possible. The second witness contains the between-sample and within-sample comparison that later explains why the variance-ratio geometry belongs in the story.

4.2 The Breakdown of Completeness The presence of several nuisance variances may suggest that incompleteness is automatic. It is not. Completeness is a property of a family of distributions, and some models with nuisance parameters do have complete sufficient statistics. For example, if $X_1, \dots, X_n$ are independent $N(0, \sigma^2)$ observations with known mean zero, then $\sum_i X_i^2$ is complete and sufficient for the unknown variance. In the common-mean model, failure follows from a different feature: the shared-mean restriction creates nontrivial contrasts with expectation zero for every value of the nuisance variances. The explicit witnesses below establish the claim.

The distinction matters. In familiar complete exponential-family examples, an unbiased function of a complete sufficient statistic is the unique uniformly minimum-variance unbiased estimator of its expectation. Here, sufficiency survives but completeness does not. Rao–Blackwellization still reduces the class of estimators under consideration, but the usual uniqueness and UMVU conclusions cannot be obtained from the Lehmann–Scheffe theorem alone.

This is not a technical nuisance at the edge of the argument. The Graybill–Deal estimator is built from $\mathbf{T}$, is unbiased, and imitates the known-variance UMVU estimator. If the statistic were complete, those facts would put the estimator much closer to a classical optimality theorem. Since the statistic is not complete, there is room for distinct functions of the same observed summaries to have the same expectation profile. The rest of the chapter explains that room carefully, because later admissibility arguments must work in the space that completeness would otherwise have collapsed.

Theorem – Incompleteness of the natural common-mean statistic

In the normal common mean problem with $k \geq 2$, the natural sufficient statistic

$$\mathbf{T} = (\bar{X}_1, S_1^2, \dots, \bar{X}_k, S_k^2)$$

is not complete.

Proof

It is enough to exhibit one nonzero measurable function of $\mathbf{T}$ whose expectation is zero for every parameter value. Define

$$g(\mathbf{T}) = \bar{X}_1 - \bar{X}_2.$$

This is a function of the statistic $\mathbf{T}$. Since both samples have the same mean μ ,

$$\mathbb{E}_{\mu, \sigma_1^2, \dots, \sigma_k^2} \{g(\mathbf{T})\} = \mathbb{E}_{\mu, \sigma_1^2, \dots, \sigma_k^2} (\bar{X}_1) - \mathbb{E}_{\mu, \sigma_1^2, \dots, \sigma_k^2} (\bar{X}_2) = \mu - \mu = 0$$

for every μ and every choice of positive variances $\sigma_1^2, \dots, \sigma_k^2$.

But $g(\mathbf{T})$ is not almost surely zero. Indeed, under the model,

$$\bar{X}_1 - \bar{X}_2 \sim N \left(0, \frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2} \right),$$

whose variance is $\sigma_1^2/n_1 + \sigma_2^2/n_2 > 0$. The inequality follows from the model assumption that all variances are strictly positive, so the distribution is nondegenerate. Thus $\mathbb{P}_{\mu, \sigma_1^2, \dots, \sigma_k^2} (\bar{X}_1 - \bar{X}_2 = 0) = 0$, so $g(\mathbf{T})$ is not equal to zero almost surely. We have found a nonzero function of the sufficient statistic with expectation zero throughout the model, which violates completeness. ■

The preceding contrast proves the theorem, so a second witness is not needed for the formal conclusion. The contrast $g(\mathbf{T}) = \bar{X}_1 - \bar{X}_2$ is still too thin to explain why the obstruction later reappears as a problem about random weights. It only says that two sample means can differ even though they have the same expectation. The Graybill–Deal estimator depends more heavily on the sample variances, because those variances determine its plug-in precision weights. A useful witness should therefore involve both the disagreement between sample means and the within-sample evidence about their sampling variability.

The next witness has exactly that form. It compares the squared difference between the two sample means with the usual unbiased estimate of the variance of that difference. In the common-mean model those two quantities have the same expectation for every variance pair, although they are different random quantities. Section 4.3 turns this balance into the geometric picture.

Lemma — A variance-based witness

For $k \geq 2$, define

$$h(\mathbf{T}) = (\bar{\mathbf{X}}_1 - \bar{\mathbf{X}}_2)^2 - \frac{\mathbf{S}_1^2}{n_1} - \frac{\mathbf{S}_2^2}{n_2}.$$

Then

$$\mathbb{E}_{\mu, \sigma_1^2, \dots, \sigma_k^2} \{h(\mathbf{T})\} = 0$$

for every common mean and every positive variance vector, but $h(\mathbf{T})$ is not almost surely zero.

Proof

Since the two sample means are independent and have the same expectation,

$$\mathbb{E}\{(\bar{\mathbf{X}}_1 - \bar{\mathbf{X}}_2)^2\} = \mathbb{V}(\bar{\mathbf{X}}_1 - \bar{\mathbf{X}}_2) = \frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2},$$

while $\mathbb{E}(\mathbf{S}_i^2) = \sigma_i^2$. Hence $\mathbb{E}\{h(\mathbf{T})\} = 0$.

It remains to show that the witness is not almost surely zero. Set

$$V = \frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}.$$

The normal contrast $\bar{\mathbf{X}}_1 - \bar{\mathbf{X}}_2$ is independent of $(\mathbf{S}_1^2, \mathbf{S}_2^2)$. Also, $\mathbb{V}(Z^2) = 2V^2$ when $Z \sim N(0, V)$, and

$$\mathbb{V}(\mathbf{S}_i^2) = \frac{2\sigma_i^4}{n_i - 1}.$$

Independence therefore gives

$$\begin{aligned} \mathbb{V}\{h(\mathbf{T})\} &= 2V^2 + \frac{2\sigma_1^4}{n_1^2(n_1 - 1)} + \frac{2\sigma_2^4}{n_2^2(n_2 - 1)} \\ &> 0. \end{aligned}$$

The final inequality follows because the variances are positive and $n_1, n_2 \geq 2$. Hence $h(\mathbf{T})$ cannot be almost surely zero. ■

The proof uses only standard normal-sample facts, but the interpretation is worth spelling out. The random variable $(\bar{\mathbf{X}}_1 - \bar{\mathbf{X}}_2)^2$ measures the observed separation between two estimates of the same mean. Its expectation is large when at least one sample mean is noisy, because noisy means scatter apart even when their population means agree. The quantity $\mathbf{S}_1^2/n_1 + \mathbf{S}_2^2/n_2$ estimates that

same sampling variance from the within-sample data. Thus $\mathbf{h}(\mathbf{T})$ subtracts an internal estimate of the expected between-sample separation from the observed between-sample separation.

At a fixed parameter value, the two sides need not be equal. A particular data set may have sample means that are farther apart than the sample variances predict, or closer together than they predict. The point is that the common-mean assumption makes the average separation exactly match the average estimated sampling variance. Since $\mathbf{S}_i^2/\sigma_i^2$ has a distribution determined by the degrees of freedom, while the unstandardized $\mathbf{S}_i^2$ changes scale with σ_i^2 , both parts of $\mathbf{h}(\mathbf{T})$ move with the nuisance variances in the same first-moment way. They can cancel in expectation for every variance vector without being equal as random variables. That is the concrete mechanism behind the nonzero kernel element.

The next section specializes this witness to the equal-size two-sample case, where $n_1 = n_2 = n$, and places its expectation balance beside the scale-free variance-ratio geometry.

4.3 A Geometric View of Incompleteness The definition of completeness is algebraic: if $\mathbb{E}_{\theta}\{g(\mathbf{T})\} = 0$ for every parameter value, then $g(\mathbf{T}) = 0$ almost surely. The proof above has already shown that this implication fails. The geometric point is to identify what the failure is made of. There is a parameter space indexed by the common mean and the nuisance variances. There are random functions of the sufficient statistic. Between them is the expectation operation, which turns each random function into a function on the parameter space. Completeness says that this operation has no nonzero function hidden at expectation zero.

The equal-size two-sample case displays the same obstruction with the least notation. It also connects directly to the Graybill–Deal weight. In this case, known variances would determine the correct point between $\overline{\mathbf{X}}_1$ and $\overline{\mathbf{X}}_2$. Unknown variances replace that fixed point by a random variance-ratio weight. The witness $\mathbf{h}(\mathbf{T})$ does not estimate the weight itself. It explains why the same sample variances that create the random weight also create directions in the space of functions of $\mathbf{T}$ that expectation cannot see.

The figure is therefore a summary of a chain already in place. Panel 1 shows how the nuisance variances split into a total scale and a relative variance ratio. Panel 2 shows why the variance-based witness has mean zero at every point of that parameter space. Panel 3 abstracts the calculation into the expectation operator and its kernel. The panels answer three separate questions: where the model varies, why $\mathbf{h}(\mathbf{T})$ is centered everywhere, and why that centering is exactly a failure of completeness.

Consider the equal-size two-sample case and write

$$\boldsymbol{\rho} = \frac{\sigma_1^2}{\sigma_1^2 + \sigma_2^2}, \quad \hat{\boldsymbol{\rho}} = \frac{\mathbf{S}_1^2}{\mathbf{S}_1^2 + \mathbf{S}_2^2}.$$

We introduce $\boldsymbol{\rho}$ and $\hat{\boldsymbol{\rho}}$ here for the rest of the chapter: $\boldsymbol{\rho}$ is the fixed variance weight determined by the parameter, and $\hat{\boldsymbol{\rho}}$ is its sample-variance analogue. In the weighted-average representation

$$\boldsymbol{\delta}_{\phi} = \overline{\mathbf{X}}_1 + (\overline{\mathbf{X}}_2 - \overline{\mathbf{X}}_1)\phi,$$

ρ is exactly the ideal known-variance weight on $\overline{X}_2$. The Graybill–Deal estimator has the same representation with $\phi = \hat{\rho}$, so $\hat{\rho}$ is its observable estimate of the ideal weight. This is the same reduction emphasized in the review literature: in the two-population equal-size problem, performance of many weighted-average rules can be read as performance in estimating $\sigma_1^2/(\sigma_1^2 + \sigma_2^2)$ from variance information [12, 13]. The witness from the preceding lemma becomes

$$h(\mathbf{T}) = (\overline{X}_1 - \overline{X}_2)^2 - \frac{S_1^2}{n} - \frac{S_2^2}{n}$$

in this setting. On the event $\overline{X}_1 \neq \overline{X}_2$, this same quantity can be written as

$$h(\mathbf{T}) = (\overline{X}_1 - \overline{X}_2)^2 \left(1 - \frac{S_1^2/n + S_2^2/n}{(\overline{X}_1 - \overline{X}_2)^2} \right).$$

The original difference form defines the witness everywhere, including the zero-contrast event. The factored form gives the interpretation. The witness is positive when the observed squared separation of the two sample means is larger than the within-sample estimate of its sampling variance, and it is negative when that separation is smaller. Thus $h(\mathbf{T})$ asks whether the distance between the two sample means is surprising relative to the variability reported by the two samples.

This is the direct link to the Graybill–Deal weight. The weight $\hat{\rho}$ uses the same two sample variances to decide where the estimate should fall between $\overline{X}_1$ and $\overline{X}_2$. The witness uses their sum, scaled by n , to judge how much evidence the observed separation carries. In other words, the Graybill–Deal rule uses the sample-variance ratio to choose the direction and size of the pull along the line segment, while $h(\mathbf{T})$ measures whether the length of that segment is large or small compared with the within-sample noise. The two objects are therefore different functions of the data, but they are responding to the same random variance information.

Panel 1: parameter space. Every positive variance pair has a unique representation

$$(\sigma_1^2, \sigma_2^2) = a(\rho, 1 - \rho), \quad a > 0,$$

with

$$0 < \rho < 1.$$

Fixing ρ and varying a traces a ray in the variance plane. The ideal weight remains fixed along that ray, while the total noise level $\sigma_1^2 + \sigma_2^2 = a$ changes. Thus ρ tells the estimator how to divide trust between the two sample means, and a tells how noisy the two sample means are in total. Those are different roles. If both variances are multiplied by the same positive constant, the best known-variance weight does not move, although the absolute risk scale changes.

The observable ratio $\hat{\rho}$ has the same invariance to common scale. Writing $S_i^2 = \sigma_i^2 W_i$, where the law of W_i depends only on $n - 1$, gives

$$\hat{\rho} = \frac{\rho W_1}{\rho W_1 + (1 - \rho) W_2}.$$

The common multiplier a has disappeared. This cancellation is why ratio-based estimator classes arise later. It also explains why the diagram draws rays rather than isolated points: all points on a ray share the same ideal weight and the same distributional problem for estimating that weight. Panel 1 has not yet proved incompleteness. It has organized the variance parameters into the coordinates that the estimator and the later subclass arguments actually use.

Panel 2: expectation over the variance plane. The two parts of $h(T)$ measure different forms of variation. The squared contrast $(\bar{X}_1 - \bar{X}_2)^2$ measures variation between the sample means. The sum $(S_1^2 + S_2^2)/n$ measures variation within the samples. These are observable quantities from the same data, but they answer different questions. The first asks how far apart the two estimates of μ landed. The second asks how far apart they were expected to land if the within-sample variance estimates are trusted. Under the common-mean model,

$$\mathbb{E}\{(\bar{X}_1 - \bar{X}_2)^2\} = \frac{\sigma_1^2 + \sigma_2^2}{n}, \quad \text{and} \quad \mathbb{E}\left\{\frac{S_1^2 + S_2^2}{n}\right\} = \frac{\sigma_1^2 + \sigma_2^2}{n}.$$

Both expectations equal a/n on a ray with total scale a . Their difference therefore has expectation zero at every point of that ray and at every point of every other ray:

$$\mathbb{E}_{\theta}\{h(T)\} = 0 \quad \text{for every } \theta.$$

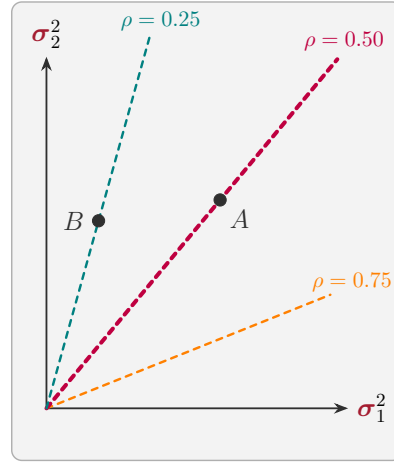
The horizontal surface in panel 2 records these means. The vertical shapes at a and B represent distributions of $h(T)$, not additional expectation surfaces. Those distributions may have different spreads and shapes because the variance parameters differ. Each remains centered at zero. The lemma in Section 4.2 showed that the spread is positive, so $h(T)$ does not collapse to the constant zero random variable.

This is the point where the nuisance variances matter most. Completeness would not fail merely because $h(T)$ has mean zero at one parameter value. Many nonconstant random variables are centered at a single distribution. The witness is damaging because the centering persists for every value of μ and every positive variance pair. The common mean removes any bias from $\bar{X}_1 - \bar{X}_2$, and the sample variances supply unbiased estimates of the two pieces of its

Why Completeness Fails: The Geometry of the Zero-Mean Witness $h(T)$

Parameter Space

Variance Plane & Ratio Laws

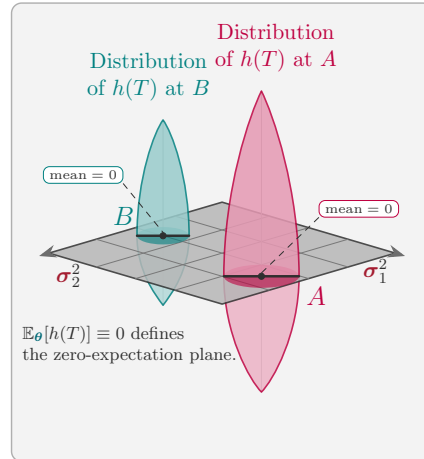


Each point is a variance pair $\theta = (\sigma_1^2, \sigma_2^2)$. Rays hold the ratio ρ fixed while the total scale changes.

↓ assign each θ the height $\mathbb{E}_{\theta}\{h(T)\}$

Expectation Surface

$\mathbb{E}\{h(T)\} = 0$ everywhere

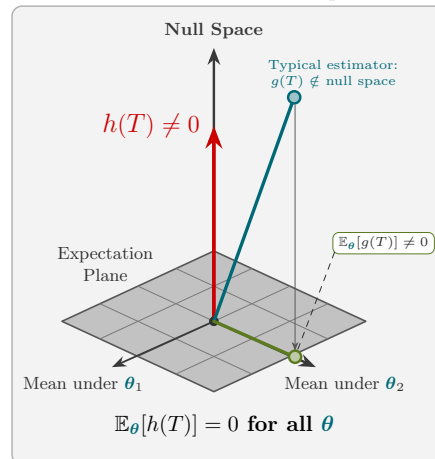


Every parameter point is assigned height $\mathbb{E}_{\theta}\{h(T)\} = 0$. The distributions can differ while sharing this zero mean.

↓ the zero surface is $\mathcal{E}(h) \equiv 0$

Function Space

The Nontrivial Null Space



Because $h(T) \neq 0$ but $\mathcal{E}(h) = 0$, the expectation map has a nonzero kernel direction. This is incompleteness.

variance. Those two facts hold across the whole parameter space. Panel 2 therefore shows a flat expectation profile, not a local coincidence.

Panel 3: the expectation operator. Let $\mathcal{F}$ be the class of integrable measurable functions of $\mathbf{T}$, and let $\mathcal{G}$ be the class of real-valued functions on the parameter space. The expectation operator $\mathcal{E} : \mathcal{F} \rightarrow \mathcal{G}$ sends q to its expectation profile, defined by $[\mathcal{E}(q)](\boldsymbol{\theta}) = \mathbb{E}_{\boldsymbol{\theta}}\{q(\mathbf{T})\}$. The output $\mathcal{E}(q)$ is the entire expectation profile of $q(\mathbf{T})$, one number for each parameter value. The kernel of $\mathcal{E}$ consists of the functions whose expectation profile is identically zero:

$$\ker(\mathcal{E}) = \{q \in \mathcal{F} : \mathcal{E}(q) \equiv 0\}.$$

Panel 3 is a schematic finite-dimensional view of this map. The two horizontal directions record expectations at two selected parameter values. A function such as $g(\mathbf{T})$ with nonzero expectations has a nonzero horizontal image. The vertical direction represents functions that are invisible to every expectation coordinate. The witness $\mathbf{h}(\mathbf{T})$ points in that direction because it is a nonzero random variable but $\mathcal{E}(\mathbf{h})$ is the zero function. Thus $\mathbf{h}$ is a nonzero element of $\ker(\mathcal{E})$. Completeness would force every element of this kernel to be zero almost surely. This nonzero kernel element is the exact reason completeness fails.

This operator language also clarifies what the figure is not saying. It is not claiming that the data contain no information about the variances, or that all unbiased estimators based on $\mathbf{T}$ have the same risk. It says something more specific. When a function of $\mathbf{T}$ lies in the kernel of $\mathcal{E}$, adding that function to an unbiased estimator leaves all first moments unchanged. Risk can still change, because risk involves second moments and cross terms. The later admissibility problem is hard precisely because first-moment identities leave many risk comparisons unresolved.

Taken together, the panels turn the definition of completeness into a spatial picture. Panel 1 supplies all parameter points. Panel 2 assigns height zero to every one of those points because the expectation profile of $\mathbf{h}(\mathbf{T})$ vanishes. Panel 3 identifies the source of that flat profile: the expectation operator has a nonzero kernel direction represented by $\mathbf{h}(\mathbf{T})$. A complete family would have no such direction. Its kernel would contain only the zero random variable.

The consequence for unbiased estimation is immediate. If $U(\mathbf{T})$ is unbiased for $\boldsymbol{\mu}$, then

$$U(\mathbf{T}) + c \mathbf{h}(\mathbf{T})$$

is also unbiased for every fixed c for which the expectation exists. This statement concerns expectation only. It does not say that the estimators have the same variance or risk. It also does not produce a rule that dominates the Graybill–Deal estimator. It shows why the usual uniqueness conclusion from the Lehmann–Scheffe theorem is unavailable [10, 12]. Unbiasedness and sufficiency leave a family of possible functions of $\mathbf{T}$. The later chapters use variance, risk, equivariance, Bayes arguments, and asymptotic criteria to study that remaining freedom.

4.4 Why Exact Admissibility Is Hard The preceding section identifies the failure of the classical theorem. The next question is what a direct admissibility proof would have to replace it with. Exact admissibility asks whether any estimator, biased or unbiased, linear or nonlinear,

smooth or irregular, can uniformly reduce risk. The risk of $\hat{\mu}_{\text{GD}}$ depends on the unknown variance vector through nonlinear functions of the sample variances. Even when $k = 2$, exact variance formulas involve special functions or infinite-series representations [8, 12]. Thus an optimality result confined to unbiased estimators would leave open the possibility of a biased dominator.

The difference between these two tasks is substantial. An unbiasedness theorem controls $\mathbb{E}_{\theta}(\delta)$. An admissibility theorem controls $\mathbb{E}_{\theta}\{(\delta - \mu)^2\}$ against every competing rule at every parameter value. A biased estimator may move its mean away from μ and still reduce risk if it lowers variance enough. A nonequivariant estimator may behave like $\hat{\mu}_{\text{GD}}$ in most regions and switch behavior in a thin part of the sample space. A discontinuous rule may be hard to rule out by smooth calculus. Completeness would have removed many unbiased alternatives from consideration, but it would still not have handled all biased competitors. Once completeness fails, even the unbiased part of the problem retains extra directions.

Unrestricted perturbations. Here is what a direct attack would look like in the equal-size two-sample case. Write a competitor as $\delta = \hat{\mu}_{\text{GD}} + h$, where h is an arbitrary measurable function of the data. Domination would require

$$\mathbf{R}(\mu, \sigma_1^2, \sigma_2^2; \hat{\mu}_{\text{GD}} + h) - \mathbf{R}(\mu, \sigma_1^2, \sigma_2^2; \hat{\mu}_{\text{GD}}) \leq 0 \quad \text{for every } (\mu, \sigma_1^2, \sigma_2^2),$$

with strict inequality somewhere. Expanding the square gives the pointwise identity

$$\{(\hat{\mu}_{\text{GD}} + h) - \mu\}^2 - (\hat{\mu}_{\text{GD}} - \mu)^2 = h^2 + 2h(\hat{\mu}_{\text{GD}} - \mu).$$

Taking expectation and substituting the equal-size two-sample form of $\hat{\mu}_{\text{GD}}$ gives

$$\mathbb{E}_{\mu, \sigma_1^2, \sigma_2^2} \left[h^2 + 2h \left\{ \bar{X}_1 + (\bar{X}_2 - \bar{X}_1) \frac{S_1^2}{S_1^2 + S_2^2} - \mu \right\} \right] \leq 0.$$

For unrestricted h , this inequality involves the joint distribution of all sample means and sample variances under every common mean and every variance pair.

The expression displays why a direct proof is not a matter of checking the variance of $\hat{\mu}_{\text{GD}}$ alone. The perturbation appears through h^2 , which is always nonnegative, and through the cross term $2h(\hat{\mu}_{\text{GD}} - \mu)$, whose sign depends on both the data rule and the parameter. A successful dominator would have to make the expected cross term negative enough to pay for the added square, uniformly over all $(\mu, \sigma_1^2, \sigma_2^2)$. The random weight in $\hat{\mu}_{\text{GD}}$ makes that cross term depend on the sample variances in a nonlinear way.

Restricted weighted averages. To expose one tractable part of the problem, now narrow the competitor class to the natural weighted-average form

$$\delta_{\phi} = \bar{X}_1 + (\bar{X}_2 - \bar{X}_1)\phi(S_1^2, S_2^2)$$

with ϕ depending only on the sample variances. This restriction removes the dependence on μ and yields the cleaner comparison

$$\frac{\sigma_1^2 + \sigma_2^2}{n} \mathbb{E}_{\sigma_1^2, \sigma_2^2} \left[\left(\phi(S_1^2, S_2^2) - \frac{\sigma_1^2}{\sigma_1^2 + \sigma_2^2} \right)^2 - \left(\frac{S_1^2}{S_1^2 + S_2^2} - \frac{\sigma_1^2}{\sigma_1^2 + \sigma_2^2} \right)^2 \right] \leq 0.$$

This narrowing illustrates the variance-ratio problem, but it does not settle admissibility outside the restricted class. The expectation is taken over two independent scaled chi-square variables. The unknown ideal weight $\sigma_1^2/(\sigma_1^2+\sigma_2^2)$ sits inside the loss, while the observable weight is a nonlinear random ratio. Showing that no measurable ϕ improves the second squared error everywhere is already a difficult variance-ratio decision problem. Allowing arbitrary h is harder still. This is the structural obstacle that the methods of Chapters 5 through 8 are each designed, in different ways, to circumvent.

This restricted calculation also explains why Chapter 5 begins with classes of weighted averages. If the estimator is required to stay on the line segment between $\bar{X}_1$ and $\bar{X}_2$, then the decision is encoded in the single random weight ϕ . If ϕ is allowed to inspect only variance ratios, or variance ratios normalized by D^2 , the problem becomes a precise question about invariant summaries. Sinha and Mouqadem's classes C_0, C_1, C_2 , and C formalize these choices, and Pal and Sinha record the resulting restricted admissibility and extended-admissibility conclusions [13, 12]. Those results are informative because they rule out natural improvements that preserve the weighted-average structure. They do not rule out every measurable perturbation.

The random fractional weights prevent a simple pointwise comparison. An unrestricted competitor may leave the weighted-average line or use information that does not reduce to a variance ratio. Numerical evidence can test many ratios and competitor families, but exact admissibility requires ruling out every measurable competitor over the continuous, unbounded parameter space.

4.5 What Has Been Ruled Out The most tempting direct route is therefore unavailable. The natural sufficient statistic remains useful, but it is not complete: nonzero functions of the sample means and sample variances have expectation zero throughout the parameter space. The standard Rao–Blackwell–Lehmann–Scheffe argument consequently cannot certify $\hat{\mu}_{GD}$ as the unique UMVU estimator. Nor would such an UMVU statement, by itself, settle admissibility against biased estimators. UMVU optimality compares variances only within the unbiased class. Admissibility must also exclude biased rules whose variance reduction may more than offset their squared bias.

An unrestricted admissibility proof is therefore more than a longer variance calculation. It must exclude domination by every measurable perturbation, for every common mean and every positive nuisance-variance vector, while the key risk comparison remains entangled with random variance-ratio weights. The figure in Section 4.3 marks the point where the classical path breaks: the expectation operator has a nonzero kernel. The algebra after the figure marks the point where the full decision-theoretic path becomes hard: risk domination requires second-moment inequalities over an unbounded parameter space.

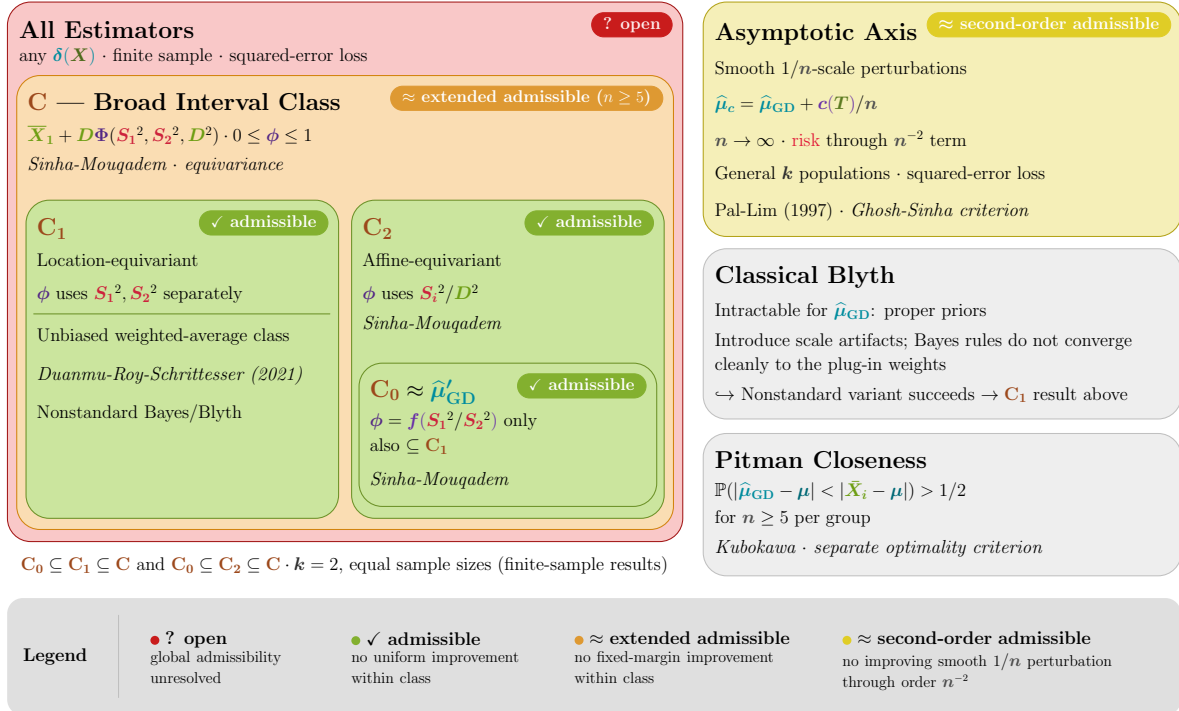
In the face of these difficulties, the path to admissibility splits into several branches. Each branch gives up some part of the full problem in order to make a precise theorem possible. One branch restricts the class of competitors. Another changes the proof technology and uses average-risk comparisons. Another changes the scale and studies asymptotic perturbations. A final branch compares $\hat{\mu}_{GD}$ with alternative estimators or with criteria other than ordinary squared-error admissibility. These branches do not merge into a complete global admissibility proof. They do, however, explain

different parts of the problem and show where a possible dominator would still have to live.

The following evidence map records this split. At this point in the book, most entries should be read as claims to be established later rather than as results already proved in the text.

Admissibility evidence for the Graybill-Deal estimator

Two independent axes · finite-sample restricted classes (left) · asymptotic perturbations (right)



Chapter 5 addresses the nested finite-sample classes on the left side of the diagram. It narrows the universe to structured equivariant subclasses and asks what can be proved when an estimator must remain a weighted average of $\bar{X}_1$ and $\bar{X}_2$, when its weight must lie between 0 and 1, and when that weight may inspect only specified invariant coordinates of (S_1^2, S_2^2, D^2) . This restriction converts an unrestricted search into a sequence of precise variance-ratio questions.

Chapter 6 addresses the classical Blyth box on the right. It explains why ordinary proper-prior Bayes arguments are hard to make converge cleanly to $\hat{\mu}_{GD}$. Chapter 7 then addresses the nonstandard Bayes route that feeds back into the **C₁**-type restricted conclusion on the left: infinitesimal prior masses make it possible to retain a Blyth-style comparison without introducing the same scale artifacts. Chapter 8 addresses the asymptotic box, where the question is whether a smooth $1/n$ -scale correction can improve $\hat{\mu}_{GD}$ at second order. Chapter 9 addresses the Pitman-closeness box and related comparisons, placing other estimators and other optimality criteria beside admissibility. Chapter 10 returns to this map after the promised results have been developed. The red outer box remains open throughout: none of these branches proves ordinary finite-sample admissibility against all estimators.

Bypassing the Roadblock I: Nested Subclasses and Equivariance

5.1 The Strategy of Restriction Exact admissibility for $\hat{\mu}_{\text{GD}}$ is an all-estimators statement in a model with nuisance variances. To prove it directly one would have to rule out every measurable rule, including biased, nonequivariant, discontinuous, and highly nonlinear procedures, over every possible value of $(\mu, \sigma_1^2, \sigma_2^2)$. That is too large a battlefield for the usual complete-class or Rao–Blackwell arguments.

The first way around this obstruction is to change the question without changing the statistical object. Instead of asking whether *any* estimator can dominate $\hat{\mu}_{\text{GD}}$, one asks whether a competitor exists inside a smaller subclass of estimators. Is $\hat{\mu}_{\text{GD}}$ admissible among estimators with the same symmetry? Among estimators that always lie between the two sample means? Among estimators whose weight is allowed to use only certain invariant summaries of the data?

These questions do not, themselves, settle the global problem. A restricted admissibility theorem says that a proposed improvement must violate at least one of the structural principles built into the subclass. If those principles are exactly the principles that make $\hat{\mu}_{\text{GD}}$ attractive, then the theorem narrows the search for a counterexample in a meaningful way.

This restriction strategy should not be discouraging; It is often how decision theory makes progress. By stepping back from broad classical tools and focusing on specific constraints, we actually gain a much richer, more nuanced understanding of the estimator in question. Once a meaningful symmetry or shape constraint has been identified, if a definitive result can be proven, then we can attempt to expand the proof from there.

In our case, the underlying structure makes perfect sense. The known-variance estimator operates simply as a weighted average of $\bar{X}_1$ and $\bar{X}_2$. Because $\hat{\mu}_{\text{GD}}$ imitates this by swapping in sample variances, the natural next question is whether any better rule exists among other data-dependent weighted averages of those exact means.

5.2 Admissibility Versus Extended Admissibility Before the subclasses are introduced, it is important to keep two decision-theoretic claims separate. The literature uses both, and they do not carry the same force.

Claim	What it rules out	What may remain possible
Admissibility of δ_0 in a class $\mathcal{D}$	No $\delta \in \mathcal{D}$ has $\mathbf{R}(\theta, \delta) \leq \mathbf{R}(\theta, \delta_0)$ for every θ , with strict improvement somewhere.	Nothing inside the class can uniformly dominate δ_0 , although estimators may still trade risk across parameter values.
Extended admissibility of δ_0 in $\mathcal{D}$	No $\delta \in \mathcal{D}$ improves by a fixed positive margin: $\mathbf{R}(\theta, \delta) \leq \mathbf{R}(\theta, \delta_0) - \epsilon$ for all θ , for some $\epsilon > 0$.	A delicate dominator is not excluded if its improvement is positive somewhere but shrinks toward zero along a boundary or tail of the parameter space.

Thus the phrase “extended admissible in $\mathcal{C}$ ” should be read cautiously. It is strong evidence against a simple across-the-board improvement, but it is weaker than admissibility in $\mathcal{C}$. By contrast, “admissible in $\mathcal{C}_0$ ” or “admissible in $\mathcal{C}_2$ ” means ordinary no-domination admissibility inside that specified class.

5.3 The Equal-Sample Two-Population Reduction The most developed restricted results concern the case $k = 2$ with equal sample sizes. The two-sample equal-size setting isolates the essential difficulty while keeping the notation manageable. Write

$$D = \bar{X}_2 - \bar{X}_1.$$

Any estimator that lies on the line segment between the two sample means can be written as

$$\delta = \bar{X}_1 + D\phi,$$

where $0 \leq \phi \leq 1$. If $\phi = 0$, the estimator is $\bar{X}_1$. If $\phi = 1$, it is $\bar{X}_2$. Intermediate values move the estimate toward the second sample mean.

The Graybill–Deal rule has exactly this form:

$$\hat{\mu}_{\text{GD}} = \bar{X}_1 + D \frac{S_1^2}{S_1^2 + S_2^2}.$$

Thus the admissibility question becomes a question about the random weight. A competitor in this family chooses a different point between $\bar{X}_1$ and $\bar{X}_2$. This reduces the all-estimators problem to a simpler weight-selection problem. That reduction is the bypass.

A domination result could still, in principle, come from a rule that leaves the interval between the two sample means, uses a discontinuous switching rule, introduces bias in a helpful way, or depends on the data through a statistic outside the restricted class. The restricted theorems tell us that a simple and natural weighted-average improvement is not available.

5.4 Defining the Subclasses Sinha and Mouqadem considered nested classes in which the weight ϕ is constrained to lie between 0 and 1 and to depend on different invariant combinations of the data [13]. In the notation used by Pal and Sinha's review, the classes are as follows.

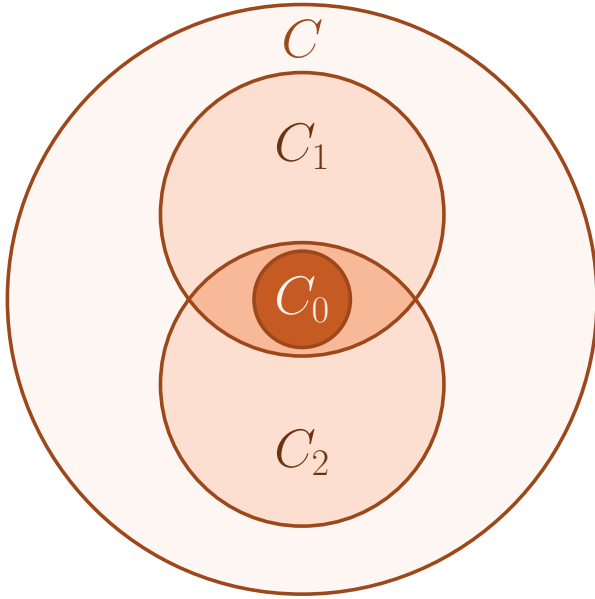
Definition – Sinha–Mouqadem subclasses

For the equal-sample two-population problem, define

$$\begin{aligned} \mathbf{C}_0 &= \{ \bar{\mathbf{X}}_1 + \mathbf{D}\phi_0(\mathbf{S}_2^2/\mathbf{S}_1^2) : 0 \leq \phi_0 \leq 1 \}, \\ \mathbf{C}_1 &= \{ \bar{\mathbf{X}}_1 + \mathbf{D}\phi_1(\mathbf{S}_1^2, \mathbf{S}_2^2) : 0 \leq \phi_1 \leq 1 \}, \\ \mathbf{C}_2 &= \{ \bar{\mathbf{X}}_1 + \mathbf{D}\phi_2(\mathbf{S}_1^2/\mathbf{D}^2, \mathbf{S}_2^2/\mathbf{D}^2) : 0 \leq \phi_2 \leq 1 \}, \\ \mathbf{C} &= \{ \bar{\mathbf{X}}_1 + \mathbf{D}\phi(\mathbf{S}_1^2, \mathbf{S}_2^2, \mathbf{D}^2) : 0 \leq \phi \leq 1 \}. \end{aligned}$$

The Graybill–Deal weight $\mathbf{S}_1^2/(\mathbf{S}_1^2 + \mathbf{S}_2^2)$ belongs to these classes in the appropriate ways. The inclusions are

$$\mathbf{C}_0 \subset \mathbf{C}_1 \subset \mathbf{C}, \quad \mathbf{C}_0 \subset \mathbf{C}_2 \subset \mathbf{C}.$$



Notice the nested structure. A theorem in $\mathbf{C}_0$ rules out competitors that use only the most classical invariant ratio. A theorem in $\mathbf{C}_2$ rules out a different and larger affine-equivariant class. A theorem in $\mathbf{C}$ would be much closer to a full answer inside interval-valued weighted averages, because $\mathbf{C}$ allows the weight to react to both the variances and the observed distance between the means.

Each class represents a different answer to the question, “What information may the weight use?”:

- In $\mathbf{C}_0$, the weight may use only the ratio $\mathbf{S}_2^2/\mathbf{S}_1^2$. This is the closest analogue of the known-variance rule, because when sample sizes are equal the relevant unknown precision comparison is determined by the variance ratio.
- In $\mathbf{C}_1$, the weight may use the two sample variances separately. This is broader than $\mathbf{C}_0$ because it can respond to the absolute scale of the observed within-sample variation, not merely to the ratio.
- In $\mathbf{C}_2$, the weight may use $\mathbf{S}_1^2/\mathbf{D}^2$ and $\mathbf{S}_2^2/\mathbf{D}^2$. This brings in the separation between the sample means through the normalizing factor $\mathbf{D}^2$. Large disagreement between $\bar{\mathbf{X}}_1$ and $\bar{\mathbf{X}}_2$ may therefore influence how the variance information is interpreted.

- In $\mathcal{C}$, the weight may use $\mathbf{S}_1^2$, $\mathbf{S}_2^2$, and $\mathbf{D}^2$ directly. This is the broadest interval-valued weighted-average class in the Sinha–Mouqadem framework.

Example 2 ($\mathcal{C}_0$ versus $\mathcal{C}_2$ weights). The classes $\mathcal{C}_0$ and $\mathcal{C}_2$ are easiest to understand by watching two candidate weights make different decisions on the same data. Suppose first that

$$\bar{X}_1 = 10, \quad \bar{X}_2 = 14, \quad \mathbf{S}_1^2 = 1, \quad \mathbf{S}_2^2 = 3.$$

Then $\mathbf{D} = \bar{X}_2 - \bar{X}_1 = 4$. The Graybill–Deal rule writes the estimate as $\bar{X}_1 + \mathbf{D}\hat{\phi}_{\text{GD}}$, where

$$\hat{\phi}_{\text{GD}} = \frac{\mathbf{S}_1^2}{\mathbf{S}_1^2 + \mathbf{S}_2^2} = \frac{1}{1 + 3} = 0.25.$$

Thus

$$\hat{\mu}_{\text{GD}} = 10 + 4(0.25) = 11.$$

The interpretation is concrete: sample 2 has the larger observed variance, so the estimator moves only 25% of the way from $\bar{X}_1$ toward $\bar{X}_2$.

A $\mathcal{C}_0$ competitor. A rule in $\mathcal{C}_0$ may look only at the variance ratio

$$\mathbf{r} = \frac{\mathbf{S}_2^2}{\mathbf{S}_1^2} = 3.$$

It cannot look separately at the absolute sizes of the variances, and it cannot look at how far apart $\bar{X}_1$ and $\bar{X}_2$ are. For example, consider the ratio-only weight

$$\phi_0(\mathbf{r}) = \frac{1}{1 + \mathbf{r}/3}.$$

At $\mathbf{r} = 3$, this gives

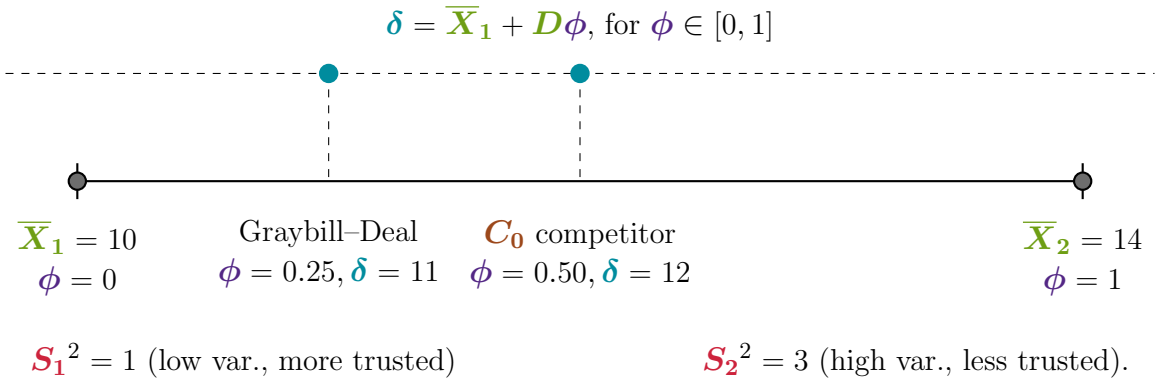
$$\phi_0(3) = \frac{1}{1 + 1} = 0.50,$$

so the corresponding estimate is

$$\bar{X}_1 + \mathbf{D}\phi_0(3) = 10 + 4(0.50) = 12.$$

This competitor is easy to motivate: it shrinks the observed variance ratio from 3 down to the observed error-scale ratio 1, so it moves less aggressively toward the lower-variance sample. The theorem says that no ratio-only adjustment of this kind, no matter how cleverly calibrated as a function of $\mathbf{r}$, can have uniformly smaller risk than Graybill–Deal over all variance configurations.

Visualizing the Weights. The diagram below maps these two competing estimates onto the line segment connecting $\bar{X}_1$ and $\bar{X}_2$, parameterized by the weight $\phi \in [0, 1]$. Because sample 1 has a lower variance ($\mathbf{S}_1^2 = 1$) than sample 2 ($\mathbf{S}_2^2 = 3$), both rules intuitively anchor their estimates closer to $\bar{X}_1$. However, the Graybill–Deal estimator responds more sharply to the variance discrepancy, taking a smaller step ($\phi = 0.25$) to land safely at $\delta = 11$. The ratio-only $\mathcal{C}_0$ competitor dilutes the impact of the observed variance difference, resulting in a larger weight ($\phi = 0.50$) that pulls the estimate further toward the less-trusted sample to $\delta = 12$.



Both points sit closer to $\overline{X}_1$ — the lower-variance sample pulls more weight.

What extra information C_2 gets. Now compare two data sets with the same observed variance ratio:

	$\overline{X}_1$	$\overline{X}_2$	S_1^2	S_2^2
Run A	10	14	1	3
Run B	10	11	1	3

For both runs, $r = 3$, so every C_0 rule must use the same weight in both runs. Graybill–Deal uses 0.25 in both, and the ratio-only competitor above uses 0.50 in both:

	D	$\hat{\mu}_{GD}$	$\overline{X}_1 + D\phi_0(3)$
Run A	4	11	12
Run B	1	10.25	10.50

A C_2 rule can distinguish these two runs because it may use

$$\left(\frac{S_1^2}{D^2}, \frac{S_2^2}{D^2} \right) = \begin{cases} \left(\frac{1}{16}, \frac{3}{16} \right), & \text{Run A,} \\ (1, 3), & \text{Run B.} \end{cases}$$

So C_2 can notice that Run A has a large separation between the sample means relative to the within-sample variances, while Run B has a much smaller separation.

To make the extra freedom visible, define a simple separation-sensitive weight

$$\phi_2(a, b) = \frac{1}{4} + \frac{0.15}{1 + a + b}, \quad a = \frac{S_1^2}{D^2}, \quad b = \frac{S_2^2}{D^2}.$$

This is not proposed as a good estimator; it is just a concrete member of the C_2 style of rule. It gives

$$\phi_2 \left(\frac{1}{16}, \frac{3}{16} \right) = 0.25 + \frac{0.15}{1.25} = 0.37,$$

but

$$\phi_2(1, 3) = 0.25 + \frac{0.15}{5} = 0.28.$$

Thus the $\mathcal{C}_2$ estimates would be

	ϕ_2	$\bar{X}_1 + D\phi_2$
Run A	0.37	11.48
Run B	0.28	10.28

This illustrates exactly what the $\mathcal{C}_2$ theorem rules out. It says that even after allowing the weight to respond to the observed separation of the sample means through the affine-invariant ratios above, no rule in this class can uniformly dominate $\hat{\mu}_{\text{GD}}$.

5.5 Equivariance Properties

Definition – Equivariance

An estimator $\hat{\mu}$ of μ is **affine equivariant** if $\hat{\mu}(ax + b) = a\hat{\mu}(x) + b$ for all $a > 0$ and $b \in \mathbb{R}$, and **location equivariant** if $\hat{\mu}(x + b) = \hat{\mu}(x) + b$ for all $b \in \mathbb{R}$. The classes $\mathcal{C}_0$ and $\mathcal{C}_2$ are affine equivariant; $\mathcal{C}_1$ and $\mathcal{C}$ impose only location equivariance, permitting scale-dependent weights.

This matters because equivariance rules out artificial estimators that exploit arbitrary coordinate choices. A proof of admissibility inside an equivariant class says that no competitor respecting the same structural logic uniformly improves the estimator.

Affine equivariance says that the estimator has no hidden dependence on the units of measurement. If temperatures are recorded in Celsius or Fahrenheit, or if lengths are recorded in centimeters or meters, the transformed estimate is exactly the transformation of the original estimate. This is the stronger of the two constraints.

Location equivariance is weaker. It says that shifting all observations by b shifts the estimate by b , but it does not force the rule to behave identically under changes of scale. A location-equivariant class can therefore contain estimators whose behavior depends on whether the data are measured in small or large units unless the rule is built carefully. This extra freedom may create possible improvements, but it also permits procedures that are less compelling from an invariance standpoint.

The restricted program therefore bypasses the nuisance-parameter roadblock by using symmetry as a substitute for completeness. Completeness would say, roughly, that the distributional family is rich enough to force uniqueness of unbiased functions. Equivariance instead says: even if the family is not complete, we can still ask whether a competitor exists among rules that obey the same transformation laws as the original problem.

5.6 Restricted Admissibility Results

Sinha and Mouqadem proved the following restricted admissibility theorem [13, 12].

Definition – Extended admissibility

An estimator δ_0 is **extended admissible** in a class $\mathcal{D}$ if no competitor in $\mathcal{D}$ improves on it by a fixed positive risk margin everywhere. Equivalently, there is no $\delta \in \mathcal{D}$ and no $\epsilon > 0$ such that

$$\mathbf{R}(\theta; \delta) \leq \mathbf{R}(\theta; \delta_0) - \epsilon \quad \text{for every } \theta.$$

This is weaker than ordinary admissibility: it rules out a uniform fixed-margin improvement, but it does not rule out an improvement whose margin shrinks toward zero along parts of the parameter space.

Theorem – Sinha–Mouqadem restricted admissibility

For $k = 2$ with equal sample sizes, the Graybill–Deal estimator is admissible in $\mathcal{C}_0$ and in $\mathcal{C}_2$. For $n \geq 5$, it is extended admissible in the broader class $\mathcal{C}$.

The proof uses two technical lemmas from Sinha and Mouqadem [13]. Lemma 2.1 is the risk-decomposition identity for weights depending on the sample-variance ratio; it converts common-mean risk into squared-error risk for estimating the ideal variance weight. Lemma 2.4 is the analogous identity when the weight may also use the normalized squared gap $(\mathbf{S}_1^2/D^2, \mathbf{S}_2^2/D^2)$.

Lemma – Risk decomposition for variance weights

Assume $k = 2$, equal sample sizes n , and normal samples with common mean μ . Let

$$\hat{\mu} = \bar{X}_1 + D\phi, \quad D = \bar{X}_2 - \bar{X}_1,$$

where ϕ is a measurable function of $(\mathbf{S}_1^2, \mathbf{S}_2^2)$. Then, with

$$\rho = \frac{\sigma_1^2}{\sigma_1^2 + \sigma_2^2},$$

the risk satisfies

$$\mathbf{R}(\sigma_1^2, \sigma_2^2; \hat{\mu}) = \frac{\sigma_1^2 \sigma_2^2}{n(\sigma_1^2 + \sigma_2^2)} + \frac{\sigma_1^2 + \sigma_2^2}{n} \mathbb{E} \left[\phi - \frac{\sigma_1^2}{\sigma_1^2 + \sigma_2^2} \right]^2.$$

Proof of the lemma

Write

$$\hat{\mu} - \mu = \{\bar{X}_1 - \mu + \rho D\} + D(\phi - \rho).$$

The first term is the error of the known-variance weighted mean. Its variance is

$$\mathbb{V}\{\bar{X}_1 - \mu + \rho D\} = \frac{\sigma_1^2 \sigma_2^2}{n(\sigma_1^2 + \sigma_2^2)}.$$

Also,

$$\text{Cov}\{\bar{X}_1 - \mu + \rho D, D\} = -\frac{\sigma_1^2}{n} + \frac{\sigma_1^2}{\sigma_1^2 + \sigma_2^2} \frac{\sigma_1^2 + \sigma_2^2}{n} = 0.$$

Normal theory gives independence of $(\bar{X}_1, \bar{X}_2)$ and (S_1^2, S_2^2) , so ϕ is independent of the sample means. The cross term in the squared error therefore has expectation zero. Finally,

$$\mathbb{E}\{D^2(\phi - \rho)^2\} = \mathbb{V}(D) \mathbb{E}(\phi - \rho)^2 = \frac{\sigma_1^2 + \sigma_2^2}{n} \mathbb{E}(\phi - \rho)^2.$$

Adding the two terms gives the identity. ■

Proof

Write every rule in the interval class as

$$\hat{\mu} = \bar{X}_1 + D\hat{\phi}, \quad D = \bar{X}_2 - \bar{X}_1, \quad 0 \leq \hat{\phi} \leq 1.$$

For fixed σ_1^2, σ_2^2 , normal theory gives $D \perp (S_1^2, S_2^2)$, $\mathbb{E}D = 0$, and

$$\mathbb{V}(D) = (\sigma_1^2 + \sigma_2^2)/n.$$

If the rule is in C_0 , then $\hat{\phi} = \phi(S_2^2/S_1^2)$. Expanding

$$\hat{\mu} - \mu = (\bar{X}_1 - \mu) + D\phi$$

and using the covariance identity $\mathbb{E}\{(\bar{X}_1 - \mu)D\} = -\sigma_1^2/n$, the Lemma 2.1 risk decomposition gives

$$R(\sigma_1^2, \sigma_2^2; \hat{\mu}) = \frac{\sigma_1^2 \sigma_2^2}{n(\sigma_1^2 + \sigma_2^2)} + \frac{\sigma_1^2 + \sigma_2^2}{n} \mathbb{E} \left[\phi(F) - \frac{\sigma_1^2}{\sigma_1^2 + \sigma_2^2} \right]^2,$$

where $F = S_2^2/S_1^2$. Thus risk comparisons inside C_0 are exactly risk comparisons for estimating $\sigma_1^2/(\sigma_1^2 + \sigma_2^2)$ from the sample-variance ratio.

The Graybill–Deal weight

$$\phi_{GD}(F) = \frac{1}{1 + F} = \frac{S_1^2}{S_1^2 + S_2^2}$$

is a proper Bayes rule in this reduced problem for the prior used by Sinha and Mouqadem; that prior assigns positive mass to every interval in the variance-ratio parameter. If a C_0 competitor had no larger risk everywhere and smaller risk somewhere, the same competitor would have smaller Bayes risk under that prior, contradicting Bayes optimality. Hence $\hat{\mu}_{GD}$ is admissible in C_0 .

For C_2 , set

$$U = \frac{S_1^2}{D^2}, \quad V = \frac{S_2^2}{D^2}.$$

Lemma 2.4 gives the analogous identity

$$R(\sigma_1^2, \sigma_2^2; \hat{\mu}) = \frac{\sigma_1^2 \sigma_2^2}{n(\sigma_1^2 + \sigma_2^2)} + \frac{\sigma_1^2 + \sigma_2^2}{n} \mathbb{E} \left[\phi(U, V) - \frac{\sigma_1^2}{\sigma_1^2 + \sigma_2^2} \right]^2.$$

The class $\mathcal{C}_2$ therefore reduces to a second invariant estimation problem, now based on $(\mathbf{U}, \mathbf{V})$. Their admissibility argument for this reduced problem again rules out any invariant weight whose squared-error risk is everywhere no larger than the Graybill–Deal weight and strictly smaller somewhere. Translating back through the displayed identity proves admissibility in $\mathcal{C}_2$.

Finally suppose, toward a contradiction, that $\hat{\mu}_{\text{GD}}$ were not extended admissible in $\mathcal{C}$. Then for some $\epsilon > 0$ there would be $\hat{\mu} = \bar{\mathbf{X}}_1 + \mathbf{D}\phi \in \mathcal{C}$ with

$$\mathbf{R}(\sigma_1^2, \sigma_2^2; \hat{\mu}) \leq \mathbf{R}(\sigma_1^2, \sigma_2^2; \hat{\mu}_{\text{GD}}) - \epsilon \quad \text{for all } \sigma_1^2, \sigma_2^2.$$

After rescaling by the variance ratio and the statistic $\mathbf{F} = \mathbf{S}_2^2/\mathbf{S}_1^2$, Sinha and Mouqadem show that this would force an improvement of the $\mathcal{C}_0$ reduced risk by a positive amount after an arbitrarily small smoothing in the scale direction. For $n \geq 5$, the expectation of the reciprocal F -ratio needed in that smoothing is finite, and the positive margin survives. This contradicts the $\mathcal{C}_0$ admissibility just proved. Hence no fixed positive-margin improvement exists in $\mathcal{C}$, which is exactly extended admissibility.

This reduction from $\mathcal{C}$ back to $\mathcal{C}_0$ is the nontrivial technical step in the argument. The idea is that a fixed positive risk gap in the broad class is stable under averaging the competing weight over the nuisance scale while holding the variance ratio fixed; this averaging produces a ratio-only weight. The proof has to control the tails of the induced F -ratio distribution so that the averaging does not erase the positive margin, which is where the $n \geq 5$ moment condition enters. ■

They also gave admissible estimators in $\mathcal{C}_1$ of the form

$$\bar{\mathbf{X}}_1 + \mathbf{D} \frac{\mathbf{S}_1^2 + c_1}{\mathbf{S}_1^2 + \mathbf{S}_2^2 + c_1 + c_2}, \quad c_1, c_2 > 0.$$

The first part of the result is easy to interpret. In $\mathcal{C}_0$, no estimator whose weight is an arbitrary function of $\mathbf{S}_2^2/\mathbf{S}_1^2$ can uniformly dominate the Graybill–Deal weight. Thus the most obvious way to improve $\hat{\mu}_{\text{GD}}$, namely to replace $\mathbf{S}_1^2/(\mathbf{S}_1^2 + \mathbf{S}_2^2)$ by a better calibrated function of the observed variance ratio, fails. The plug-in weight is not merely a convenient first guess inside this ratio-only class; it is admissible there.

The $\mathcal{C}_2$ result is complementary rather than redundant. In $\mathcal{C}_2$, the rule may also react to the observed squared distance between the two sample means through the ratios $\mathbf{S}_i^2/\mathbf{D}^2$. One might expect this extra information to help: when the two sample means are far apart relative to the within-sample variation, perhaps the weight should be more conservative; when they are close, perhaps it should behave differently. Sinha and Mouqadem’s theorem says that even this affine-equivariant use of the observed separation does not produce a uniform risk improvement over $\hat{\mu}_{\text{GD}}$.

The extended-admissibility part has a more cautious meaning. It says that no rule in $\mathcal{C}$ beats $\hat{\mu}_{\text{GD}}$ by a fixed positive amount everywhere in the variance space. So the theorem blocks a blunt, uniform improvement.

It does not block every imaginable improvement. A remaining competitor would have to be more delicate: its gain could be positive at some parameter values, while shrinking toward zero along other parts of the variance space.

That distinction is the reason the theorem is incomplete. It strongly suggests that if $\hat{\mu}_{\text{GD}}$ is dominated inside $\mathcal{C}$, the domination must be delicate. A successful competitor could not simply shave a visible amount of risk everywhere. Its improvement would have to approach zero near some boundary or tail region of the nuisance-variance space.

5.7 What the C1 Estimators Demonstrate

The displayed $\mathcal{C}_1$ family

$$\bar{X}_1 + D \frac{S_1^2 + c_1}{S_1^2 + S_2^2 + c_1 + c_2}$$

is worth pausing over.

Theorem – Sinha–Mouqadem C1 admissible estimators

For any $c_1, c_2 > 0$, the estimator

$$\bar{X}_1 + D \frac{S_1^2 + c_1}{S_1^2 + S_2^2 + c_1 + c_2}$$

is admissible in $\mathcal{C}_1$.

Proof

Let a general member of $\mathcal{C}_1$ be

$$\hat{\mu} = \bar{X}_1 + D \phi(S_1^2, S_2^2), \quad 0 \leq \phi \leq 1.$$

As above, D is independent of (S_1^2, S_2^2) , and expansion of squared error gives

$$\mathbf{R}(\sigma_1^2, \sigma_2^2; \hat{\mu}) = \frac{\sigma_1^2 \sigma_2^2}{n(\sigma_1^2 + \sigma_2^2)} + \frac{\sigma_1^2 + \sigma_2^2}{n} \mathbb{E} \left[\phi(S_1^2, S_2^2) - \frac{\sigma_1^2}{\sigma_1^2 + \sigma_2^2} \right]^2.$$

Thus admissibility in $\mathcal{C}_1$ reduces to admissibility of the corresponding estimator of $\sigma_1^2/(\sigma_1^2 + \sigma_2^2)$ from the two independent sample variances.

Take independent inverse-gamma priors on σ_1^2 and σ_2^2 with the same shape parameter and positive scale constants proportional to c_1 and c_2 . Under the normal sampling model, the posterior scales are the prior scales plus the observed sums of squares, so the posterior mean of the relative variance weight is

$$\phi_{c_1, c_2} = \frac{S_1^2 + c_1}{S_1^2 + S_2^2 + c_1 + c_2}.$$

This is precisely the weight used in the estimator. The prior has a strictly positive density on the whole variance space, and the reduced squared risk is continuous. Therefore a competitor in $\mathcal{C}_1$ that improved the displayed estimator everywhere and strictly

somewhere would also have strictly smaller integrated Bayes risk under this prior, contradicting the Bayes optimality of the posterior mean. Hence the estimator is admissible in $\mathcal{C}_1$. ■

The estimator resembles $\hat{\mu}_{\text{GD}}$, but it adds positive constants to the sample-variance terms. These constants act like a stabilizing prior or shrinkage device for the variance weight. When $\mathbf{S}_1^2$ and $\mathbf{S}_2^2$ are small, the constants keep the weight from reacting too violently to random variance fluctuations. When the sample variances are large relative to $\mathbf{c}_1$ and $\mathbf{c}_2$, the rule behaves more like Graybill–Deal.

This result answers a different problem. Sinha and Mouqadem do not prove that $\hat{\mu}_{\text{GD}}$ itself is admissible in $\mathcal{C}_1$. They construct a shifted family of estimators that are admissible in $\mathcal{C}_1$. The construction shows that admissibility in this location-equivariant class is not empty or degenerate. There are natural weighted-average procedures, close in spirit to $\hat{\mu}_{\text{GD}}$, that can be protected against domination in $\mathcal{C}_1$.

This does not necessarily constitute evidence against $\hat{\mu}_{\text{GD}}$. Admissibility is not a uniqueness statement: several different procedures can be admissible in the same class because none uniformly dominates another. The shifted-weight estimators show that the class $\mathcal{C}_1$ contains natural protected rules; they do not imply that the unshifted Graybill–Deal rule is inadmissible.

But this does raise obvious questions: is $\hat{\mu}_{\text{GD}}$ itself admissible in the same unbiased weighted-average class? Classical arguments did not fully settle that question in the same way. Much later, the general nonstandard Bayes characterization of Duanmu, Roy, and Schriitesser, when applied to this common mean problem, yielded admissibility of $\hat{\mu}_{\text{GD}}$ in the $\mathcal{C}_1$ class

$$\bar{\mathbf{X}}_1 + D\phi(\mathbf{S}_1^2, \mathbf{S}_2^2), \quad 0 \leq \phi \leq 1,$$

for the equal-size two-sample problem [4]. Chapter 6 begins the Bayes/Blyth story, and Chapter 7 explains why that result belongs naturally to its nonstandard extension, and why it still remains a restricted-class theorem rather than a full admissibility theorem.

The distinction is important. Sinha and Mouqadem’s $\mathcal{C}_1$ theorem concerns the shifted weights with positive constants $\mathbf{c}_1$ and $\mathbf{c}_2$. It does not by itself give admissibility of the unshifted Graybill–Deal estimator in $\mathcal{C}_1$. The later Duanmu–Roy–Schriitesser result concerns $\hat{\mu}_{\text{GD}}$ itself in that same class, for the equal-size two-sample problem. Thus the two results have the same ambient class but different estimators.

5.8 Open Questions The restricted-admissibility program gives strong evidence, but it also leaves a list of doors still open.

- (a) Can extended admissibility in $\mathcal{C}$ be strengthened to exact admissibility in $\mathcal{C}$? This would still not solve the all-estimators problem, but it would close the broad interval-weighted class.
- (b) If $\hat{\mu}_{\text{GD}}$ is dominated globally, must the dominating rule leave the interval between $\bar{\mathbf{X}}_1$ and $\bar{\mathbf{X}}_2$, violate affine or location equivariance, or introduce a very delicate bias? The restricted results suggest that at least one of these departures may be necessary.

- Ⓒ How far do the two-sample equal-size arguments extend to unequal sample sizes or to $k > 2$ populations? The geometry of the weight vector becomes more complex, and the clean one-dimensional variance-ratio intuition weakens.
- Ⓓ Can the proof techniques be connected to a Bayes or complete-class argument that applies outside the restricted classes? This question leads directly to Blyth's method and to modern nonstandard Bayes characterizations.

In summary, within the most natural neighborhoods of the Graybill–Deal rule, there is no obvious domination, but this does not completely rule out the possibility that a dominating estimator can exist. The remaining admissibility problem is about whether an extremely subtle estimator, outside or on the boundary of these natural classes, can exploit the nuisance variances in a way that the restricted theory cannot see.

This is why the next bypass is not simply “make the equivariant class bigger.” Enlarging the class eventually brings back the same all-estimators difficulty that Chapter 4 exposed. Bayes and Blyth arguments are more indirect: instead of trying to describe every possible competitor by symmetry, they ask whether a hypothetical improvement would show up after risk is averaged against priors that put mass across the parameter space. In light of this, we will seek to refine the proof technology rather than merely widening the subclass.

Chapter 6:

Blyth's Method

6.1 Why Turn to Bayes Arguments? The subclass strategy explored in the previous chapter narrows the set of possible competitors. A Bayes strategy, rather, changes the proof technology. Instead of attempting to compare $\hat{\mu}_{\text{GD}}$ pointwise against every estimator directly, it averages risk over parameter values and asks whether $\hat{\mu}_{\text{GD}}$ can be obtained, or at least approximated, as a Bayes rule.

This is a natural bypass because admissibility and Bayes optimality are intimately related. If a rule is the unique Bayes rule under a proper prior with enough support, then it is typically admissible. The intuition is simple. If another estimator had risk no larger everywhere and smaller somewhere, then its average risk under a prior that sees the improvement would also be smaller. That would contradict Bayes optimality.

The difficulty is that the common mean problem is not a compact, one-parameter textbook example. The parameter space contains the common mean and nuisance variances:

$$\theta = (\mu, \sigma_1^2, \sigma_2^2).$$

The variance coordinates are unbounded, and $\hat{\mu}_{\text{GD}}$ contains random ratios of sample variances. A direct proper-prior Bayes proof is therefore not waiting on the surface. The Bayes framework is promising precisely because it can sometimes turn an impossible pointwise comparison into an average-risk comparison, but here the averaging itself becomes technically hard.

6.2 Generalized Bayes Rules

Definition – Generalized Bayes estimator

Bayesian decision theory averages risk over a prior. Under squared-error loss, the Bayes estimator is the posterior mean. If the prior is improper, the posterior mean may still be well-defined; in that case it is called a **generalized Bayes estimator**.

Generalized Bayes estimators often sit near admissibility results. In many regular problems, an estimator that is a suitable limit of Bayes rules under increasingly diffuse proper priors is admissible. This connection is one reason decision theorists keep returning to Bayes arguments even when the goal is a frequentist risk statement.

For the common mean problem, the tempting improper prior would be flat in the common mean and somehow noninformative over the variances. But different choices over the nuisance variance space induce different marginal weighting behavior, and the resulting posterior mean need not reduce to $\hat{\mu}_{\text{GD}}$ in any clean finite-dimensional way.

To see the issue, compare the known-variance and unknown-variance stories. If σ_1^2 and σ_2^2 are known, a flat prior on μ gives the posterior mean

$$\frac{\sigma_1^{-2} \bar{X}_1 + \sigma_2^{-2} \bar{X}_2}{\sigma_1^{-2} + \sigma_2^{-2}},$$

which is the inverse-variance weighted estimator. In that case the Bayes calculation, likelihood calculation, and linear unbiased calculation all point to the same rule.

With unknown variances, one might hope that a noninformative variance prior would simply replace each σ_i^2 by S_i^2 and produce $\hat{\mu}_{\text{GD}}$. But posterior averaging does not usually commute with taking ratios. The posterior mean of an inverse-variance weight is not generally the plug-in weight obtained by inserting sample variances. The nuisance-variance prior changes the posterior distribution of the relative precision, and small differences in the tail behavior of that prior can change the resulting estimator.

Thus generalized Bayes reasoning bypasses the completeness failure only if one can find the right limiting prior and then control the risk limit. The common mean problem makes both steps fragile.

6.3 Blyth's Method Blyth's method proves admissibility by constructing a sequence of priors under which the target rule is asymptotically Bayes, while the priors continue to put nonnegligible mass on every ordinary region of the parameter space. A useful classical form of the method is given below.

Definition – Blyth's method

Let Θ be a nondegenerate convex subset of Euclidean space, and suppose the risk function $\mathbf{R}(\theta, \delta_0)$ is continuous. To prove that δ_0 is admissible, it is enough to construct priors π_m , with distribution functions F_{π_m} , and exact Bayes rules δ_m satisfying three conditions.

- (a) The Bayes risks $r(\pi_m, \delta_0)$ and $r(\pi_m, \delta_m)$ are finite for each m .
- (b) For every nondegenerate convex set $C \subset \Theta$, there are constants $K > 0$ and M such that

$$\int_C dF_{\pi_m}(\theta) \geq K \quad \text{for all } m \geq M.$$

Thus the prior sequence cannot lose all mass from a fixed ordinary region.

- (c) The excess Bayes risk of the target rule vanishes:

$$\lim_{m \rightarrow \infty} \{r(\pi_m, \delta_0) - r(\pi_m, \delta_m)\} = 0.$$

These conditions force admissibility.

Suppose, toward a contradiction, that δ_0 is inadmissible. Then some estimator δ' satisfies

$$\mathbf{R}(\theta, \delta') \leq \mathbf{R}(\theta, \delta_0)$$

for all θ , with strict inequality at some point θ_0 . By continuity of the risk functions, the strict improvement persists on a convex neighborhood C of θ_0 : for some $\epsilon_1 > 0$,

$$\mathbf{R}(\theta, \delta_0) - \mathbf{R}(\theta, \delta') \geq \epsilon_1 \quad (\theta \in C).$$

Therefore, for all large m ,

$$\begin{aligned} r(\pi_m, \delta_0) - r(\pi_m, \delta') &\geq \int_C \{R(\theta, \delta_0) - R(\theta, \delta')\} dF_{\pi_m}(\theta) \\ &\geq \int_C \epsilon_1 dF_{\pi_m}(\theta) \\ &\geq \epsilon_1 K > 0. \end{aligned}$$

Since δ_m is the exact Bayes rule under π_m ,

$$r(\pi_m, \delta_0) - r(\pi_m, \delta_m) \geq \epsilon_1 K$$

for all large m . This contradicts the assumed convergence of the excess Bayes risk to zero. Hence no such dominating δ' exists, and δ_0 is admissible.

One point about the prior sequence deserves emphasis before working through an example. “Spreading the priors out” should not be confused with simply letting an ordinary proper prior become more diffuse. The condition seeks to exclude any sequence of ordinary probability priors that become so diffuse that they assign vanishing mass to every fixed compact region. Blyth’s argument instead needs enough mass to remain on each ordinary region where a possible strict risk improvement might occur, and one way to arrange this is to use finite but unnormalized prior measures rather than normalized diffuse priors. The following normal-mean example makes the mechanism explicit.

Example 3 (Blyth’s method for a normal mean). Let $\mathbf{X} \sim N(\theta, 1)$, and estimate θ under squared-error loss

$$L(\theta, \delta) = (\delta - \theta)^2.$$

The target estimator is the usual estimator

$$\delta_0(\mathbf{X}) = \mathbf{X}.$$

Equivalently, $\mathbf{X}$ may be viewed as a sample mean already standardized to have variance one. Its risk is

$$R(\theta, \delta_0) = E_\theta[(\mathbf{X} - \theta)^2] = 1.$$

The tempting diffuse prior $N(0, m)$ is not, by itself, suitable for the mass condition, because the probability assigned to any fixed bounded interval tends to zero as $m \rightarrow \infty$. Blyth’s construction instead keeps the normal shape but drops the normalizing constant depending on m . Define the finite prior measure

$$\pi_m(\theta) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{\theta^2}{2m}\right), \quad m = 1, 2, \dots$$

This is $\sqrt{m}$ times the $N(0, m)$ density, so its total mass is

$$\int_{-\infty}^{\infty} \pi_m(\theta) d\theta = \sqrt{m}.$$

First check the mass condition. If $\mathcal{C} \subset \mathbb{R}$ is any nondegenerate convex set, then for $m \geq 1$,

$$\pi_m(\theta) \geq \pi_1(\theta).$$

Therefore

$$\int_{\mathcal{C}} \pi_m(\theta) d\theta \geq \int_{\mathcal{C}} \pi_1(\theta) d\theta = K > 0.$$

Next compute the Bayes risk of δ_0 :

$$r(\pi_m, \delta_0) = \int_{-\infty}^{\infty} R(\theta, \delta_0) \pi_m(\theta) d\theta = \int_{-\infty}^{\infty} \pi_m(\theta) d\theta = \sqrt{m}.$$

This is finite for each fixed m .

Because π_m is a constant multiple of a $N(0, m)$ prior, the exact Bayes rule has the usual normal posterior mean:

$$\delta_m(\mathbf{X}) = \frac{m}{m+1} \mathbf{X}.$$

Its frequentist risk is

$$\begin{aligned} R(\theta, \delta_m) &= E_{\theta} \left[\left(\frac{m}{m+1} \mathbf{X} - \theta \right)^2 \right] \\ &= \left(\frac{m}{m+1} \right)^2 + \left(\frac{m}{m+1} \theta - \theta \right)^2 \\ &= \frac{m^2 + \theta^2}{(m+1)^2}. \end{aligned}$$

Hence

$$\begin{aligned} r(\pi_m, \delta_m) &= \int_{-\infty}^{\infty} \frac{m^2 + \theta^2}{(m+1)^2} \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{\theta^2}{2m}\right) d\theta \\ &= \sqrt{m} \left(\frac{m}{m+1} \right)^2 + \sqrt{m} \left(\frac{m}{(m+1)^2} \right) \\ &= \sqrt{m} \left(\frac{m}{m+1} \right). \end{aligned}$$

The Bayes risk of the exact Bayes rule is also finite.

Finally,

$$\begin{aligned} r(\pi_m, \delta_0) - r(\pi_m, \delta_m) &= \sqrt{m} - \sqrt{m} \left(\frac{m}{m+1} \right) \\ &= \frac{\sqrt{m}}{m+1} \longrightarrow 0. \end{aligned}$$

All three Blyth conditions are satisfied; see Figure 6.1. Therefore $\delta_0(\mathbf{X}) = \mathbf{X}$ is an admissible estimator of θ under squared-error loss in the $N(\theta, 1)$ model.

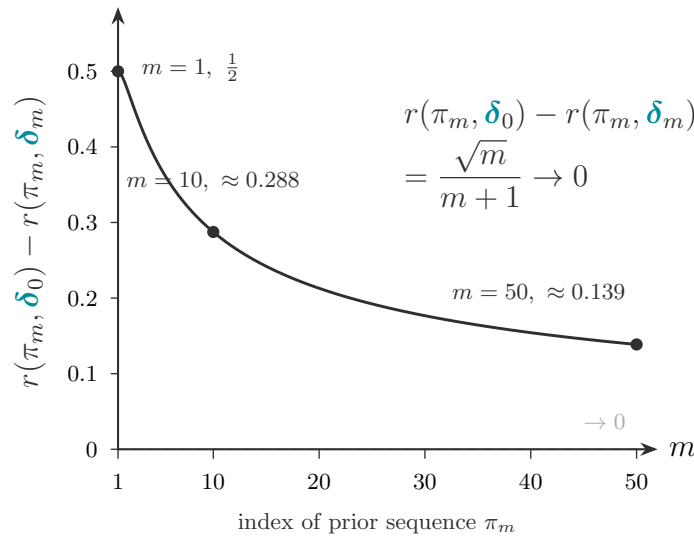


Figure 6.1. Excess Bayes risk of $\delta_0(X) = X$ under the prior sequence π_m . The quantity $r(\pi_m, \delta_0) - r(\pi_m, \delta_m) = \sqrt{m}/(m+1)$ is strictly decreasing in m and converges to zero, satisfying the third Blyth condition. The convergence is genuine but slow: at $m = 50$ the excess has fallen only to approximately 0.14. Blyth's argument requires only that the limit is zero, not that it is attained quickly.

For an ordinary single-parameter normal mean, the method is similar: the usual estimator behaves like the posterior mean under a flat prior, and the prior sequence above approximates that flat prior while retaining enough mass on every fixed interval.

The proof therefore has two moving parts.

- (a) The priors must spread through the parameter space in a way that prevents a pointwise improvement from being averaged away.
- (b) The estimator under study must be close enough to the corresponding Bayes rules that the Bayes-risk gap tends to zero at the right rate.

For ordinary one-parameter normal mean problems, both parts can be made clean. Diffuse normal priors or compactly supported priors approximate the flat prior, and the sample mean is exactly or nearly the associated Bayes estimator. For Graybill–Deal, neither part is clean because the nuisance variances must be spread out as well as μ , and because the plug-in sample-variance weight is not obviously the posterior mean under a simple variance prior.

6.4 The Wall of Intractability For $\hat{\mu}_{\text{GD}}$, a Blyth proof would need to control Bayes risks over μ and all nuisance variances. The estimator is a ratio involving random sample variances, so the integrals involve both Gaussian mean parts and chi-square variance parts. The weights change with the nuisance parameters in a nonlinear way.

Here, the mass condition is harder than in the one-dimensional normal-mean example for a specific reason. A prior sequence must spread not only along the μ -axis but also across the unbounded variance quadrant $(0, \infty)^2$. If ordinary probability priors are made increasingly diffuse over that quadrant, the prior mass assigned to any fixed variance rectangle $[a, b] \times [c, d]$ can vanish. This is exactly the failure mode ruled out by Blyth's condition 2: a strict improvement located in such a fixed variance region would be averaged away before the Bayes-risk comparison could detect it.

The most tempting calculation is also a trap. Conditional on the sample variances, a weighted-average estimator $\bar{X}_1 + D\phi(\mathbf{S}_1^2, \mathbf{S}_2^2)$ has a risk comparison that can be rewritten as a comparison between $\phi(\mathbf{S}_1^2, \mathbf{S}_2^2)$ and the unknown ideal weight

$$\frac{\sigma_1^2}{\sigma_1^2 + \sigma_2^2}$$

up to a positive scale factor. This turns a common-mean estimation problem into a variance-ratio estimation problem. That is a useful simplification, but it concentrates the nuisance parameters into difficult ratio rather than eliminating it them.

Here is a concrete way to see the obstruction. Take the equal-sample two-population case and put independent inverse-gamma priors $\sigma_i^2 \sim \text{IG}(\mathbf{a}, \mathbf{b})$, with fixed $\mathbf{a}, \mathbf{b} > 0$, on the two variances. If one makes the most optimistic shortcut and updates the variances only through the within-sample sums of squares, then the posterior precisions $\mathbf{U}_i = 1/\sigma_i^2$ have gamma laws

$$\mathbf{U}_i \mid \mathbf{S}_i^2 \sim \text{Gamma}\left(\mathbf{a} + \frac{\nu}{2}, \mathbf{b} + \frac{\nu \mathbf{S}_i^2}{2}\right), \quad \nu = n - 1,$$

where the second argument is the rate. The corresponding posterior mean would involve

$$E\left[\frac{\mathbf{U}_2}{\mathbf{U}_1 + \mathbf{U}_2} \mid \mathbf{S}_1^2, \mathbf{S}_2^2\right],$$

not the plug-in Graybill–Deal weight

$$\frac{\mathbf{S}_1^2}{\mathbf{S}_1^2 + \mathbf{S}_2^2}.$$

With $\mathbf{b} > 0$, this posterior quantity is not even scale invariant in the same way: multiplying both sample variances by the same constant leaves the Graybill–Deal weight unchanged, but it changes the two gamma rates through the fixed additive term $\mathbf{b}$. Setting $\mathbf{b} = 0$ removes that particular scale defect, but then one is already moving toward an improper or limiting prior, and the expectation of a random ratio still does not collapse automatically to the ratio with $\mathbf{S}_i^2$ inserted. The full Bayes calculation is harder still, because μ and the variances are coupled through the likelihood terms $n(\bar{X}_i - \mu)^2/\sigma_i^2$.

Example 4 (A two-population Blyth-style numerical obstruction). The normal-mean Blyth example above worked because the diffuse prior produced Bayes rules that visibly converged to the estimator X . Here is the analogous temptation for Graybill–Deal, and why it is harder.

Take two populations with $n_1 = n_2 = 10$, so $\nu = n - 1 = 9$. Suppose the summaries from one experiment are

$$\bar{X}_1 = 10, \quad \bar{X}_2 = 12, \quad \mathbf{S}_1^2 = 1, \quad \mathbf{S}_2^2 = 4.$$

The Graybill–Deal form is

$$\hat{\mu}_{\text{GD}} = \overline{X}_1 + (\overline{X}_2 - \overline{X}_1) \frac{\mathbf{S}_1^2}{\mathbf{S}_1^2 + \mathbf{S}_2^2}.$$

Here the weight on the move from $\overline{X}_1$ toward $\overline{X}_2$ is

$$\frac{1}{1+4} = 0.200,$$

so

$$\hat{\mu}_{\text{GD}} = 10 + 2(0.200) = 10.400.$$

The first proper-prior attempt. Try independent inverse-gamma priors

$$\sigma_i^2 \sim \text{IG}(2, 1), \quad i = 1, 2.$$

This is not a Blyth sequence yet; it is the simplest proper-prior trial one might write down. If we make the optimistic shortcut of updating each variance only through its within-sample sum of squares, then the posterior precisions $U_i = 1/\sigma_i^2$ satisfy

$$U_i \mid \mathbf{S}_i^2 \sim \text{Gamma}\left(2 + \frac{9}{2}, 1 + \frac{9\mathbf{S}_i^2}{2}\right),$$

with the second argument interpreted as the rate. Thus

$$U_1 \mid \mathbf{S}_1^2 = 1 \sim \text{Gamma}(6.5, 5.5), \quad U_2 \mid \mathbf{S}_2^2 = 4 \sim \text{Gamma}(6.5, 19).$$

The Bayes analogue of the Graybill–Deal weight is not obtained by inserting posterior mean variances into the ratio. It is the posterior expectation of the random precision weight,

$$E\left[\frac{U_2}{U_1 + U_2} \mid \mathbf{S}_1^2 = 1, \mathbf{S}_2^2 = 4\right].$$

A direct one-dimensional numerical integration over the ratio $U_1/(U_1 + U_2)$ gives approximately

$$0.239.$$

So this posterior-weight rule would estimate

$$10 + 2(0.239) = 10.478,$$

not 10.400.

Trying to make the prior more diffuse. A natural next move is to reduce the inverse-gamma scale parameter. Keep the shape 2, but replace the scale 1 by b . The shortcut posterior rates become

$$b + \frac{9\mathbf{S}_i^2}{2}.$$

For the same observed variances, the posterior weight behaves as follows:

b	posterior weight	resulting estimate
1.0	0.239	10.478
0.1	0.217	10.435
0.0 (formal limit)	0.215	10.429

The proper-prior estimates move toward $\hat{\mu}_{\text{GD}}$, but they do not land on it. This is already different from the $N(\theta, 1)$ Blyth example, where the Bayes rule $\frac{m}{m+1}X$ visibly converges to X . The shortcut is favorable to the attempted Bayes explanation: the full posterior also includes the terms $n(\bar{X}_i - \mu)^2/\sigma_i^2$, which couple the two variances through the unknown common mean and make the calculation less tractable, not closer to the plug-in Graybill–Deal weight.

The scale problem. Graybill–Deal is scale invariant in the observed variances. If both sample variances are multiplied by 10, the weight is still

$$\frac{10}{10 + 40} = 0.200.$$

But with the proper prior $\text{IG}(2, 1)$, the shortcut posterior weight changes from 0.239 to about 0.217, because the fixed prior scale 1 becomes less important relative to the larger within-sample sums of squares. This exposes the first of two visible failures in the numerical record: *scale contamination*. The Bayes weight responds to the absolute magnitude of the observed variances through the fixed prior parameter, whereas the Graybill–Deal ($\hat{\mu}_{\text{GD}}$) weight responds only to their ratio. The second failure is a *convergence shortfall*: even as the prior is made increasingly diffuse (as in the formal $b = 0$ limit), the Bayes weight does not converge to the plug-in weight – the gap between 0.215 and 0.200 persists.

A classical Blyth proof would need to repair both failures simultaneously. One must construct a sequence of proper priors over $(\mu, \sigma_1^2, \sigma_2^2)$ that drives the Bayes-rule gap to zero, while keeping enough prior mass on every ordinary region of the variance space to detect potential risk improvements. While the one-dimensional normal-mean proof makes these two tasks look almost automatic, in the common mean problem they pull against each other. Making the variance prior diffuse enough to reduce prior-scale artifacts is not the same thing as showing that the exact posterior mean converges to the plug-in $\hat{\mu}_{\text{GD}}$ rule at the rate Blyth's method needs. The next chapter explains why a nonstandard extension of Blyth's method can achieve exactly this in a restricted Graybill–Deal class.

This is the natural stopping point for classical Blyth theory. It tells us what a proof of admissibility should try to average, and it explains why ordinary proper-prior sequences are hard to make work in the common-mean problem. Chapter 7 keeps the same Bayes/Blyth philosophy but changes the underlying number system: infinitesimal prior masses make it possible to spread mass over the variance space without losing every ordinary region.

Chapter 7:

Hyperreals and Nonstandard Blyth's Method

Classical Blyth arguments ask for a sequence of priors whose Bayes rules approach the estimator under study while still assigning enough mass to every ordinary region where a strict risk improvement might occur. Chapter 6 showed why that request is unusually delicate for $\hat{\mu}_{\text{GD}}$: the variance coordinates are unbounded, the Bayes weight is a posterior expectation of a random ratio, and ordinary diffuse priors can lose the very regions they need to detect.

Bayes averaging still points in the right direction. The difficulty is that the averaging device must do two jobs that ordinary proper priors have trouble doing at the same time. It must be diffuse enough that the posterior rule is, up to a negligible error, $\hat{\mu}_{\text{GD}}$. In particular, the prior must not introduce an ordinary variance scale. It must also keep enough mass on every bounded ordinary region to expose a hypothetical strict risk improvement. A usual proper prior has only ordinary real mass to spend, so spreading it farther into the tails can make its local mass too small for the Blyth comparison. A formal improper prior has the opposite defect: it may suggest the right limiting rule, but it does not by itself supply a legitimate Bayes risk.

The nonstandard route gives a precise language for the phrase “a positive amount of mass too small to affect the limiting Bayes rule.” Ordinary real analysis has no positive number smaller than every positive real scale. One can only imitate such a number indirectly, usually by taking a sequence of smaller and smaller positive real numbers and then proving a limiting statement. The hyperreal number system adds new numbers for this purpose. It contains the ordinary real line, but it also contains positive infinitesimals. These are numbers larger than zero and smaller than 1, 1/10, 1/100, and in fact smaller than every ordinary positive real number.

That one extra kind of number changes the Bayes construction. A hyperreal prior can assign a positive infinitesimal mass to many parameter values. Each such mass is too small to disturb the limiting form of the Bayes rule at ordinary scales, yet it is still positive inside the enlarged number system. A hyperfinite sum of many such masses can then have a finite total mass. Thus the proof can treat the prior as an internal probability distribution while still making it look, from the ordinary real point of view, like the diffuse or improper prior suggested by the formal calculation.

This chapter uses nonstandard analysis only as much as this Bayes argument requires. The reader does not need a general theory of hyperreal calculus. Instead, four ideas carry the load. Hyperreals provide infinitesimal and unlimited numbers. The star map copies ordinary objects, such as Θ , D , risks, and priors, into the enlarged setting. Internal objects are the objects on which the copied finite arguments may be used. Hyperfinite sets may be externally infinite, but inside the nonstandard setting they behave like finite sets of the form $\{1, \dots, N\}$. This permits Bayes-risk sums with the same algebraic form as finite sums.

Two logical principles explain why the construction is more than a metaphor. Transfer says that ordinary finite statements, once written in the right first-order form, remain true after the objects are starred. Saturation supplies internal objects that satisfy many finite requirements at once. In this chapter, that means we may replace a limiting sequence of increasingly diffuse ordinary priors

by one internal prior with infinitesimal masses. The prior is proper inside the hyperreal universe, and it still places mass on the ordinary regions needed for Blyth's comparison.

With that orientation, the nonstandard approach keeps the same decision-theoretic idea as Blyth's method but permits a different kind of prior. The proof works inside a hyperreal extension where infinitesimal positive masses and hyperfinite sums are legitimate internal objects. This gives a way for the Blyth mass condition and the Bayes-risk comparison to coexist in problems where classical proper priors pull those demands apart.

7.1 From Classical Blyth to Hyperreal Priors None of this means Blyth's method is conceptually flawed; it merely indicates that the classical version is too weak for this problem. Duanmu and Roy first developed a nonstandard Bayes framework for extended admissibility, including a nonstandard analogue of Blyth's method [5]. Duanmu, Roy, and Schrittmesser then sharpened this line into an exact admissibility characterization and a Graybill–Deal application [4]. In that form, a nonstandard version of Blyth's method is sound and complete for admissibility. The argument operates in a nonstandard extension of the real line. We only need a small amount of vocabulary here; Goldblatt is a standard source for the hyperreal background and the roles of transfer, saturation, and hyperfinite sets [6].

Definition – Hyperreal number

A **hyperreal** number is an element of a nonstandard extension ${}^*\mathbb{R}$ of the real line. The extension contains the ordinary real numbers, but also contains unlimited numbers larger than every ordinary real number and infinitesimal numbers smaller in absolute value than every ordinary positive real number.

The hyperreals arise because they supply a richer notion of prior distribution. In a standard Bayesian analysis, every parameter point must receive positive **real** prior mass, which forces a trade-off between coverage and tail behavior. The hyperreals dissolve this constraint by making a third option available: mass that is positive, yet negligibly small in a precise sense.

Definition – Positive infinitesimal

A **positive infinitesimal** is a hyperreal number that is positive but smaller than every ordinary positive real number. We write $\beta^* > 0$ for one such infinitesimal.

Assigning infinitesimal mass to each parameter point produces a prior that is, in the standard sense, improper, yet remains coherent within the hyperreal framework. To reason about such priors rigorously one needs a nonstandard setting large enough to accommodate infinite index sets that still behave like finite ones.

Definition – Star map and internal object

The **star map** sends an ordinary object x to its nonstandard extension *x . Thus a parameter space Θ , a class of procedures D , a risk function r , and a prior class R become ${}^*\Theta$, *D , *r , and *R . Objects that arise inside the nonstandard universe in this way, or are built there by internal definitions, are called **internal**. Objects such as the set of all standard copies

$\{*\theta : \theta \in \Theta\}$ are usually **external**.

This distinction is not a formality. Hyperfinite summation, transferred finite convexity, and hyperpriors all require internal sets. The set of all standard parameters is typically external, so the proof does not put a prior directly on that external set. Instead it uses saturation to find an internal hyperfinite set that contains every standard parameter copy.

Definition – Hyperfinite set

A **hyperfinite** set is an internal set that is in internal bijection with $\{1, \dots, N\}$ for some hyperinteger $N \in {}^*\mathbb{N}$. If N is unlimited, the set is externally infinite, but internally it supports the same algebraic manipulations as a finite set.

Hyperfinite sets are the combinatorial backbone of the construction. If $\tilde{\Theta} = \{\theta_1, \dots, \theta_N\}$ is hyperfinite and Π is an internal probability mass function on it, then the hyper-Bayes risk of an internal procedure Δ is the hyperfinite sum

$${}^*r(\Pi, \Delta) = \sum_{i=1}^N {}^*r(\theta_i, \Delta) \Pi(\{\theta_i\}).$$

This is a literal internal finite-sum formula with N allowed to be an unlimited hyperinteger.

Definition – Transfer and saturation

Transfer says that every first-order statement true in the ordinary superstructure is true of its starred version in the nonstandard universe. **Saturation** says, in the form used here, that a family of internal requirements with the finite intersection property has a simultaneous solution, provided the family is not too large for the chosen nonstandard universe.

Together, transfer and saturation ensure that the hyperfinite Bayes analysis is a rigorous argument. Transfer licenses finite arguments after they have been written in first-order form, and saturation supplies internal hyperfinite objects satisfying many finite requirements at once. One must still keep internal and external sets separate; this bookkeeping is where the nonstandard proof does real work beyond a change of notation.

Definition – Hyperprior and standard procedures

Let $\mathbf{R}$ be a class of ordinary priors containing all finitely supported priors. A **hyperprior** is an element $\Pi \in {}^*\mathbf{R}$. If $\mathbf{D}$ is the ordinary class of procedures, write

$$\sigma\mathbf{D} = \{*\delta : \delta \in \mathbf{D}\}.$$

Thus $\sigma\mathbf{D}$ is the external standard-copy class inside ${}^*\mathbf{D}$. A procedure Δ_0 is Π -*Bayes among $\sigma\mathbf{D}$ if

$${}^*r(\Pi, \Delta_0) \leq {}^*r(\Pi, \Delta) \quad (\Delta \in \sigma\mathbf{D}).$$

Definition – Determining family

A family $\mathcal{B}$ of nonempty subsets of Θ is **determining** if every strict risk improvement at one parameter point contains a fixed positive improvement on at least one member of the family: whenever $r(\theta_0, \delta_1) < r(\theta_0, \delta_0)$ for some θ_0 , there are $B \in \mathcal{B}$ and $\epsilon > 0$ such that

$$r(\theta, \delta_1) < r(\theta, \delta_0) - \epsilon \quad (\theta \in B).$$

Definition – Nonstandard Blyth condition

The procedure δ_0 satisfies the **nonstandard Blyth condition** with respect to a determining family $\mathcal{B}$ if there are a hyperprior Π and a positive hyperreal $\tilde{\rho}$ such that, for every $B \in \mathcal{B}$, there is an ordinary constant $C_B > 0$ with

$$\tilde{\rho} \leq C_B \Pi(*B),$$

and

$$\frac{*r(\Pi, *\delta_0) - \inf_{\Delta \in \sigma D} *r(\Pi, \Delta)}{\tilde{\rho}} \approx 0.$$

This is the nonstandard version of the classical Blyth ratio. The numerator is the excess hyper-Bayes risk of δ_0 . The denominator measures the smallest scale of prior mass needed to see a real improvement on ordinary parameter regions. Requiring the ratio to be infinitesimal says that any possible strict improvement would create more Bayes-risk gain than the residual Bayes-risk gap allows.

Lemma – Finite witness for finitely many competitors

Let $G = \{\delta_1, \dots, \delta_m\} \subset D$ be finite, and let $\text{conv}(G)$ be its randomized convex hull. If δ_0 is admissible among $\text{conv}(G)$ and is not risk-equivalent to a member of that convex hull, then there is a finite set $F \subset \Theta$ such that for every $\delta \in \text{conv}(G)$ there is $\theta \in F$ with

$$r(\theta, \delta_0) < r(\theta, \delta).$$

Proof

Identify $\text{conv}(G)$ with the ordinary simplex $\{p \in [0, 1]^m : \sum_i p_i = 1\}$. For each θ , the map $\delta \mapsto r(\theta, \delta)$ is linear, hence continuous, on this simplex.

For each $\delta \in \text{conv}(G)$, admissibility and non-equivalence give a parameter θ_δ such that

$$r(\theta_\delta, \delta_0) < r(\theta_\delta, \delta).$$

By continuity in the simplex coordinate, the same strict inequality remains true throughout some open neighbourhood U_δ of δ . The simplex is compact, so finitely many such neighbourhoods $U_{\delta^{(1)}}, \dots, U_{\delta^{(k)}}$ cover it. Taking

$$F = \{\theta_{\delta^{(1)}}, \dots, \theta_{\delta^{(k)}}\}$$

gives the claimed finite witness. ■

Lemma – Finite Bayes separation

Suppose Θ_0 is finite and $G = \{\delta_1, \dots, \delta_m\}$ is finite. If δ_0 is admissible on Θ_0 among $\text{conv}(G)$, then there is a prior π on Θ_0 , positive at every point of Θ_0 , such that

$$\sum_{\theta \in \Theta_0} r(\theta, \delta_0) \pi(\theta) \leq \sum_{\theta \in \Theta_0} r(\theta, \delta) \pi(\theta) \quad (\delta \in \text{conv}(G)).$$

Proof

This is the finite complete-class theorem used by Duanmu, Roy, and Schritteser; it is the finite-parameter version of the standard Bayes/admissibility equivalence for convex randomized decision problems [4]. The proof is finite-dimensional, so we spell out the geometry.

Write a risk vector as $v(\delta) = (r(\theta, \delta))_{\theta \in \Theta_0} \in \mathbb{R}^{\Theta_0}$, and let

$$K = \{v(\delta) - v(\delta_0) : \delta \in \text{conv}(G)\}.$$

The set K is compact and convex because it is the continuous image of a simplex. Admissibility of δ_0 on Θ_0 says that K has no point in the negative orthant except possibly the origin:

$$K \cap \{x : x_\theta \leq 0 \text{ for all } \theta, x \neq 0\} = \emptyset.$$

Equivalently, K is disjoint from every compact slice

$$N_a = \{x : x_\theta \leq 0 \text{ for all } \theta, \sum_{\theta} x_\theta \leq -a\}, \quad a > 0.$$

Separating K from these slices and then using the finite-dimensional alternative theorem for convex cones gives a vector $\lambda \in (0, \infty)^{\Theta_0}$ such that

$$\sum_{\theta \in \Theta_0} \lambda_\theta \{r(\theta, \delta) - r(\theta, \delta_0)\} \geq 0 \quad (\delta \in \text{conv}(G)).$$

The strict positivity of λ is the part that uses ordinary admissibility, and it gives more than weak non-domination on a face. If every separating normal vanished on some coordinate, the corresponding parameter point could be removed from the finite game without changing the lower support, and iterating this reduction would produce either a strictly positive normal on the remaining coordinates or a competitor improving δ_0 on all coordinates of Θ_0 , contrary to admissibility. Normalizing λ to have total mass one gives the desired prior. $\blacksquare$

Theorem – Duanmu–Roy–Schritteser nonstandard Bayes characterization

Let $\mathcal{D}$ be a convex class of procedures, and suppose $\delta_0 \in \mathcal{D}$. If δ_0 is admissible among $\mathcal{D}$, then there is a hyperprior $\Pi \in {}^*\mathcal{R}$ with hyperfinite support such that

$$\Pi(\{^*\theta\}) > 0 \quad (\theta \in \Theta)$$

and $^*\delta_0$ is Π - * Bayes among $\sigma\mathcal{D}$. Conversely, any such hyperprior Bayes optimality condition implies admissibility of δ_0 among the standard procedures.

Proof

The forward implication is the theorem of Duanmu, Roy, and Schritteser [4]. The following proof records the mechanism and the points at which nonstandard analysis is used.

Assume first that δ_0 is admissible. For any finite $F \subset \Theta$ and finite $G \subset \mathcal{D}$, the request “find a hyperfinite $\tilde{\Theta} \subset {}^*\Theta$ and a hyperfinite $\tilde{\mathcal{D}} \subset {}^*\mathcal{D}$ with ${}^*F \subset \tilde{\Theta}$ and ${}^*G \subset \tilde{\mathcal{D}}$ ” is finitely satisfiable. Saturation therefore gives internal hyperfinite sets

$$\Theta \subset \tilde{\Theta} \subset {}^*\Theta, \quad \sigma\mathcal{D} \subset \tilde{\mathcal{D}} \subset {}^*\mathcal{D}.$$

The inclusions are external shorthand for saying that every standard copy is contained in the corresponding internal set.

We also need $\tilde{\Theta}$ to witness admissibility against the hyperfinite competitor hull. The finite-witness lemma says that for every finite ordinary competitor set G , some finite ordinary set of parameters witnesses that every non-equivalent member of $\text{conv}(G)$ fails to dominate δ_0 . Risk-equivalent competitors are harmless ties. Thus, after quotienting by equality of risk vectors, this is precisely the finite admissibility statement needed for the finite Bayes theorem. The statement is first-order once G is finite, and its transfer gives the corresponding hyperfinite assertion. Enlarging $\tilde{\Theta}$, if necessary, by the transferred finite witness for $\tilde{\mathcal{D}}$, we may assume that $^*\delta_0$ is admissible on $\tilde{\Theta}$ among the internal convex hull $\text{conv}(\tilde{\mathcal{D}})$, with ties treated as ties.

Now transfer the finite Bayes separation lemma. Since $\tilde{\Theta}$ and $\tilde{\mathcal{D}}$ are hyperfinite, the transferred finite theorem applies internally to the restricted hyperfinite decision problem. Hence there is an internal hyperprior Π on $\tilde{\Theta}$, positive on every point of $\tilde{\Theta}$, such that

$${}^*r(\Pi, {}^*\delta_0) \leq {}^*r(\Pi, \Delta) \quad (\Delta \in \text{conv}(\tilde{\mathcal{D}})).$$

Because $\sigma\mathcal{D} \subset \tilde{\mathcal{D}}$, this inequality holds in particular for every standard procedure copy ${}^*\delta$. Because $\{^*\theta : \theta \in \Theta\} \subset \tilde{\Theta}$ and the transferred finite prior is positive on its whole support, $\Pi(\{^*\theta\}) > 0$ for every standard θ . These positive masses may be infinitesimal; that is exactly what lets the prior cover a noncompact parameter space while still behaving internally like a proper prior.

This is the step classical Bayes arguments cannot imitate on an unbounded parameter space. A countably additive proper prior can spread out so much that ordinary regions

receive vanishing mass along a limiting sequence; an improper prior can keep the formal mass but no longer has an ordinary Bayes risk. The hyperprior keeps both features at once inside the nonstandard universe: it is an internal prior for which hyperfinite Bayes-risk sums make sense, and it still places positive mass on each standard parameter value.

Conversely, suppose such a hyperprior exists and ${}^*\delta_0$ is hyper-Bayes optimal among the standard procedures. If a standard rule δ' dominated δ_0 , then

$$r(\theta, \delta') \leq r(\theta, \delta_0) \quad \text{for every standard } \theta,$$

with strict inequality at some θ_0 . The hyper-Bayes risk difference is

$${}^*r(\Pi, {}^*\delta_0) - {}^*r(\Pi, {}^*\delta') = \sum_{\theta \in \text{supp}(\Pi)} \{ {}^*r(\theta, {}^*\delta_0) - {}^*r(\theta, {}^*\delta') \} \Pi(\theta).$$

The weak inequality is a statement about all ordinary parameters and therefore transfers to all points of ${}^*\Theta$; every summand is nonnegative. At ${}^*\theta_0$, the risk improvement is a positive ordinary real number, and the hyperprior mass is positive; its contribution is therefore strictly positive. The whole hyperfinite sum is strictly positive, contradicting Π -*Bayes optimality of ${}^*\delta_0$. Therefore no standard dominator exists, and δ_0 is admissible. ■

Theorem – Nonstandard Blyth criterion

If $\mathcal{B}$ is a determining family and δ_0 satisfies the nonstandard Blyth condition with respect to $\mathcal{B}$, then δ_0 is admissible. Conversely, every admissible procedure satisfies the nonstandard Blyth condition with respect to every determining family.

Proof

This is the Blyth part of Duanmu–Roy–Schritteser's Theorem 4.7, specialized to the notation of this chapter [4]. We include the short argument because it explains exactly why the mass condition matters.

The converse is immediate from the preceding characterization. If δ_0 is admissible, choose the hyperprior Π for which ${}^*\delta_0$ is exactly *Bayes among σD . Let

$$\tilde{\rho} = \min\{\Pi(\{\theta\}) : \theta \in \text{supp}(\Pi)\}.$$

The minimum exists because the support is hyperfinite. Since every standard point has mass at least $\tilde{\rho}$, every nonempty ordinary $B \in \mathcal{B}$ contains some θ_B with $\Pi({}^*B) \geq \Pi(\{\theta_B\}) \geq \tilde{\rho}$. Thus the mass comparison holds with $C_B = 1$. The Bayes-risk numerator is zero, so the Blyth ratio is zero.

For the admissibility direction, suppose the nonstandard Blyth condition holds but $\delta' \in D$ dominates δ_0 . Since $\mathcal{B}$ is determining, there are $B \in \mathcal{B}$ and $\epsilon > 0$ such that

$$r(\theta, \delta') < r(\theta, \delta_0) - \epsilon \quad (\theta \in B).$$

Transfer gives the same strict inequality on *B . Therefore

$${}^*r(\Pi, {}^*\delta_0) - {}^*r(\Pi, {}^*\delta') \geq \epsilon \Pi({}^*B).$$

Since ${}^*\delta' \in \sigma D$, the left side is no larger than the Blyth excess numerator. The mass condition $\tilde{\rho} \leq C_B \Pi({}^*B)$ gives

$$\frac{{}^*r(\Pi, {}^*\delta_0) - \inf_{\Delta \in \sigma D} {}^*r(\Pi, \Delta)}{\tilde{\rho}} \geq \frac{\epsilon}{C_B},$$

which is a positive ordinary number. This contradicts the requirement that the same ratio be infinitesimal. Hence no dominator exists. $\blacksquare$

This result bypasses a foundational problem in classical decision theory. Ordinary proper priors may be too coarse for noncompact parameter spaces: they can assign too little mass to regions where a hypothetical improvement occurs. Improper priors are suggestive but often lack a fully legitimate Bayes-risk interpretation. The nonstandard framework introduces priors with infinitesimal masses. Such priors can be spread over enormous parameter spaces while still giving every standard point positive attention. In that enriched setting, the Bayes/admissibility link becomes exact.

For the Graybill–Deal problem, Duanmu, Roy, and Schritteser revisit the two-sample C_1 class of unbiased weighted-average estimators

$$\hat{\mu} = \bar{X}_1 + D \hat{\phi}(S_1^2, S_2^2), \quad 0 \leq \hat{\phi} \leq 1.$$

Within that class, the risk comparison reduces to the performance of $\hat{\phi}$ as an estimator of the ideal variance-ratio weight. They use a hyperprior built from inverse-gamma distributions with infinitesimal scale parameter. Under this hyperprior, the exact Bayes weight is a stabilized version of the Graybill–Deal weight, and the excess Bayes risk of $\hat{\mu}_{GD}$ is controlled by an infinitesimal order term. The nonstandard Blyth criterion then yields admissibility within the class.

Here the hyperprior is not the hyperfinite-support prior constructed in the previous characterization theorem. It is an element of the starred ordinary prior class obtained by taking a familiar inverse-gamma family and replacing its scale parameter by a positive infinitesimal. The nonstandard Blyth criterion allows such hyperpriors; the hyperfinite-support theorem explains why this kind of Bayes reasoning is complete in principle, but the Graybill–Deal application uses a more analytic hyperprior because it gives explicit posterior calculations.

Theorem – Duanmu–Roy–Schritteser relative admissibility

For the two-sample common-mean problem, the Graybill–Deal estimator is admissible among the C_1 class of unbiased weighted-average procedures

$$\bar{X}_1 + D \phi(S_1^2, S_2^2), \quad 0 \leq \phi \leq 1.$$

Proof

Let

$$\hat{\mu}_\phi = \overline{X}_1 + D \phi(S_1^2, S_2^2), \quad 0 \leq \phi \leq 1.$$

Since D is independent of the sample variances and $\mathbb{E}D = 0$, every rule in this class is unbiased. For two weights ϕ_0, ϕ_1 , the risk difference is

$$\begin{aligned} & R(\sigma_1^2, \sigma_2^2; \hat{\mu}_{\phi_0}) - R(\sigma_1^2, \sigma_2^2; \hat{\mu}_{\phi_1}) \\ &= \frac{\sigma_1^2 + \sigma_2^2}{n} \mathbb{E} \left[\left(\phi_0 - \frac{\sigma_1^2}{\sigma_1^2 + \sigma_2^2} \right)^2 - \left(\phi_1 - \frac{\sigma_1^2}{\sigma_1^2 + \sigma_2^2} \right)^2 \right]. \end{aligned}$$

Thus the problem is exactly the reduced problem of estimating the ideal variance-ratio weight $\sigma_1^2/(\sigma_1^2 + \sigma_2^2)$ from the two sample variances.

Duanmu, Roy, and Schritteser put on (σ_1^2, σ_2^2) the product inverse-gamma hyperprior with ordinary shape parameter $0 < \alpha < 1/2$ and positive infinitesimal scale β^* . Under this hyperprior the exact Bayes weight is

$$\phi_\Pi = \frac{S_1^2 + 2\beta^*/(n-1)}{S_1^2 + S_2^2 + 4\beta^*/(n-1)},$$

a stabilized Graybill–Deal weight. This formula is the posterior mean of the ideal weight $\sigma_1^2/(\sigma_1^2 + \sigma_2^2)$ under the inverse-gamma hyperprior after observing (S_1^2, S_2^2) . The factor $2/(n-1)$ comes from the inverse-gamma posterior scale $\beta^* + (n-1)S_i^2/2$: multiplying the two posterior scale terms by the common factor $2/(n-1)$ gives $S_i^2 + 2\beta^*/(n-1)$. The same algebra is used again in Duanmu–Roy–Schritteser's Claim 5.2. Hence the corresponding rule minimizes hyper-Bayes risk in the nonstandard reduced problem.

The role of the infinitesimal scale is visible in this display. The exact hyper-Bayes rule is not literally $\hat{\mu}_{GD}$; it is the same plug-in ratio with infinitesimal stabilizing constants added to the two observed variance terms. Thus the proof does not assert that the ordinary Graybill–Deal weight is an exact posterior mean under a classical proper prior. It shows something subtler: the gap between $\hat{\mu}_{GD}$ and the exact hyper-Bayes rule is small enough, relative to the hyperprior mass on every ordinary region, to satisfy the nonstandard Blyth condition.

It remains to verify the nonstandard Blyth ratio. Let $\mathcal{B}$ be the family of bounded ordinary open subsets $O \subset (0, \infty)^2$ whose closures stay inside the variance space. This is a determining family because the risk functions in this reduced problem are continuous: a strict pointwise improvement remains strict on a small ordinary neighbourhood. For any $O \in \mathcal{B}$, choose ordinary constants $0 < \epsilon_O < M_O < \infty$ such that

$$\epsilon_O \leq \sigma_i^2 \leq M_O \quad ((\sigma_1^2, \sigma_2^2) \in O, i = 1, 2),$$

and let $\lambda(O)$ be its ordinary Lebesgue area. On O , the inverse-gamma density

$$f_{\alpha, \beta^*}(x) = \frac{(\beta^*)^\alpha}{\Gamma(\alpha)} x^{-(\alpha+1)} \exp\{-\beta^*/x\}$$

satisfies

$$\exp\{-\beta^*/\sigma_i^2\} \geq \frac{1}{2}$$

because β^* is infinitesimal and $\sigma_i^2 \geq \epsilon_O$. Also $\mathbf{x}^{-(\alpha+1)} \geq M_O^{-(\alpha+1)}$ on O . Therefore the hyperprior mass of O is bounded below by

$$\Pi(*O) \geq C_O(\beta^*)^{2\alpha}, \quad C_O = \frac{M_O^{-2(\alpha+1)}\lambda(O)}{4\Gamma(\alpha)^2} > 0.$$

Thus every ordinary test region keeps positive hyperprior mass of order $(\beta^*)^{2\alpha}$.

Now compare the Graybill–Deal weight

$$\phi_{\text{GD}} = \frac{S_1^2}{S_1^2 + S_2^2}$$

with the exact hyper-Bayes weight ϕ_{Π} . Substituting the two weights into the risk-difference identity above and integrating first over the posterior distribution of (σ_1^2, σ_2^2) , then over the predictive distribution of (S_1^2, S_2^2) , gives the excess hyper-Bayes risk. More explicitly, after conditioning on (S_1^2, S_2^2) , the posterior distribution is the product

$$\sigma_i^2 \mid S_i^2 \sim \text{IG}\left(\alpha + \frac{n-1}{2}, \beta^* + \frac{S_i^2}{2}\right), \quad i = 1, 2.$$

The posterior expected loss is a quadratic function of the chosen weight, and its minimizer is exactly ϕ_{Π} . Expanding that quadratic at its minimizer gives a nonnegative posterior excess term. The inverse-gamma posterior means and the algebra in Duanmu–Roy–Schrittesser's Claim 5.2 reduce that term, up to an ordinary constant depending only on n and α , to the bound

$$*r(\Pi, *\hat{\mu}_{\text{GD}}) - *r(\Pi, \Delta_{\Pi}) \leq C_1 \mathbb{E}_{\bar{\Pi}} \left[\frac{4(\beta^*)^2}{4\beta^* + S_1^2 + S_2^2} \right],$$

for some ordinary $C_1 > 0$, where $\bar{\Pi}$ denotes the predictive distribution of the two sample variances under the hyperprior. The remaining integral is controlled by the predictive change of variables

$$U_i = \frac{2\beta^*}{2\beta^* + S_i^2}.$$

Under the predictive distribution, U_1 and U_2 are independent beta variables. Moreover

$$\frac{4\beta^*}{4\beta^* + S_1^2 + S_2^2} = \frac{U_1}{1 + U_1/U_2} \leq 1.$$

Hence

$$*r(\Pi, *\hat{\mu}_{\text{GD}}) - *r(\Pi, \Delta_{\Pi}) \leq C'\beta^*,$$

for some ordinary $C' > 0$. Since Δ_{Π} is the exact hyper-Bayes rule in the enlarged internal reduced problem, while the Blyth infimum is only over standard procedure copies, we also have

$${}^*r(\Pi, {}^*\hat{\mu}_{\text{GD}}) - \inf_{\Delta \in \sigma C_1} {}^*r(\Pi, \Delta) \leq {}^*r(\Pi, {}^*\hat{\mu}_{\text{GD}}) - {}^*r(\Pi, \Delta_{\Pi}) \leq C'\beta^*.$$

Now set the Blyth scale

$$\tilde{\rho} = (\beta^*)^{2\alpha}.$$

The mass estimate gives

$$\tilde{\rho} \leq C_O^{-1} \Pi({}^*O) \quad (O \in \mathcal{B}),$$

which is the required mass condition, with an O -dependent ordinary constant. Combining this with the excess-risk estimate gives

$$\frac{{}^*r(\Pi, {}^*\hat{\mu}_{\text{GD}}) - \inf_{\Delta \in \sigma C_1} {}^*r(\Pi, \Delta)}{\tilde{\rho}} \leq \frac{C'\beta^*}{(\beta^*)^{2\alpha}} = C'(\beta^*)^{1-2\alpha}.$$

Equivalently, the stronger region-normalized estimate

$$\frac{{}^*r(\Pi, {}^*\hat{\mu}_{\text{GD}}) - \inf_{\Delta \in \sigma C_1} {}^*r(\Pi, \Delta)}{\Pi({}^*O)} \leq \frac{C'}{C_O} (\beta^*)^{1-2\alpha}$$

also holds for each ordinary region O . Since $0 < \alpha < 1/2$, the exponent $1 - 2\alpha$ is positive, so both displayed ratios are infinitesimal. Therefore the excess Bayes risk of $\hat{\mu}_{\text{GD}}$ is infinitesimal on the Blyth scale $\tilde{\rho}$, and $\tilde{\rho}$ is controlled by the hyperprior mass assigned to every ordinary determining region.

This is exactly the nonstandard Blyth condition for the determining family of ordinary bounded open regions in the variance space. By the nonstandard Blyth characterization, no rule in the stated C_1 unbiased class can dominate $\hat{\mu}_{\text{GD}}$. Thus $\hat{\mu}_{\text{GD}}$ is admissible within that class. ■

In plain terms, the proof has four parts. It builds a hyperprior with infinitesimal scale β^* . It shows that the excess Bayes risk of $\hat{\mu}_{\text{GD}}$ is at most of order β^* . It also shows that every ordinary variance region receives hyperprior mass at least of order $(\beta^*)^{2\alpha}$. Since $1 - 2\alpha > 0$, the ratio of these two orders is infinitesimal, which is exactly the Blyth comparison needed inside C_1 .

This is a major modern advance for two reasons. First, it gives a clean Bayes/Blyth explanation for a restricted admissibility result that had been difficult to access by ordinary finite-dimensional methods. Second, it shows that Blyth's method, in the right nonstandard form, is complete for admissibility.

It does not, however, prove admissibility among all estimators under ordinary squared-error loss. The class restriction still matters. A global competitor, if one exists, might be biased, might not be expressible through a bounded weight $\hat{\phi}(S_1^2, S_2^2)$, might depend on D^2 or other features in a

nonclassical way, or might violate the interval constraint. The modern theorem therefore sharpens the boundary of the open problem without closing it.

7.2 Questions Raised by the Bayes Bypass The Bayes and Blyth viewpoint reframes the Graybill–Deal question. Instead of asking only, “Can we calculate the risk difference against every competitor?” it asks, “Can we find priors under which $\hat{\mu}_{\text{GD}}$ is exactly or approximately Bayes, and can those priors see every possible improvement?”

That reframing raises several questions.

- (a) Is there a classical proper-prior sequence, without infinitesimals, that proves full admissibility of $\hat{\mu}_{\text{GD}}$ under ordinary squared-error loss?
- (b) If $\hat{\mu}_{\text{GD}}$ is not an exact generalized Bayes estimator under a simple improper prior, is it a limit of Bayes estimators under a more delicate variance prior construction?
- (c) Can the nonstandard $\mathcal{C}_1$ -class proof be widened to the full class $\mathcal{C}$, to unequal sample sizes, or to $k > 2$ populations?
- (d) If a global dominating estimator exists, what feature prevents it from being detected by the current Bayes/Blyth arguments?

The distinction between relative and full admissibility is easy to miss in a literature review. A relative admissibility theorem may be mathematically deep and still leave the unrestricted theorem open. The safest phrasing is: the strongest classical question is not settled, but the space in which a counterexample could live has become increasingly constrained.

The conceptual lesson is therefore subtle. Bayes methods do not magically solve the nuisance-parameter problem, but they identify the right kind of obstruction. One must choose a prior sequence, or a nonstandard prior, that spreads mass over the variance space in exactly the right way while keeping the Bayes rule close to $\hat{\mu}_{\text{GD}}$. That is why the Bayes bypass is both powerful and incomplete: it tells us what a proof should look like, and it also shows why the proof is hard.

The next bypass changes the scale of the argument while keeping the same proof philosophy. If exact finite-sample Bayes control is too delicate, one can ask whether the obstruction already appears in large-sample risk expansions. Chapter 8 takes that route. It cannot see arbitrary finite-sample competitors, but it can see smooth local corrections to $\hat{\mu}_{\text{GD}}$ at the first order where a plug-in variance penalty should matter. In that sense, the asymptotic chapter tests whether the unresolved finite-sample gap is visible in the nearest large-sample neighborhood of the estimator.

Chapter 8:

Bypassing the Roadblock II: Asymptotic Successes and Second-Order Admissibility

8.1 Shifting the Goalposts Another way around the finite-sample wall is to let sample sizes grow. Instead of asking whether a competitor can reduce exact risk at every finite sample size, asymptotic decision theory compares risk expansions. The leading term captures first-order efficiency; the next term captures the more delicate second-order behavior.

Pal and Lim use this approach to study the Graybill–Deal estimator [11]. Their paper begins from the same tension that runs through this book: the exact admissibility of $\hat{\mu}_{\text{GD}}$ among all estimators is open, despite restricted admissibility results such as those of Sinha and Mouqadem. Rather than claim to settle the finite-sample problem, Pal and Lim ask a more local asymptotic question: can a smooth $1/n$ -scale correction uniformly improve $\hat{\mu}_{\text{GD}}$ at the second-order risk level?

8.2 Second-Order Admissibility Second-order admissibility is the right asymptotic question only after the first-order question has already been settled. In the present problem, the known-variance precision-weighted estimator has risk of order n^{-1} . The Graybill–Deal estimator uses random sample variances in place of the unknown variance constants, but it still matches that known-variance benchmark through the leading n^{-1} term. Thus a competitor cannot be interesting at first order unless it gives up the usual large-sample efficiency. Any smooth improvement that preserves the first-order picture must therefore appear in the next retained term, namely the n^{-2} term of the risk expansion.

This explains both parts of the definition below. The comparison ignores terms smaller than n^{-2} , because Pal and Lim are asking whether the first available correction to the Graybill–Deal risk can be made uniformly negative. It also requires strict improvement somewhere at that order, because equality through second order would leave the two estimators indistinguishable for the asymptotic question being asked.

Definition – Second-order better

Let $n_i = n\lambda_i$, where $\lambda_i > 0$ is fixed and $n \rightarrow \infty$. For two estimators δ_1 and δ_2 , say that δ_1 is **second-order better** than δ_2 if

$$\mathbf{R}(\theta, \delta_1) \leq \mathbf{R}(\theta, \delta_2)$$

through the n^{-2} term of the risk expansion for every θ , with strict inequality at that order for at least one θ .

Definition – Second-order admissibility

An estimator is **second-order admissible** if no estimator in the class under study is second-order better than it.

This is weaker than exact admissibility but still informative. It says that any uniform improvement must hide beyond the first two terms of the asymptotic risk expansion. If $\hat{\mu}_{\text{GD}}$ failed even this test, then one could write down a smooth local correction that improves its risk at the first nontrivial asymptotic order. Pal and Lim prove that no such correction exists in their perturbation class.

8.3 The Pal–Lim Setup

For $i = 1, \dots, k$, let

$$\mathbf{X}_{ij} \stackrel{\text{iid}}{\sim} N(\boldsymbol{\mu}, \boldsymbol{\sigma}_i^2), \quad j = 1, \dots, n_i,$$

with all $\boldsymbol{\sigma}_i^2$ unknown. As in the earlier chapters,

$$\bar{\mathbf{X}}_i = \frac{1}{n_i} \sum_{j=1}^{n_i} \mathbf{X}_{ij}, \quad \mathbf{S}_i^2 = \frac{1}{n_i - 1} \sum_{j=1}^{n_i} (\mathbf{X}_{ij} - \bar{\mathbf{X}}_i)^2.$$

Then

$$\bar{\mathbf{X}}_i \sim N\left(\boldsymbol{\mu}, \frac{\boldsymbol{\sigma}_i^2}{n_i}\right), \quad \frac{(n_i - 1)\mathbf{S}_i^2}{\boldsymbol{\sigma}_i^2} \sim \chi_{n_i-1}^2,$$

and the sample means and sample variances are mutually independent across groups. Pal and Lim write $\mathbf{S}_i$ for what this textbook calls $\mathbf{S}_i^2$. This is a difference in notation for the sample variance, not a switch to the sample standard deviation, and it matters in the Taylor expansions below. In this chapter we keep the textbook convention.

The Graybill–Deal estimator is

$$\hat{\mu}_{\text{GD}} = \frac{\sum_{i=1}^k (n_i / \mathbf{S}_i^2) \bar{\mathbf{X}}_i}{\sum_{i=1}^k n_i / \mathbf{S}_i^2}.$$

Writing $n_i = n \lambda_i$, the same estimator can be written without the common factor n as

$$\hat{\mu}_{\text{GD}} = \frac{\sum_{i=1}^k (\lambda_i / \mathbf{S}_i^2) \bar{\mathbf{X}}_i}{\sum_{i=1}^k \lambda_i / \mathbf{S}_i^2}.$$

Pal and Lim consider the smooth perturbation class

$$\hat{\mu}_{\mathbf{c}} = \hat{\mu}_{\text{GD}} + \frac{1}{n} \mathbf{c}(\mathbf{T}),$$

where

$$\mathbf{T} = (\bar{\mathbf{X}}_1, \mathbf{S}_1^2, \dots, \bar{\mathbf{X}}_k, \mathbf{S}_k^2)$$

and $\mathbf{c}$ is continuously differentiable, with enough local smoothness for the Taylor expansion used in the proof. The restriction to this class is justified by the fact that $\hat{\mu}_{\text{GD}}$ is first-order efficient: the interesting local competitors are not $O(1)$ changes to the estimator, but $O(n^{-1})$ corrections.

Example 5 (Numerical size of first- and second-order terms). The phrase “second-order” can feel abstract until the two scales are put next to actual risk values. Take two populations with equal sample sizes, common mean $\mu = 0$, and variances

$$\sigma_1^2 = 1, \quad \sigma_2^2 = 4.$$

The known-variance precision-weighted estimator is

$$\hat{\mu}_{\text{known}} = \frac{(n/1)\bar{X}_1 + (n/4)\bar{X}_2}{n + n/4}.$$

Its risk is the reciprocal of the total precision:

$$\mathbf{R}(\hat{\mu}_{\text{known}}) = \frac{1}{n/\sigma_1^2 + n/\sigma_2^2} = \frac{1}{n + n/4} = \frac{0.8}{n}.$$

This $0.8/n$ term is the first-order benchmark. It is the risk level no reasonable large-sample estimator can improve by an $O(n^{-1})$ amount.

Plugging in the sample variances. Graybill–Deal replaces the true variances by the sample variances:

$$\hat{\mu}_{\text{GD}} = \frac{(n/S_1^2)\bar{X}_1 + (n/S_2^2)\bar{X}_2}{n/S_1^2 + n/S_2^2}.$$

Because S_1^2 and S_2^2 fluctuate, $\hat{\mu}_{\text{GD}}$ has slightly more risk than the impossible known-variance rule. A numerical calibration for this variance configuration gives:

n	$0.8/n$	risk of $\hat{\mu}_{\text{GD}}$	excess risk	$n^2(\text{excess})$	$\frac{\text{excess}}{0.8/n}$
20	0.0400	0.0413	0.0013	0.52	0.0325
50	0.0160	0.0163	0.00026	0.65	0.0163
100	0.0080	0.00806	0.000056	0.56	0.0070

The first two risk columns are close. The n^2 column gives the main diagnostic: multiplying the excess by n^2 leaves a roughly constant number. That is what it means, numerically, for the plug-in cost to live at the n^{-2} scale. The final column compares that cost with the first-order benchmark itself. It decreases like $1/n$, so the extra risk from estimating the variances is small relative to the leading risk term even though it is the first correction visible in the expansion.

Why perturbations are scaled as $1/n$. Now imagine a smooth competitor of the Pal-Lim form

$$\hat{\mu}_c = \hat{\mu}_{\text{GD}} + \frac{1}{n}c(\mathbf{T}).$$

If c is typically of size 1, then the estimator itself is being moved by about $1/n$. For the same three sample sizes, that is

n	$1/n$	$(1/n)^2$
20	0.050	0.0025
50	0.020	0.0004
100	0.010	0.0001

The squared size of such a perturbation is right in the same neighborhood as the excess-risk column above. So a $1/n$ -scale correction is neither too large nor too small: it is exactly the size at which a competitor could hope to change the second-order risk without disrupting first-order efficiency.

What the theorem says in this numerical picture. For $n = 50$, the first-order benchmark is about 0.0160, while the plug-in cost of using sample variances is only about 0.00026. A successful second-order competitor would have to shave risk at this smaller scale, not by changing the leading 0.0160 term. Pal and Lim's theorem says that, within their smooth perturbation class, no choice of $\mathbf{c}(\mathbf{T})$ can make that n^{-2} -level improvement uniformly over the parameter space. The table is only a numerical illustration, but it shows why the theorem is aimed at the right layer of the risk expansion.

8.4 Comparing Two Smooth Perturbations

Take two estimators in this class,

$$\hat{\mu}_c = \hat{\mu}_{\text{GD}} + \frac{1}{n}\mathbf{c}(\mathbf{T}), \quad \hat{\mu}_d = \hat{\mu}_{\text{GD}} + \frac{1}{n}\mathbf{d}(\mathbf{T}),$$

and set

$$\mathbf{g}(\mathbf{T}) = \mathbf{d}(\mathbf{T}) - \mathbf{c}(\mathbf{T}).$$

The risk difference is

$$\begin{aligned} \mathbf{R}(\hat{\mu}_d) - \mathbf{R}(\hat{\mu}_c) &= \frac{2}{n}E \left[\left\{ \hat{\mu}_{\text{GD}} + \frac{1}{n}\mathbf{c}(\mathbf{T}) - \mu \right\} \mathbf{g}(\mathbf{T}) \right] + \frac{1}{n^2}E\{\mathbf{g}^2(\mathbf{T})\} \\ &= \mathbf{A} + \mathbf{B}. \end{aligned}$$

This identity is just the algebra of squared-error loss: $(a + \epsilon)^2 - a^2 = 2a\epsilon + \epsilon^2$, with $\epsilon = n^{-1}\mathbf{g}(\mathbf{T})$.

The technical work is to expand $\mathbf{g}(\mathbf{T})$ around the true parameter point

$$\omega = (\mu, \dots, \mu, \sigma_1^2, \dots, \sigma_k^2).$$

Pal and Lim use a multivariate Taylor expansion in the centered variables

$$\bar{X}_i - \mu, \quad S_i^2 - \sigma_i^2.$$

The normal model is used in a direct way here. Odd centered sample-mean terms vanish, and the sample means are independent of the sample variances. Many cross terms in the Taylor expansion are therefore exactly zero; other terms are smaller than the n^{-2} order being retained.

8.5 The Role of η

The only first-order Taylor contribution that survives in the leading part of $\mathbf{A}$ is the derivative of $\mathbf{g}$ in the common-mean direction. Pal and Lim show that, after discarding terms of order $o(n^{-2})$,

$$\mathbf{A}_1 = \frac{2}{n^2}\eta \mathbf{g}^{(\mu)}(\omega),$$

where

$$\eta = E \left[\frac{\sum_{i=1}^k \sigma_i^2 / S_i^2}{\sum_{i=1}^k \lambda_i / S_i^2} \right].$$

This η is a positive finite scale factor. Indeed,

$$\min_i \left(\frac{\sigma_i^2}{\lambda_i} \right) \leq \eta \leq \max_i \left(\frac{\sigma_i^2}{\lambda_i} \right),$$

so η is bounded away from zero and infinity for fixed $(\sigma_1^2, \dots, \sigma_k^2, \lambda_1, \dots, \lambda_k)$, and it does not depend on n .

The rest of the retained terms are simpler. Pal and Lim obtain

$$A_2 = \frac{2}{n^2} c(\omega) g(\omega), \quad B = \frac{1}{n^2} g^2(\omega),$$

again up to $o(n^{-2})$ terms. Hence the second-order risk difference is

$$R(\hat{\mu}_d) - R(\hat{\mu}_c) = \frac{1}{n^2} \left\{ g^2(\omega) + 2c(\omega)g(\omega) + 2\eta g^{(\mu)}(\omega) \right\} + o(n^{-2}).$$

Thus $\hat{\mu}_d$ is second-order better than $\hat{\mu}_c$ precisely when the bracketed expression is nonpositive for every ω , with strict inequality somewhere. This is the point at which the problem becomes the same kind of one-dimensional second-order admissibility calculation studied by Ghosh and Sinha.

8.6 The Ghosh–Sinha Criterion For fixed variance vector $\sigma^2 = (\sigma_1^2, \dots, \sigma_k^2)$, the remaining differential inequality is in the μ direction:

$$g^2(\mu, \sigma^2) + 2c(\mu, \sigma^2)g(\mu, \sigma^2) + 2\eta g_\mu(\mu, \sigma^2) \leq 0.$$

Following Ghosh and Sinha, Pal and Lim state the necessary and sufficient tail-integral conditions under which this inequality forces $g \equiv 0$. The inequality should be read as a one-dimensional comparison problem with μ as the independent variable and the variance vector held fixed. The quadratic term in g , the drift term involving c , and the derivative term involving g_μ form a Riccati-type differential inequality. After the usual change of variables, this is equivalent to a Sturm–Liouville-type comparison problem, and the exponential integrals below are its scale functions.

Those scale functions measure whether a nonzero solution can persist all the way to $+\infty$ or $-\infty$ while keeping the risk-difference inequality nonpositive. Divergence at both ends says that both tails of the μ -axis act as natural boundaries: the comparison equation has no room for a nonzero global solution satisfying the required sign condition. This is why the two tail conditions below are exactly the obstruction to a second-order improvement.

For every fixed σ^2 , every $\mu_0 \in \mathbb{R}$, and $\eta > 0$, the conditions are

$$\int_{\mu_0}^{\infty} \frac{1}{\eta} \exp \left\{ -\frac{1}{\eta} \int_{\mu_0}^{\mu} c(u, \sigma^2) du \right\} d\mu = \infty,$$

and

$$\int_{-\infty}^{\mu_0} \frac{1}{\eta} \exp \left\{ \frac{1}{\eta} \int_{\mu}^{\mu_0} c(u, \sigma^2) du \right\} d\mu = \infty.$$

When these two integrals diverge, any competitor whose second-order risk difference is everywhere nonpositive must have $g = 0$. In other words, the competitor cannot genuinely improve $\hat{\mu}_c$ at second order. The integrals are not arbitrary side conditions. They are the scale functions for a one-dimensional Riccati-type comparison problem obtained from the differential inequality in the μ direction. Divergence at both tails means that a nonzero solution cannot remain global while keeping the second-order risk inequality nonpositive; the only global solution compatible with both tails is the trivial one.

8.7 Applying the Criterion to Graybill–Deal

Theorem – Pal–Lim second-order admissibility

Within the perturbation class

$$\hat{\mu}_c = \hat{\mu}_{\text{GD}} + \frac{1}{n} c(T),$$

the Graybill–Deal estimator $\hat{\mu}_{\text{GD}}$ is second-order admissible under squared-error loss.

Proof

Let a competitor in the same local class be

$$\hat{\mu}_g = \hat{\mu}_{\text{GD}} + \frac{1}{n} g(T).$$

Pal and Lim’s second-order expansion gives, up to $o(n^{-2})$,

$$\mathbf{R}(\omega; \hat{\mu}_g) - \mathbf{R}(\omega; \hat{\mu}_{\text{GD}}) = \frac{1}{n^2} \left\{ g^2(\omega) + 2\eta(\omega)g^{(\mu)}(\omega) \right\},$$

where $\omega = (\mu, \sigma_1^2, \dots, \sigma_k^2)$ and $\eta(\omega) > 0$ is the first-order information factor appearing in their expansion. Thus a second-order dominator would require

$$g^2(\omega) + 2\eta(\omega)g^{(\mu)}(\omega) \leq 0 \quad \text{for all } \omega,$$

with strict inequality somewhere.

The Ghosh–Sinha criterion says that this differential inequality has only the zero solution, and hence cannot yield strict improvement, provided the two tail integrals

$$\int_{\mu_0}^{\infty} \frac{1}{\eta} \exp \left\{ -\frac{1}{\eta} \int_{\mu_0}^{\mu} c(u, \sigma^2) du \right\} d\mu$$

and

$$\int_{-\infty}^{\mu_0} \frac{1}{\eta} \exp \left\{ \frac{1}{\eta} \int_{\mu}^{\mu_0} c(u, \sigma^2) du \right\} d\mu$$

both diverge. For $\hat{\mu}_{\text{GD}}$, the baseline perturbation is $\mathbf{c} \equiv 0$. The exponential factors are therefore equal to one, and the conditions reduce to

$$\int_{\mu_0}^{\infty} \frac{1}{\eta} d\mu = \infty, \quad \int_{-\infty}^{\mu_0} \frac{1}{\eta} d\mu = \infty.$$

For fixed variance vector, $0 < \eta < \infty$ and η does not shrink to zero along the μ -axis. The reason is the bound displayed earlier: η is bounded below by $\min_i \sigma_i^2 / \lambda_i > 0$, which depends only on the fixed variance vector and allocation constants, not on μ . Thus both integrals are infinite half-line lengths multiplied by a positive constant. In the ODE comparison interpretation, both ends of the μ -axis are therefore natural boundaries: a nonzero g would force the transformed solution to blow up or change sign before it could satisfy the inequality globally. The criterion applies and forces $g = 0$. Therefore no nonzero smooth $1/n$ -scale perturbation can have everywhere no larger second-order risk and somewhere smaller second-order risk than $\hat{\mu}_{\text{GD}}$. ■

This result is one of the strongest positive pieces of asymptotic evidence for $\hat{\mu}_{\text{GD}}$. It says that, after the known first-order efficiency has been accounted for, the next-order risk geometry still does not reveal a smooth local correction that uniformly improves the estimator.

8.8 What the Result Does and Does Not Prove The Pal–Lim theorem should be read with exactly the right amount of optimism. It is not an exact finite-sample admissibility theorem. The competitor class is asymptotic and smooth, and the comparison keeps risk terms only through order n^{-2} . A finite-sample dominator could, in principle, differ from $\hat{\mu}_{\text{GD}}$ in a way that is not captured by a smooth $n^{-1}\mathbf{c}(\mathbf{T})$ perturbation, or it could improve risk only through terms smaller than the second-order expansion detects.

But the theorem still has force. It rules out the most natural asymptotic route to improvement: a calibrated smooth correction to the Graybill–Deal weight. Together with the restricted admissibility results of Chapter 5 and the Bayes/Blyth perspectives of Chapters 6 and 7, it tightens the picture. If $\hat{\mu}_{\text{GD}}$ is globally inadmissible, the improvement cannot be a simple invariant weight adjustment, cannot appear as a smooth $1/n$ -scale second-order correction, and has not yet been exposed by the known Bayes-style arguments. That is why the remaining exact problem is so delicate.

The closest unresolved connection is with Chapter 5. The finite-sample bypass protects $\hat{\mu}_{\text{GD}}$, or nearby rules, inside classes whose members are still built from the sample means and sample variances in controlled ways. The second-order bypass starts from a different end: it allows smooth perturbations of the asymptotic weight and proves that none can improve the second-order risk uniformly. A direct bridge between the two would identify which finite-sample weight classes have Pal–Lim limits, and which Pal–Lim perturbations can be realized by honest finite-sample rules. Such a bridge would turn the present negative evidence into a more geometric statement: any global dominator would have to leave both the protected finite-sample classes and the smooth second-order tangent directions at the same time.

8.9 Open Questions The second-order result narrows the search for a counterexample, but it also leaves several concrete questions open.

- Ⓐ Can the Pal–Lim perturbation class be enlarged to include less smooth or more global changes to $\hat{\mu}_{\text{GD}}$ while preserving a tractable second-order risk comparison?
- Ⓑ If a finite-sample dominator exists, does its improvement first appear only at order smaller than n^{-2} , or does it escape the asymptotic expansion by being nonsmooth or boundary-sensitive?
- Ⓒ Can the second-order calculation be connected directly to the restricted classes in Chapter 5? More concretely, which subclasses of $\mathcal{C}_0$, $\mathcal{C}_1$, or $\mathcal{C}_2$ have smooth Pal–Lim limits, and which smooth Pal–Lim perturbations come from finite-sample estimators in those classes? A positive answer would show that the finite-sample and asymptotic bypasses are describing the same forbidden directions. A negative answer would locate a gap where a counterexample might still hide.
- Ⓓ Do analogous second-order admissibility results hold under unequal sample sizes, more than two populations, or alternative loss functions that still respect the common-mean structure?

At this point the three bypasses have done different jobs. Restricted admissibility controls natural finite-sample subclasses; Bayes and Blyth methods explain what a prior-based admissibility proof would have to accomplish; and second-order asymptotics rules out the most local smooth corrections. None is the missing global theorem by itself. Chapter 9 therefore steps back from proof technology and organizes the current state: which claims are about $\hat{\mu}_{\text{GD}}$, which are about alternative criteria or losses, and where the open all-estimators question still sits.

Alternative Estimators and Competing Criteria

9.1 A Brief Historical Thread By the early 1980s, the basic surprise was already clear. Graybill and Deal had given a compelling plug-in precision-weighted estimator, and later work had clarified when it improves on the individual sample means in variance. But that was not an admissibility theorem. The unresolved question was stronger: could any measurable estimator, possibly biased and nonequivariant, uniformly lower squared-error risk?

The community's response from 1982 onward was not a single straight attack but a sequence of controlled bypasses. Sinha and Mouqadem isolated natural weighted-average subclasses for the equal-sample two-population problem and proved ordinary admissibility in $\mathcal{C}_0$ and $\mathcal{C}_2$, together with extended admissibility in the larger class $\mathcal{C}$ for $n \geq 5$ [13, 12]. This was a major narrowing result: a counterexample could not simply be a better invariant recalibration of the sample-variance ratio.

At the same time, other authors changed the benchmark rather than the estimator class. Brown and Cohen, and related small-sample work reviewed by Pal and Sinha, constructed alternatives that improve particular sample-mean comparisons or use additional structure [2, 12]. Zacks and Kubokawa studied randomized, equivariant, minimax, and maximum-variance-scaled losses, showing that different losses can make different estimators, including the grand mean, decision-theoretically natural [12, 9]. Kubokawa then showed that under Pitman closeness, $\hat{\mu}_{\text{GD}}$ is closer than the individual sample means at smaller sample sizes than the variance criterion suggests [3].

The 1990s brought an asymptotic route. Pal and Lim asked whether a smooth $1/n$ -scale perturbation could improve $\hat{\mu}_{\text{GD}}$ at the second-order risk level, and showed that this local improvement is unavailable [11]. By 2005, Pal and Sinha's review could therefore summarize a wide set of strong partial results, but not a global all-estimators admissibility theorem [12].

The reason the problem resisted these attempts is structural. The common mean creates just enough linkage among the samples to make the plug-in estimator compelling, while the nuisance variances destroy completeness and keep risk comparisons tied to random ratios of scaled chi-square variables. Restricted classes, alternative losses, Pitman closeness, and asymptotic expansions each remove one layer of difficulty. None removes all of them at once. The 2021 nonstandard Bayes result of Duanmu, Roy, and Schritteser is the latest major advance in this thread: it proves admissibility of $\hat{\mu}_{\text{GD}}$ in a specific unbiased $\mathcal{C}_1$ -type class by using infinitesimal priors, while still leaving the unrestricted ordinary squared-error problem open [4].

9.2 Small Sample Size Alternatives The $n_i \geq 11$ threshold naturally led researchers to ask whether other unbiased estimators perform better for small samples. Cohen and Sackrowitz, and Brown and Cohen, constructed alternatives that improve on individual sample means under smaller sample sizes in two-population settings [12, 2]. These estimators modify the random weight rather than using the raw inverse sample-variance weight.

The point is easiest to see in the two-population case. A weighted-average rule has the form

$$\hat{\mu}_\phi = \bar{X}_1 + (\bar{X}_2 - \bar{X}_1)\phi,$$

where ϕ is a statistic built from the variance estimates and, in some constructions, from the observed difference $\bar{X}_1 - \bar{X}_2$. The Graybill–Deal rule uses the plug-in choice

$$\phi_{\text{GD}} = \frac{S_1^2}{S_1^2 + S_2^2}$$

when the sample sizes are equal. Small-sample alternatives keep the same weighted-average shape but temper this random weight. They may shrink it toward a symmetric value, truncate it, or add stabilizing constants before forming the ratio. Such changes can reduce the volatility created by estimating variances from few observations.

This does not say that $\hat{\mu}_{\text{GD}}$ has failed. It says that the comparison has changed. A small-sample modification may improve a variance comparison with an individual sample mean, while the same modification may no longer be the natural plug-in analogue of the known-variance estimator. The rest of the chapter keeps track of this distinction: the estimator, the parameter restrictions, and the criterion all matter.

9.3 Incorporating Prior Knowledge If one knows in advance that the variances are ordered, for example $\sigma_1^2 \leq \sigma_2^2$, then the unrestricted Graybill–Deal estimator need not be optimal.

Theorem – Ordered-variance improvement

In the two-population equal-sample setting, prior knowledge such as $\sigma_1^2 \leq \sigma_2^2$ can be used to construct an unbiased estimator whose variance is no larger than that of $\hat{\mu}_{\text{GD}}$ throughout the ordered variance space [13, 12].

Proof

Assume equal sample sizes and $\sigma_1^2 \leq \sigma_2^2$. The ordered-variance estimator reviewed by Pal and Sinha is

$$\hat{\mu}^* = \bar{X}_1 + D \min \left\{ \frac{1}{2}, \frac{S_1^2}{S_1^2 + S_2^2} \right\}.$$

The weight is a function of the sample variances only, so it is independent of D . Hence

$$\mathbb{E}(\hat{\mu}^* - \mu) = \mathbb{E}(\bar{X}_1 - \mu) + \mathbb{E}D \mathbb{E} \left[\min \left\{ \frac{1}{2}, \frac{S_1^2}{S_1^2 + S_2^2} \right\} \right] = 0,$$

and the modified rule is unbiased.

For any variance-measurable weight ϕ , the same expansion used earlier gives

$$\mathbf{R}(\sigma_1^2, \sigma_2^2; \bar{X}_1 + D\phi) = \frac{\sigma_1^2 \sigma_2^2}{n(\sigma_1^2 + \sigma_2^2)} + \frac{\sigma_1^2 + \sigma_2^2}{n} \mathbb{E} \left[\phi - \frac{\sigma_1^2}{\sigma_1^2 + \sigma_2^2} \right]^2.$$

Under $\sigma_1^2 \leq \sigma_2^2$, the ideal weight $\sigma_1^2/(\sigma_1^2 + \sigma_2^2)$ is at most $1/2$. The ordered estimator is the pointwise projection of the Graybill–Deal random weight

$$\phi_{\text{GD}} = \frac{S_1^2}{S_1^2 + S_2^2}$$

onto the interval $[0, 1/2]$. Projection onto a closed interval cannot increase squared distance to any point in that interval, so

$$\left[\min \left\{ \frac{1}{2}, \phi_{\text{GD}} \right\} - \frac{\sigma_1^2}{\sigma_1^2 + \sigma_2^2} \right]^2 \leq \left[\phi_{\text{GD}} - \frac{\sigma_1^2}{\sigma_1^2 + \sigma_2^2} \right]^2$$

pointwise. Taking expectations in the risk identity proves that $\mathbf{R}(\sigma_1^2, \sigma_2^2; \hat{\mu}^*) \leq \mathbf{R}(\sigma_1^2, \sigma_2^2; \hat{\mu}_{\text{GD}})$ throughout the ordered parameter space, with strict improvement where the Graybill–Deal weight exceeds $1/2$ with positive probability and the ideal weight is below $1/2$. This event has positive probability whenever $\sigma_1^2 < \sigma_2^2$, because $(n-1)S_i^2/\sigma_i^2$ has a continuous χ_{n-1}^2 law and the event $S_1^2 > S_2^2$ contains an open set of positive density. ■

Extra structural knowledge changes the decision problem. A rule designed for the fully unrestricted variance space should not be expected to exhaust the information in a restricted variance space.

9.4 Randomized Rules and the Grand Mean Zacks studied randomized and equivariant estimators based on tests comparing variance information and sample means [12]. The grand-mean result below is included for the same reason as the Pitman-closeness result: it shows how much the answer depends on the criterion. Once the loss is scaled by the largest variance, a symmetric rule can beat a rule that tries to estimate the precision ratio.

Theorem – Grand mean under maximum-variance loss

In the equal-sample two-population setting, the simple grand mean is minimax under loss scaled by the maximum variance. Zacks’ two-population result and Kubokawa’s later k -population result also establish admissibility under this loss [12, 9].

Proof

For two equal-size samples, the grand mean is

$$\bar{X}_G = \frac{\bar{X}_1 + \bar{X}_2}{2}.$$

Under the maximum-variance-scaled loss

$$L(\mu, \hat{\mu}; \sigma_1^2, \sigma_2^2) = \frac{(\hat{\mu} - \mu)^2}{\max(\sigma_1^2, \sigma_2^2)},$$

its risk is

$$\mathbf{R}(\sigma_1^2, \sigma_2^2; \bar{X}_G) = \frac{1}{\max(\sigma_1^2, \sigma_2^2)} \mathbb{V} \left(\frac{\bar{X}_1 + \bar{X}_2}{2} \right) = \frac{\sigma_1^2 + \sigma_2^2}{4n \max(\sigma_1^2, \sigma_2^2)} \leq \frac{1}{2n}.$$

The upper bound is attained when $\sigma_1^2 = \sigma_2^2$.

To see that this maximum risk cannot be improved, restrict the parameter space to the submodel $\sigma_1^2 = \sigma_2^2 = \sigma^2$. In that submodel the problem is ordinary estimation of a normal mean from $2n$ independent observations with common variance σ^2 . Under squared error divided by σ^2 , the usual full-sample mean has constant risk

$$\frac{\mathbb{V}\{(\bar{X}_1 + \bar{X}_2)/2\}}{\sigma^2} = \frac{1}{2n}.$$

The standard normal-location minimax theorem says that no estimator on this equal-variance submodel can have maximum scaled risk below this value. One way to see the lower bound is to average the risk over a flat normal prior for μ with variance tending to infinity; the Bayes risk tends to $1/(2n)$.

No estimator for the larger unequal-variance problem can have maximum risk below the minimax risk on this smaller submodel. Since the grand mean attains that lower bound, it is minimax.

The admissibility statement is a theorem from the cited literature, not a consequence of the displayed minimax calculation alone. The calculation explains why the grand mean is the right candidate under this loss: the least favorable case occurs on the equal-variance submodel, and on that submodel the grand mean is exactly the usual mean of all $2n$ observations. Kubokawa's extension uses the same normalization by $\max_{1 \leq i \leq k} \sigma_i^2$ and replaces the two-sample grand mean with the symmetric average of the k sample means. ■

These results teach a useful lesson: “best” depends on the loss function. Under ordinary squared-error loss, the natural rule is still the plug-in precision-weighted $\hat{\mu}_{GD}$. Under maximum-variance-scaled loss, the grand mean can become the right object.

9.5 Pitman Closeness Pitman closeness is a different kind of finite-sample comparison. It does not ask which estimator has smaller expected squared error; it asks which estimator is more likely to land closer to the true mean in a single experiment. Kubokawa's result therefore belongs with the competing criteria in this chapter rather than with the small-sample variance alternatives. It shows that $\hat{\mu}_{GD}$ can have a favorable closeness property even in sample-size ranges where the variance criterion is more cautious.

Theorem – Kubokawa Pitman-closeness result

In the sense of Pitman closeness, $\hat{\mu}_{\text{GD}}$ is closer than both individual sample means for $m \geq 5$ and $n \geq 5$ [3].

Proof

Let the first sample size be m and the second sample size be n . Pitman closeness of an estimator $\hat{\mu}_1$ to $\hat{\mu}_2$ means

$$\Pr_{\mu, \sigma_1^2, \sigma_2^2} \{(\hat{\mu}_1 - \mu)^2 \leq (\hat{\mu}_2 - \mu)^2\} \geq \frac{1}{2}$$

for every $\mu \in \mathbb{R}$ and every positive σ_1^2, σ_2^2 . It is enough to compare $\hat{\mu}_{\text{GD}}$ with $\bar{X}_1$, because the comparison with $\bar{X}_2$ is obtained by interchanging the two samples.

For unequal sample sizes, the two-sample Graybill–Deal rule is

$$\hat{\mu}_{\text{GD}} = \bar{X}_1 + \frac{nS_1^2}{nS_1^2 + mS_2^2}(\bar{X}_2 - \bar{X}_1).$$

The event that $\hat{\mu}_{\text{GD}}$ is at least as close to μ as $\bar{X}_1$ is

$$(\hat{\mu}_{\text{GD}} - \mu)^2 \leq (\bar{X}_1 - \mu)^2.$$

After subtracting $\bar{X}_1 - \mu$ and factoring the difference of squares, this event is equivalent to

$$\frac{nS_1^2}{nS_1^2 + mS_2^2}(\bar{X}_2 - \bar{X}_1) \left\{ 2(\bar{X}_1 - \mu) + \frac{nS_1^2}{nS_1^2 + mS_2^2}(\bar{X}_2 - \bar{X}_1) \right\} \leq 0.$$

This algebra is the bridge to Kubokawa's theorem. Kubokawa writes

$$K_1 = \frac{1}{m} \sum_{j=1}^m (X_{1j} - \bar{X}_1)^2, \quad K_2 = \frac{1}{n} \sum_{j=1}^n (X_{2j} - \bar{X}_2)^2.$$

The book's unbiased variances satisfy

$$K_1 = \frac{m-1}{m} S_1^2, \quad K_2 = \frac{n-1}{n} S_2^2.$$

Kubokawa's estimator has the form

$$\bar{X}_1 + \frac{a}{1 + R\phi}(\bar{X}_2 - \bar{X}_1), \quad R = \frac{bK_2 + c(\bar{X}_1 - \bar{X}_2)^2}{K_1},$$

where a, b, c are nonnegative constants and ϕ is positive. For $a = 1$, $c = 0$, and

$$b = \frac{m-1}{n-1},$$

we get

$$R = \frac{m-1}{n-1} \frac{K_2}{K_1} = \frac{m\mathbf{S}_2^2}{n\mathbf{S}_1^2} \Rightarrow \frac{1}{1+R} = \frac{n\mathbf{S}_1^2}{n\mathbf{S}_1^2 + m\mathbf{S}_2^2},$$

which is exactly the Graybill–Deal coefficient of $\overline{X}_2 - \overline{X}_1$. The translation from Kubokawa’s notation to the notation of this book is therefore complete.

It remains to check Kubokawa’s hypotheses. His Theorem 2.1 states that the displayed estimator is Pitman closer than $\overline{X}_1$ if

$$\phi(\cdot) \geq \frac{(m-1)a}{2(n-3)b}, \quad m \geq 2, \quad n \geq 4, \quad 0 < a \leq 4/3.$$

For $\hat{\mu}_{\text{GD}}$, the substitutions are

$$a = 1, \quad c = 0, \quad b = \frac{m-1}{n-1}, \quad \phi \equiv 1.$$

With these substitutions, the lower-bound condition becomes

$$\frac{n-1}{2(n-3)} \leq 1,$$

which holds exactly when $n \geq 5$. The other hypotheses are also satisfied: $a = 1$ lies in $(0, 4/3]$, $c = 0$ is allowed, ϕ is positive, and $m \geq 5, n \geq 5$ imply $m \geq 2, n \geq 4$. The reciprocal moment used in Kubokawa’s lemma is finite because $(n-1)K_2/\sigma_2^2$ has a χ_{n-1}^2 distribution and $\mathbb{E}(1/\chi_\nu^2) < \infty$ precisely when $\nu > 2$. This is guaranteed by $n \geq 5$.

Kubokawa’s theorem therefore gives Pitman closeness of $\hat{\mu}_{\text{GD}}$ to $\overline{X}_1$. To compare $\hat{\mu}_{\text{GD}}$ with $\overline{X}_2$, apply the same theorem after swapping the labels of the two samples. The swapped application uses

$$a = 1, \quad c = 0, \quad b = \frac{n-1}{m-1}, \quad \phi \equiv 1,$$

so its lower-bound condition is

$$\frac{m-1}{2(m-3)} \leq 1,$$

which holds exactly when $m \geq 5$. The structural hypotheses and reciprocal-moment condition are the same as above with m and n interchanged. Hence, when both sample sizes are at least five, $\hat{\mu}_{\text{GD}}$ is Pitman closer than each individual sample mean. ■

This does not contradict the variance threshold; it uses a different notion of being better. The chapter’s larger lesson is the same: the theory around $\hat{\mu}_{\text{GD}}$ has several valid comparison criteria. Change the sample-size regime, the side information, the loss function, or the closeness criterion, and the decision problem changes with it.

Where the Graybill–Deal Story Stands

1959 A Natural Beginning The story began in 1959 with a question that still sounds almost too simple to cause much trouble. Suppose several independent estimates point toward the same unknown mean, but their variances are unknown and may differ. If the variances were known, the answer would be routine: weight each sample mean by its precision. Graybill and Deal proposed doing the obvious empirical thing. They replaced each unknown variance by its sample variance and used the resulting estimated precisions as weights [7].

That plug-in step produces $\hat{\mu}_{\text{GD}}$. It is easy to compute, easy to explain, and unbiased under the normal common-mean model. Its weight moves toward the sample that appears more precise. This is exactly the behavior one would ask for before seeing any of the technical difficulties.

The first twist is that an estimated precision is random. A sample can receive too much weight because its variance happened to look small. Graybill and Deal’s finite-sample analysis showed that this danger eventually comes under control, but only after enough degrees of freedom have accumulated. For two populations, $\hat{\mu}_{\text{GD}}$ has smaller variance than each individual sample mean for every positive pair of population variances if and only if both sample sizes are at least 11. Later work extended and corrected the corresponding thresholds for several populations [7, 12].

This was a strong opening result, but it did not answer the question that came to organize the subject. Smaller variance than either sample mean is a pair of comparisons. Admissibility asks for comparison with every estimator, including biased rules, nonequivariant rules, and rules with forms that bear little resemblance to a weighted average. The passage from “better than the two obvious alternatives” to “not uniformly worse than anything” is where the story becomes difficult.

1950 The Usual Shortcut Closes For many unbiased estimation problems, one hopes to condition on a complete sufficient statistic and appeal to the Lehmann–Scheffe theorem. Here the natural statistic

$$(\bar{X}_1, S_1^2, \bar{X}_2, S_2^2)$$

is minimal sufficient, but it is not complete. The common mean links the two samples, while the separate unknown variances leave enough freedom to construct nonzero unbiased functions of the statistic. The standard route to a unique best unbiased estimator therefore stops at precisely the point where it would normally become useful.

That obstruction shaped the research that followed. Investigators did not find a single theorem that settled the unrestricted problem. They learned to ask which competitors could be ruled out, one family at a time.

1982 The Restricted-Class Campaign In 1982, Sinha and Mouqadem studied natural weighted-average classes for the equal-size two-population problem [13]. These classes describe different amounts of freedom in choosing the random weight between $\bar{X}_1$ and $\bar{X}_2$.

The class $\mathcal{C}_0$ allows the weight to depend only on the observed variance ratio. Admissibility there says that no ratio-only recalibration of the Graybill–Deal weight uniformly improves $\hat{\mu}_{\text{GD}}$. The class $\mathcal{C}_2$ also lets the weight react, in an affine-equivariant way, to the observed separation between the two sample means. Admissibility in that class rules out another tempting strategy: changing the variance-based weight whenever the sample means happen to lie far apart.

The larger interval class $\mathcal{C}$ contains all rules that remain between the two sample means. For $n \geq 5$, Sinha and Mouqadem proved extended admissibility of $\hat{\mu}_{\text{GD}}$ in this class. No competitor in $\mathcal{C}$ can improve its risk by a fixed positive margin throughout the variance space. Extended admissibility still leaves room for a dominator whose gain approaches zero near a boundary or in a tail, so this result comes close to ordinary admissibility without reaching it.

Their work on $\mathcal{C}_1$ took a related path. Sinha and Mouqadem constructed admissible estimators whose weights add positive constants to the sample variances. These shifted rules showed that the class contains natural Bayes-protected procedures. They did not yet establish admissibility of the unshifted Graybill–Deal estimator itself.

By this stage, the possible hiding places for a dominator had become rather strange. A successful competitor could not be a simple correction based only on the variance ratio. It could not be one of the affine-equivariant adjustments covered by $\mathcal{C}_2$. Within the broad interval class, it could not maintain a positive risk advantage everywhere. None of this was a global proof, but the easy candidates were disappearing.

1989 Changing the Question Other lines of work tested $\hat{\mu}_{\text{GD}}$ under different standards. Brown and Cohen and later small-sample constructions changed the weight to improve selected comparisons or to use additional information [2, 12]. Under losses scaled by the largest variance, Kubokawa found settings in which the grand mean has minimax and admissibility properties [9]. Under Pitman closeness, $\hat{\mu}_{\text{GD}}$ can be closer to the common mean than either individual sample mean when both sample sizes are at least 5, well before the variance threshold of 11 [3].

These results do not compete for a single title of “best estimator.” They show that the answer depends on what is being compared. Squared-error risk, maximum-variance-scaled loss, variance, and Pitman closeness ask different questions. The Graybill–Deal rule can look protected under one criterion while another criterion favors a modified weight or the grand mean.

1997 The Asymptotic Turn In 1997, Pal and Lim approached the problem by allowing the sample sizes to grow [11]. They asked whether a smooth correction of order $1/n$ could improve the second-order risk expansion of $\hat{\mu}_{\text{GD}}$. This is the first scale at which the cost of estimating the variances appears after the leading known-variance term.

Their answer was no. Within their smooth perturbation class, $\hat{\mu}_{\text{GD}}$ is second-order admissible. A local analytic adjustment to the plug-in weight cannot produce a uniform improvement at that order. If an exact finite-sample dominator exists, it must escape this result. It may be nonsmooth, boundary-sensitive, global rather than local, or visible only beyond the terms retained in the expansion.

Pal and Sinha’s 2005 review could therefore report a curious state of affairs [12]. Several decades of work had produced strong finite-sample results, restricted admissibility theorems, alternative procedures, and asymptotic protection. The original unrestricted admissibility question remained open.

2021 A Modern Return to Bayes Ideas The most recent development treated in this book arrived in 2021. Duanmu and Roy developed a nonstandard form of the Bayes and Blyth viewpoint, using hyperreal priors and infinitesimal Bayes-risk comparisons [5]. Duanmu, Roy, and Schrittwieser then applied the general characterization to the common-mean problem [4].

For the equal-size two-population setting, their application gives admissibility of $\hat{\mu}_{\text{GD}}$ in the unbiased $\mathcal{C}_1$ -type class

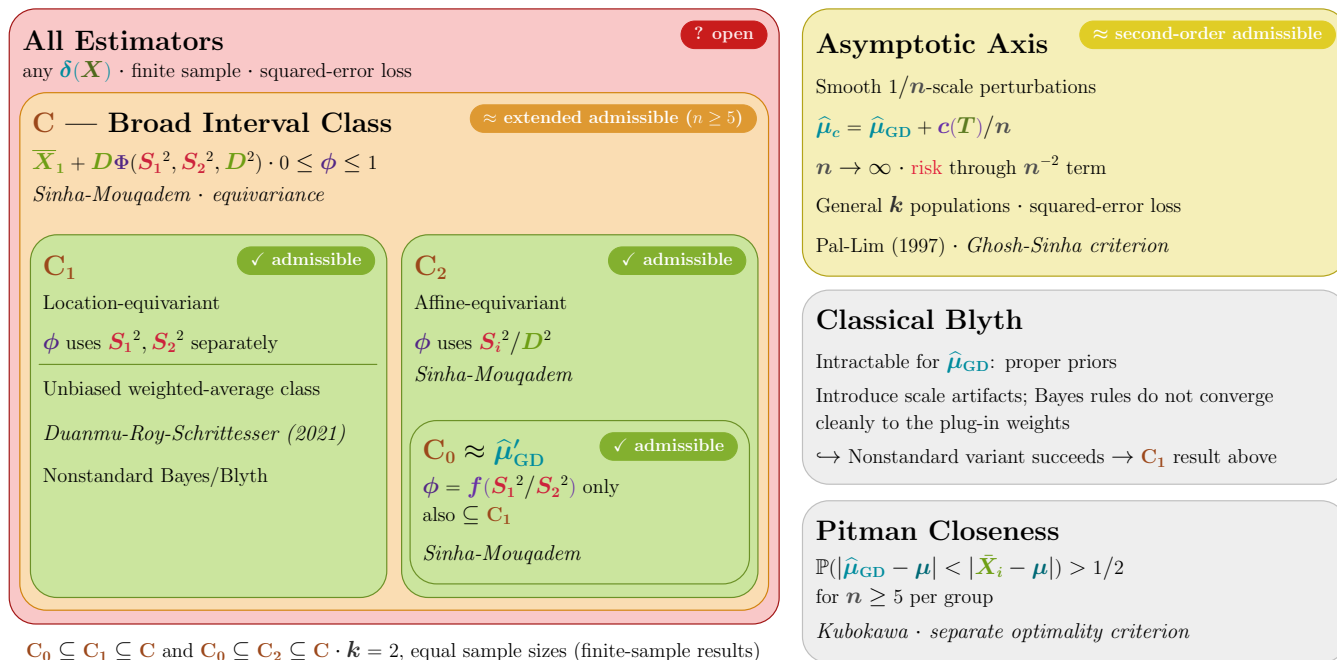
$$\overline{X}_1 + D\phi(S_1^2, S_2^2), \quad 0 \leq \phi \leq 1.$$

This closes the particular gap left by the earlier shifted-weight result: $\hat{\mu}_{\text{GD}}$ itself is protected in that class. The theorem remains a restricted-class theorem. It does not compare $\hat{\mu}_{\text{GD}}$ with every measurable estimator under ordinary squared-error loss.

The Evidence Map The results now fit into a story, but they still live on different logical levels. The figure below records those levels. The left side contains exact finite-sample statements for restricted decision classes. The right side contains asymptotic results and other methods or criteria. The outer red box is the question that has survived every chapter.

Admissibility evidence for the Graybill–Deal estimator

Two independent axes · finite-sample restricted classes (left) · asymptotic perturbations (right)



Read from the center outward, the left side shows how protection weakens as competitors gain freedom. The green regions mark classes in which the relevant admissibility statement has been proved. The red outer region represents all finite-sample estimators under ordinary squared-error loss. No theorem discussed in this book reaches that boundary.

The right side records results that run beside the finite-sample class inclusions. Second-order admissibility blocks smooth $1/n$ -scale corrections. Pitman closeness supports $\hat{\mu}_{\text{GD}}$ under a probability-of-being-closer comparison. Blyth and nonstandard Bayes methods explain how average-risk arguments can protect procedures even when ordinary proper-prior calculations do not directly settle the case.

2005 Beyond the Two-Sample Proving Ground Most of this book has stayed with two populations of equal size because that is the smallest setting in which all the main difficulties are already present. There is one variance ratio, one random weight, and one difference between sample means. Even there, the unrestricted problem remains open.

Unequal sample sizes break the symmetry. The ideal weight depends jointly on n_i and σ_i^2 , and the plug-in rule is no longer controlled by one symmetric variance ratio. Brown and Cohen's work shows how quickly small-sample improvements and confidence procedures require case-specific calibration [2].

With $k > 2$, a single variance ratio becomes a vector of relative variances. The Graybill–Deal estimator remains the natural plug-in precision-weighted average, and its variance comparisons with individual sample means have extensions with known sample-size thresholds [12]. Results under maximum-variance-scaled loss also extend to broader multi-population settings [9]. The unrestricted ordinary squared-error question persists, now with more possible directions in which a competitor can depart from $\hat{\mu}_{\text{GD}}$.

2026 The Unfinished Ending After all the machinery in the preceding chapters, the final answer is surprisingly easy to state. The known-variance precision-weighted estimator is fully understood. Its unknown-variance plug-in version is unbiased, performs well in established finite-sample comparisons, is admissible in several natural restricted classes, is extended admissible in a broad interval-weighted class, and resists smooth second-order corrections. The 2021 nonstandard Bayes application adds admissibility of $\hat{\mu}_{\text{GD}}$ in the unbiased C_1 -type class.

Yet exact admissibility among all estimators under ordinary squared-error loss has not been established by any result covered here. Nor has a dominating estimator been produced. The Graybill–Deal estimator sits between those two outcomes: protected from every direct attack discussed in this book, but not protected by a theorem broad enough to end the matter.

That unresolved point gives the subject its unusual shape. A formula proposed in 1959 still looks like the right answer. Decades of work have explained many reasons for trusting it and many reasons the proof is hard. The remaining gap is narrow in some directions and completely open in others. For now, the story ends with a well-supported estimator and a question still waiting for its final argument.

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